MUST-SCORE FACTS

Geography

Consolidated Must-Score Facts and must-score drill cards for every chapter.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

Chapters

1. Daily Read — High-Yield Revision Facts
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3. Topic 2 — Climate of India
4. Topic 3 — Drainage System
5. Topic 4 — Lakes, Waterfalls & Water Resources
6. Topic 5 — Soils
7. Topic 6 — Agriculture
8. Topic 7 — Natural Vegetation & Biodiversity Geography
9. Topic 8 — Minerals, Energy & Industry
10. Topic 9 — Transport & Communication
11. Topic 10 — Tribes & Institutions
12. Topic 11 — Population Geography
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Daily Read — High-Yield Revision Facts

1. Indian Mountain Passes (State-Wise Cheat-Sheet)

State / UT	Mountain Pass	Connects / Geographic Significance
Ladakh	**Karakoram Pass**	Ancient caravan route between India and Yarkand (Tarim Basin, China)
	Zoji La	Connects **Srinagar with Kargil and Leh**; Great Himalayas; Zojila Tunnel
	Khardung La	Gateway to the Shyok and Nubra Valleys
	Chang La	Connects Leh to Pangong Tso lake
	Fotu La	Highest point on the Srinagar-Leh highway (Zaskar Range)
	Umling La	**World's highest motorable pass/road** (19,024 ft) built by BRO
Jammu & Kashmir	**Banihal Pass**	Connects **Jammu with Srinagar**; Jawahar Tunnel under Pir Panjal Range
	Pir Panjal Pass	Traditional route from Jammu to Srinagar valley (*Peer Ki Gali*)
	Burzil Pass	Connects Srinagar with Gilgit; ancient gateway to Central Asia
Himachal Pradesh	**Rohtang Pass**	Connects Kullu Valley with Lahaul and Spiti Valleys; **Atal Tunnel** (9.02 km) under Rohtang
	Shipki La	**Sutlej River enters India** from Tibet through this pass; Zaskar Range
	Bara-lacha La	Connects **Mandi/Manali to Leh** (Zaskar Range); origin of Chandra & Bhaga rivers
	Debsa Pass	Connects Spiti and Parvati Valleys
Uttarakhand (UKPCS Top Priority!)	**Lipulekh Pass**	Tri-junction of **India, Nepal, and Tibet**; primary route for **Kailash Mansarovar Yatra** (Pithoragarh)
	Mana Pass (Dungri La)	Connects Uttarakhand with Tibet; north of Badrinath (Chamoli)
	Niti Pass	Connects Chamoli (Uttarakhand) with Tibet
	Muling La	High mountain pass in Uttarkashi north of Gangotri connecting with Tibet
	Traill's Pass	Connects **Pithoragarh with Bageshwar** (Pindari Glacier to Milam Valley)
	Sin La	Connects Darma and Byans valleys in Pithoragarh
Sikkim	**Nathu La**	Strategic border post on India-China border; Old Silk Route branch into Chumbi Valley; reopened for trade in 2006
	Jelep La	Connects **Sikkim with Lhasa (Tibet)** passing through the Chumbi Valley
Arunachal Pradesh	**Bom Di La**	Connects Arunachal Pradesh (Kameng) with Lhasa (Tibet)
	Diphu Pass	Strategic pass at the **tri-junction of India, China, and Myanmar**
	Bum La	High mountain pass near Tawang on India-China McMahon line
	Dihang Pass	Connects Arunachal Pradesh with Myanmar
	Pangsau Pass	Historic Ledo Road (Stilwell Road) connecting Assam to Myanmar

Western Ghats Gaps (Strict North-to-South Chronology)

1. **Thalghat (Kasara Ghat)**: Connects **Mumbai to Nashik / Nagpur / North India** (National Highway 160 & railway line).
2. **Bhorghat**: Connects **Mumbai to Pune / Bengaluru / South India** (Mumbai-Pune Expressway).
3. **Palghat (Palakkad Gap)**: Located between **Nilgiri Hills (North) and Anaimalai Hills (South)**; connects **Coimbatore (Tamil Nadu) to Palakkad / Kochi (Kerala)**.

4. **Shenkottai Pass**: Located between **Cardamom Hills and Agasthyamalai Hills**; connects **Kollam (Kerala) to Madurai / Tenkasi (Tamil Nadu)**.

2. Highest Peaks of Major Ranges in India

Mountain Range / Region	Highest Peak	Elevation (m)	Location / Crucial Takeaway
Trans-Himalayas (Karakoram)	**K2 (Godwin-Austen)**	**8,611 m**	**Second highest peak in the world**; highest point in Indian territory (POK)
Great Himalayas (India)	**Kanchenjunga**	**8,586 m**	**Highest peak in undisputed Indian territory**; located on Sikkim-Nepal border
Garhwal Himalayas (Uttarakhand)	**Nanda Devi**	**7,816 m**	**Highest peak located ENTIRELY within India**; Chamoli, UNESCO WHS
Garhwal (Second Highest)	**Kamet**	**7,756 m**	Chamoli district, Uttarakhand
Aravalli Range	**Guru Shikhar**	**1,722 m**	Mount Abu (Sirohi district, Rajasthan); oldest fold mountains in India
Satpura Range	**Dhupgarh**	**1,350 m**	Near Pachmarhi (Hoshangabad/Narmadapuram, Madhya Pradesh)
Vindhya Range	**Sadbhavna Shikhar (Kalumar)**	**752 m**	Damoh district, Madhya Pradesh
Western Ghats (Sahyadri)	**Anamudi**	**2,695 m**	Anaimalai Hills (Idukki, Kerala) — **HIGHEST PEAK OF PENINSULAR / SOUTH INDIA**
Nilgiri Hills	**Doda Betta**	**2,637 m**	Udhagamandalam / Ooty (Tamil Nadu); meeting point of Western & Eastern Ghats
Eastern Ghats	**Jindhagada Peak / Arma Konda**	**1,690 m**	Araku Valley (Visakhapatnam, Andhra Pradesh); Mahendragiri (1,501 m, Odisha) often given in older PCS options
Chhotanagpur Plateau	**Parasnath Peak (Sammed Shikhar)**	**1,366 m**	Giridih (Jharkhand); sacred Jain pilgrimage site (20 Tirthankaras attained Moksha)
Andaman & Nicobar Islands	**Saddle Peak**	**732 m**	North Andaman Island (Saddle Peak National Park)
Great Nicobar Island	**Mount Thullier**	**642 m**	Southernmost peak in India
Naga Hills (Purvanchal)	**Saramati Peak**	**3,826 m**	Indo-Myanmar border, Nagaland
Mizo Hills (Lushai Hills)	**Phawngpui (Blue Mountain)**	**2,157 m**	Mizoram

3. Panch Prayag & Indian River Drainage System

A. The Panch Prayag of Uttarakhand (Order along Alaknanda River)

Memorize downstream order from source towards plains: 1. **Vishnuprayag**: **Alaknanda** + **Dhauliganga** (Chamoli) 2. **Nandaprayag**: **Alaknanda** + **Nandakini** (Chamoli) 3. **Karnaprayag**: **Alaknanda** + **Pindar** (Chamoli) 4. **Rudraprayag**: **Alaknanda** + **Mandakini** (Rudraprayag) 5. **Devprayag**: **Alaknanda** + **Bhagirathi** → Official **GANGA** begins here! (Tehri Garhwal)

B. Ganga River Tributaries (Left vs Right Bank)

- **Left Bank Tributaries (West to East)**: 1. **Ramganga** (originates in Dudhatoli range, Pauri Garhwal; flows through Jim Corbett NP; meets Ganga at Kannauj) 2. **Gomati** (originates in plains at **Gomat Taal / Fulhar Lake, Pilibhit,**

UP — *only Himalayan/Ganga tributary originating in the plains!*; Lucknow & Jaunpur on its banks; joins Ganga near Saidpur, Ghazipur) 3. **Ghaghra / Karnali** (originates from Mapchachungo glacier near Mansarovar; joined by Sarda/Kali; **largest tributary by water volume**; joins Ganga at Chhapra, Bihar) 4. **Gandak** (originates in Nepal Himalayas; joins Ganga at Sonpur / Hajipur, Bihar) 5. **Kosi** (*Sorrow of Bihar*; 7 streams / Saptakoshi; notorious for course-shifting; joins at Kursela, Katihar) 6. **Mahananda** (originates in Darjeeling hills; **last left-bank tributary of Ganga in India**).

- **Right Bank Tributaries**: 1. **Yamuna** (longest tributary, 1376 km; Yamunotri glacier, Bandarpunch peak; joins at Prayagraj Triveni Sangam; its tributaries: Chambal, Sind, Betwa, Ken, Tons, Hindon) 2. **Son** (originates from **Amarkantak Plateau, MP**; flows north-eastwards through Sonbhadra UP; joins Ganga near Danapur/Patna, Bihar) 3. **Punpun** (originates in Palamu, Jharkhand; joins Ganga at Fatuha near Patna).

C. Peninsular Drainage: West-Flowing vs East-Flowing

- **West-Flowing (into Arabian Sea — form Estuaries, NO deltas)**:
- **Narmada** (1312 km): Amarkantak plateau (MP); flows through **Rift Valley** between Vindhyas (North) and Satpuras (South); forms **Dhuandhar Falls** at Marble Rocks (Bhedaghat, Jabalpur); Sardar Sarovar Dam.
- **Tapi / Tapti** (724 km): Multai (Betul district, MP); flows in rift valley south of Satpuras; Ukai Dam & Kakrapar Project (Gujarat).
- **Sabarmati**: Dhebar Lake (Aravalli Range, Udaipur); flows through Ahmedabad and Gandhinagar into Gulf of Khambhat.
- **Mahi**: Vindhyan hills (Dhar, MP); **CUTS THE TROPIC OF CANCER TWICE!**
- **Luni**: Originates near Ajmer (Aravallis) as Sagarmati; **ends in the marshy Rann of Kutch (Inland drainage, never reaches the sea)**; sweet water up to Balotra, then saline.
- **Sharavati**: Shimoga (Karnataka); forms famous **Jog Falls (Gersoppa / Mahatma Gandhi Falls)**.
- **Periyar & Bharathappuzha (Ponnani)**: Kerala.
- **East-Flowing (into Bay of Bengal — form large Deltas)**:
- **Mahanadi** (851 km): Sihawa hills (Dhamtari, Chhattisgarh); flows through Odisha; Hirakud Dam.
- **Godavari** (1465 km — **Dakshin Ganga / Vriddha Ganga**): Trimbakeshwar (Nashik, Maharashtra); **longest river in Peninsular India**; Tributaries: Pravara, Manjra, Penganga, Wainganga, Wardha, **Pranhita** (largest), Indravati, Sabari.
- **Krishna** (1401 km): Mahabaleshwar (Maharashtra); Tributaries: Koyna, Ghataprabha, Malaprabha, Bhima, **Tungabhadra** (largest), Musi (Hyderabad on its banks), Munneru.
- **Kaveri** (800 km — **Ganga of the South**): Talakaveri (Brahmagiri Range, Coorg, Karnataka); flows year-round (perennial due to SW & NE monsoons); creates river islands: **Srirangapatna, Shivanasamudra, and Srirangam**; Tributaries: Hemavati, Harangi, Shimsha, Arkavathi (left); Lakshmantirtha, Kabini, Bhavani, Noyyal, Amaravati (right).

4. Multipurpose River Valley Projects, Dams & Waterfalls

Project / Dam / Waterfall	River	State	Exam Crucial Fact
Tehri Dam	**Bhagirathi + Bhilangana**	**Uttarakhand**	**Highest dam in India** (260.5 m); rock and earth-fill embankment
Bhakra Nangal Dam	**Sutlej**	Himachal Pradesh / Punjab	Highest straight gravity dam in India; reservoir is **Gobind Sagar**
Hirakud Dam	**Mahanadi**	Odisha (Sambalpur)	**Longest earthen dam in India / world** (main dam 4.8 km, total 25.8 km)
Sardar Sarovar Dam	**Narmada**	Gujarat (Navagam)	Beneficiary states: Gujarat, MP, Maharashtra, Rajasthan; Statue of Unity nearby

Project / Dam / Waterfall	River	State	Exam Crucial Fact
Nagarjuna Sagar Dam	**Krishna**	Andhra Pradesh / Telangana border	World's largest masonry dam; creates Nagarjunasagar reservoir
Rihand Project (Govind Ballabh Pant Sagar)	**Rihand** (tributary of Son)	**Sonbhadra (Uttar Pradesh)**	**Largest man-made artificial reservoir / lake in India by volume**
Mata Tila & Rajghat Dam	**Betwa**	Lalitpur (Uttar Pradesh / MP)	Joint venture between UP and MP
Kunchikal Falls	**Varahi**	Karnataka (Shimoga)	Highest tiered waterfall in India (455 m)
Jog Falls (Gersoppa)	**Sharavati**	Karnataka (Shimoga)	Highest untiered plunge waterfall (253 m)
Dhuandhar Falls	**Narmada**	Bhedaghat, Jabalpur (MP)	"Smoke cascade" cutting through marble gorge
Chitrakote Falls	**Indravati**	Bastar (Chhattisgarh)	Known as the **"Niagara of India"** (widest horseshoe waterfall in India)
Hundru Falls	**Subarnarekha**	Ranchi (Jharkhand)	98 m drop on Chotanagpur plateau
Nohkalikai Falls	Rain-fed stream	Cherrapunji (Meghalaya)	Tallest plunge waterfall in India (340 m)

5. Major Agricultural Crops & Top Producing States

Crop Category	Specific Crop	Top Producing State (1st Rank)	Other Key States / Prelims Note
Foodgrains	**Total Foodgrains**	**Uttar Pradesh**	Followed by Madhya Pradesh, Punjab
	Wheat	**Uttar Pradesh**	Followed by MP, Punjab (Punjab highest per hectare yield)
	Rice	**West Bengal**	Followed by Uttar Pradesh, Punjab
	Maize	**Karnataka**	Followed by Madhya Pradesh, Maharashtra
	Coarse Cereals (Nutri-cereals)	**Rajasthan**	Followed by Karnataka, Maharashtra
	Total Pulses	**Madhya Pradesh**	Followed by Maharashtra, Rajasthan
	Gram (Chana)	**Maharashtra / MP**	Accounts for ~40% of total pulses
Oilseeds	**Total Oilseeds**	**Rajasthan**	Followed by Maharashtra, Madhya Pradesh
	Rapeseed & Mustard	**Rajasthan**	(>45% of national production)
	Soyabean	**Maharashtra / MP**	MP traditionally called *Soyabean State*
	Groundnut	**Gujarat**	Followed by Rajasthan, Tamil Nadu
Cash / Commercial	**Sugarcane**	**Uttar Pradesh**	Followed by Maharashtra, Karnataka (Maha highest sugar recovery)
	Cotton	**Gujarat**	Followed by Maharashtra, Telangana
	Jute & Mesta	**West Bengal**	(>75% of national production - Golden Fibre)
	Tea	**Assam**	Followed by West Bengal (Darjeeling - 1st GI tag 2004), Tamil Nadu
	Coffee	**Karnataka**	(>70% of national production - Arabica & Robusta in Kodagu/Chikkamagaluru)
	Natural Rubber	**Kerala**	(>75% of national production)
	Tobacco	**Gujarat / Andhra Pradesh**	CTRI located at Rajahmundry (AP)

6. Major Tribes of India & State-Wise Matrix (Direct Match-Ups)

State / UT	Prominent Indigenous Tribes	Key Ethnographic / Exam Fact
Uttarakhand (UKPCS Top Priority!)	**Tharu, Jaunsari, Bhotia, Buksa, Raji (Van Rawat)**	**Tharu**: Largest tribe in UK & UP; historically celebrate Diwali as mourning (*Shok Parva*); live in Terai (US Nagar). **Jaunsari**: Second largest; polyandry tradition; Dehradun (Chakrata). **Bhotia**: Transhumance (*Ritu Pravas*); seasonal migration to bugyals; Indo-Tibetan trade. **Buksa**: Claim Rajput descent (Pawar); Terai region. **Raji**: Smallest tribe (*Van Rawat* / kings of forest); cave dwellings.
Uttar Pradesh	**Tharu, Gond, Kharwar, Sahariya, Parahiya, Baiga, Agariya, Chero**	**Sonbhadra** has highest ST population & diversity in UP; Sahariya in Lalitpur; Agariya (iron smelters) in Sonbhadra.
Madhya Pradesh	**Bhil, Gond, Baiga, Sahariya, Kol, Korku, Bharia**	**Highest ST population in India**; Bhils largest tribe of India; *Bhagoria* festival; *Pithora* painting.
Rajasthan	**Bhil, Meena, Garasia, Sahariya, Damor**	Meenas are the most populous tribe in Rajasthan; Sahariya in Baran.
Odisha	**Santhal, Bonda, Dongria Kondh, Juang, Saora**	Largest number of **Particularly Vulnerable Tribal Groups (PVTGs: 13 out of 75 in India)**; Dongria Kondh worship Niyamgiri hills.
Tamil Nadu	**Toda, Badaga, Kota, Irula, Kurumba**	**Toda**: Inhabit Nilgiri Hills; pastoral tribe centered on water buffaloes; practice fraternal polyandry; barrel-shaped huts.
Andhra Pradesh / Telangana	**Chenchu, Yanadi, Koya, Lambada**	Chenchu: Primitive hunter-gatherers in Nallamala forest / Srisailam.
Sikkim	**Lepcha, Bhutia, Limbu**	Lepchas are indigenous inhabitants; practice sacred groves.
Himachal Pradesh	**Gaddi, Gujjar, Kinnaura, Pangwala**	Gaddis: Pastoral transhumance shepherds of Dhauladhar range.
Andaman & Nicobar Islands	**Great Andamanese, Onge, Jarawa, Sentinelese, Shompen, Nicobarese**	Negrito stock (Jarawa, Sentinelese, Onge - North Sentinel isolated); Shompen are Mongoloid stock.

7. Major Sea Ports & National Waterways of India

A. 13 Major Ports of India

- **Western Coast**: 1. **Deendayal Port (Kandla)** (Gulf of Kutch, Gujarat): Tidal port; free trade zone (SEZ); handles crude oil & grains. 2. **Mumbai Port** (Maharashtra): Largest natural port and harbour in India. 3. **Jawaharlal Nehru Port (JNPT / Nhava Sheva)** (Navi Mumbai): **Largest container port in India**; fully mechanized. 4. **Mormugao Port** (Goa - Zuari estuary): Premier port for iron ore exports from Goa. 5. **New Mangalore Port** (Karnataka): Handles iron ore exports from Kudremukh. 6. **Cochin Port** (Kerala - Willingdon Island in Vembanad Lake): Natural harbour; International Container Transshipment Terminal (Vallarpadam).
- **Eastern Coast**: 7. **V.O. Chidambaranar Port (Tuticorin)** (Tamil Nadu - Gulf of Mannar): Artificial deep-sea port; pearl fishery. 8. **Chennai Port** (Tamil Nadu): **Oldest artificial harbour on the east coast**; second busiest. 9. **Kamarajar Port (Ennore)** (Tamil Nadu): **First corporatized port in India** (public company). 10. **Visakhapatnam Port** (Andhra Pradesh): **Deepest landlocked and protected port in India** (Dolphin's Nose hill protects from storms). 11. **Paradip Port** (Odisha - Mahanadi delta): Deep-water port exporting iron ore to Japan. 12. **Syama Prasad Mookerjee Port (Kolkata–Haldia)** (West Bengal): **Only riverine major port in India** (on Hooghly River); twin dock system (Kolkata & Haldia).
- **Island**: 13. **Port Blair Port** (Andaman & Nicobar Islands).

B. Core National Waterways (NW) of India

- **NW-1**: **Ganga–Bhagirathi–Hooghly River System** from **Prayagraj to Haldia** (1,620 km) — **Longest National Waterway in India**; passes through UP, Bihar, Jharkhand, WB.
- **NW-2**: **Brahmaputra River** from **Sadiya to Dhubri** (891 km) in Assam.
- **NW-3**: **West Coast Canal** from **Kottapuram to Kollam** (205 km) in Kerala (includes Champakara & Udyogmandal canals).
- **NW-4**: **Krishna–Godavari Rivers & Canals** from Kakinada to Puducherry (1,078 km).
- **NW-5**: **Brahmani River, Deltaic channels of Mahanadi & East Coast Canal** (588 km) in Odisha & WB.
- **NW-16**: **Barak River** from Lakhipur to Bhanga (121 km) in Assam.

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8. Census 2011 Essential Fact Card (India vs UP vs UK)

Demographic Indicator	All-India	Uttar Pradesh	Uttarakhand
Total Population	**121.08 Crore** (17.5% of world)	**19.98 Crore** (16.51% of India - Rank 1)	**1.01 Crore** (0.83% of India - Rank 20)
Decadal Growth Rate (2001–11)	**17.70%** (Highest: Meghalaya 27.95%; Lowest/Negative: Nagaland -0.58%)	**20.22%** (Highest: Gautam Buddha Nagar 49.1%; Lowest: Kanpur Nagar 9.9%)	**18.81%** (Highest: US Nagar 33.4%; Negative: Almora -1.28%, Pauri Garhwal -1.41%)
Population Density	**382 per sq km** (Highest: Bihar 1106; Lowest: Arunachal 17)	**829 per sq km** (Rank 4 among states) • Highest: **Ghaziabad** (3,971) • Lowest: **Lalitpur** (242)	**189 per sq km** (Rank 25) • Highest: **Haridwar** (801) • Lowest: **Uttarkashi** (41)
Overall Sex Ratio	**943 females / 1000 males** (Highest: Kerala 1084; Lowest: Haryana 879)	**912 females / 1000 males** • Highest: **Jaunpur** (1,024), Azamgarh (1,019), Deoria (1,017) • Lowest: **Gautam Buddha Nagar** (851)	**963 females / 1000 males** • Highest: **Almora** (1,139), Rudraprayag (1,114) • Lowest: **Haridwar** (880), Dehradun (902)
Child Sex Ratio (0–6 yrs)	**919** (Highest: Mizoram 970; Lowest: Haryana 834)	**902** • Highest: **Balrampur** (950) • Lowest: **Baghpat** (841)	**890** • Highest: **Pithoragarh** (916) • Lowest: **Pithoragarh / Champawat**
Literacy Rate (Overall)	**73.0%** (Male: 80.9%, Female: 64.6%; Gap: 16.3%)	**67.7%** (Male: 77.3%, Female: 57.2%; Gap: 20.1%) • Highest: **Gautam Buddha Nagar** (80.1%) • Lowest: **Shravasti** (46.7%)	**78.82%** (Male: 87.4%, Female: 70.0%; Gap: 17.4%) • Highest: **Dehradun** (84.25%) • Lowest: **Haridwar** (73.43%)
SC Population %	**16.6%** (Highest %: Punjab 31.9%; Highest No.: UP)	**20.7%** of state pop (Highest No.: Sitapur; Highest %: Kaushambi 34.7%; Lowest: Baghpat)	**18.8%** of state pop (Highest %: Bageshwar; Lowest %: Dehradun)
ST Population %	**8.6%** (Highest %: Lakshadweep/Mizoram; Highest No.: MP; Nil in Punjab, Haryana)	**0.6%** of state pop (Highest No. & %: **Sonbhadra**; Lowest: Baghpat; Tharu is largest tribe)	**2.9%** of state pop (Highest %: **Udham Singh Nagar**; Tharu & Jaunsari are largest)

9. Minerals of Uttar Pradesh & Uttarakhand (state-site map)

National mineral belts stay in the Minerals chapter. Daily sheet needs the **UP / UK places** that otherwise vanish (same gap as Hulas on the Indus table).

Mineral / resource	Uttar Pradesh	Uttarakhand	Trap
Limestone / cement	**Sonbhadra** (Churk, Dalla, Ghurma); also Banda–Mirzapur belt	**Dehradun**, Tehri, Pithoragarh, Pauri belts	Do not give Rajasthan's Makrana marble as a UP fact
Coal	**Singrauli coalfield** (Sonbhadra; shared with MP)	Negligible	Jharia / Raniganj are **Jharkhand–WB**, not UP
Silica sand / glass	**Shankargarh (Prayagraj)**, Banda; feeds **Firozabad** glass	—	Firozabad = glass craft; sand is Prayagraj belt
Bauxite	**Banda, Vindhyan** pockets (small)	—	Big bauxite = Odisha / Gujarat, not UP
Andalusite / sillimanite	**Sonbhadra / Mirzapur** (refractory)	—	
Magnesite	—	**Jhiroli / Bageshwar** (Kumaon) — India's important magnesite belt	Magnesite ≠ magnesium metal plant trap
Gypsum	—	**Dehradun / Nainital** foothill pockets	Rajasthan remains India's gypsum giant
Steatite / soapstone	—	**Almora / Pithoragarh** (also used in local craft)	
Copper (old)	—	**Askot (Pithoragarh)** — historic Kumaon copper	Khetri copper is **Rajasthan**
Rock phosphate	—	**Mussoorie / Durmala–Maldeota** (Dehradun)	
Building stone / granite	Mirzapur–Sonbhadra; Lalitpur	Granite–quartzite of Lesser Himalaya	
Energy water	**Rihand / Govind Ballabh Pant Sagar** (Sonbhadra) — largest man-made lake by volume	**Tehri** (Bhagirathi–Bhilangana) — highest dam	Hydel ≠ mineral ore

> **Logic: Sonbhadra** is UP's mineral hub (coal + limestone + refractories). **Bageshwar magnesite** and **Askot copper** are the UK pair. Do not park Rajasthan or Jharkhand fields in these two states.

10. Major Minerals of India & Famous Mines Master Matrix

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Iron Ore (Hematite & Magnetite)	**Odisha** (1st in production & reserves), Chhattisgarh, Karnataka, Jharkhand	• **Odisha**: Mayurbhanj (Gorumahisani, Badampahar), Keonjhar (Joda), Sundargarh • **Chhattisgarh**: **Bailadila** (Bastar/Dantewada — high-grade hematite exported to Japan via Visakhapatnam Port), Dalli-Rajhara (Durg — feeds Bhilai Steel Plant) • **Karnataka**: **Kudremukh** (Chikkamagaluru — horse-face shaped; slurry pipeline to Mangalore), Bababudan Hills, Bellary-Hospet (Sandur) • **Jharkhand**: **Noamundi**, Gua, Kiriburu (West Singhbhum) • **Goa**: Marmagao belt	• India is rich in **Hematite** (~70%) and **Magnetite** (~30%). • Dharwar and Cuddapah rock systems are primary sources. • *Trap:* Bailadila is in **Chhattisgarh**, NOT MP or Odisha!

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Bauxite (Aluminium Ore)	**Odisha** (>50% of national production), Gujarat, Jharkhand	• **Odisha**: **Panchpatmali** (Koraput — India's largest deposit; feeds NALCO refinery at Damanjodi), Kalahandi, Rayagada, Bolangir • **Jharkhand**: **Lohardaga**, Gumla, Latehar • **Gujarat**: Jamnagar, Junagadh, Kutch • **Madhya Pradesh**: Amarkantak plateau (Shahdol), Balaghat • **Maharashtra**: Kolhapur, Ratnagiri	• Associated with **laterite rocks** of tertiary age formed via intense leaching. • NALCO (Damanjodi & Angul) and HINDALCO (Renukoot, UP) are major processors.
Copper	**Madhya Pradesh** (1st in production), Rajasthan (1st in reserves), Jharkhand	• **Madhya Pradesh**: **Malanjkhand** (Balaghat district — largest open-cast copper mine in Asia; accounts for ~50% of Indian production) • **Rajasthan**: **Khetri**, Chandmari, Kolihan (Jhunjhunu), Kho-Dariba (Alwar), Delwara (Udaipur) • **Jharkhand**: **Mosabani**, **Ghatshila**, Rakha (Singhbhum Copper Belt) • **Uttarakhand**: Askot (Pithoragarh — old workings)	• Hindustan Copper Limited (HCL, est. 1967) holds public monopoly. • India is heavily import-dependent for copper ore (~80% imported).
Manganese	**Madhya Pradesh** (1st in production), Maharashtra, Odisha	• **Madhya Pradesh**: **Balaghat** (Bharveli underground mine), Chhindwara • **Maharashtra**: Nagpur, Bhandara, Ratnagiri • **Odisha**: Sundargarh, Keonjhar, Koraput • **Karnataka**: Shimoga, Bellary	• Vital ferro-alloy: ~10 kg manganese is needed to manufacture **1 tonne of steel**. • MOIL (Manganese Ore India Limited) headquartered in Nagpur.
Coal	**Jharkhand** (1st in reserves), **Odisha / Chhattisgarh** (1st in production)	• **Jharkhand**: **Jharia** (Dhanbad — **richest and only storehouse of prime coking coal in India**), **Bokaro**, Giridih, North/South Karanpura, Ramgarh • **West Bengal**: **Raniganj** (Asansol/Bardhaman — **oldest commercial coalfield in India, started 1774 by Summer & Heatly**) • **Chhattisgarh**: **Korba** (Hasdeo valley), Tatapani-Ramkola, Chirimiri • **Odisha**: **Talcher** (Brahmani valley — feeds NTPC super thermal plant), Ib valley • **Madhya Pradesh**: **Singrauli** (Jhingurdah seam — **second thickest coal seam in the world**; shared with UP Sonbhadra), Pench-Kanhan • **Telangana**: Singareni (Godavari valley — SCCL) • **Tamil Nadu**: **Neyveli** (Cuddalore — premier deposit of **Lignite / Brown Coal** in India) • **Assam**: Makum, Nazira, Ledo (Tertiary coal; high sulphur content)	• **Gondwana Coal** (~98% of India's reserves; 250 million yrs old; low moisture, low sulphur, high ash, non-coking/coking). • **Tertiary Coal** (~2% reserves; 15–60 million yrs old; NE states & Jammu; high sulphur & volatile matter).
Petroleum & Crude Oil	**Offshore (Bombay High)** (1st in production), Rajasthan, Gujarat, Assam	• **Offshore**: **Bombay High** (1974 discovery by ONGC vessel *Sagar Samrat*; 160 km off Mumbai), Bassein, Ravva (Krishna-Godavari) • **Rajasthan**: **Barmer Basin** (Mangala, Bhagyam, Aishwarya fields — discovered by Cairn/Vedanta; largest onshore producer) • **Gujarat**: **Ankleshwar** (*Fountain of Prosperity* named by Nehru), Kalol, Mehsana, Aliabet Island (Gulf of Khambhat) • **Assam**: **Digboi** (**oldest producing oilfield & refinery in India and Asia, 1901**), Naharkatiya, Moran-Hugrijan, Surma Valley	• Refineries: **Jamnagar (Gujarat)** is the world's largest grass-root refining complex (private - Reliance). • Indian Oil's Digboi is the oldest operational refinery.

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Gold	**Karnataka** (>90% of national production), Andhra Pradesh	• **Karnataka**: **Kolar Gold Fields (KGF)** (shut down in 2001 due to depth & costs), **Hutti Gold Mines** (Raichur — **only active primary gold producing mine in India today**) • **Andhra Pradesh**: Ramagiri (Anantapur) • **Jharkhand**: Alluvial placer gold in Subarnarekha river sands	• Hutti produces ~99% of India's indigenous mined gold.
Diamond	**Madhya Pradesh** (holds >90% of diamond resources)	• **Madhya Pradesh**: **Panna** (**Majhgawan pipe mine** operated by NMDC — **the ONLY mechanized active diamond mine in Asia/India**) • Historic: **Kollur Mine** (Guntur/Golconda, Andhra Pradesh — origin of the Koh-i-Noor, Hope Diamond, and Daria-i-Noor)	• Kimberly pipe formations in Vindhyan conglomerates.
Uranium	**Jharkhand, Andhra Pradesh**	• **Jharkhand**: **Jaduguda** (East Singhbhum — India's first uranium mine, 1967), Bhatin, Narwapahar, Turamdih, Banduhurang (all run by UCIL) • **Andhra Pradesh**: **Tummalapalle** (Kadapa/YSR district — massive global-scale reserves discovered in 2011) • **Meghalaya**: Domiasiat / Mawthabah (Kasi Hills - sandstone deposit) • **Rajasthan**: Rohil (Sikar)	• Operated under Department of Atomic Energy (DAE) via Uranium Corporation of India Ltd (UCIL, Jaduguda).
Thorium	**Kerala, Odisha, Tamil Nadu**	• **Kerala**: **Monazite beach sands** of Chavara, Kollam, and Alappuzha • **Odisha**: Ganjam coast (Chhatrapur beach sands) • **Tamil Nadu**: Manavalakurichi (Kanyakumari)	• Derived from **Monazite sand** (contains thorium phosphate, cerium, and rare earths). • Indian Rare Earths Limited (IREL) processes beach sand minerals. India holds the world's largest thorium reserves.
Lead & Zinc / Silver	**Rajasthan** (near 100% monopoly)	• **Rajasthan**: **Zawar Mines** (Udaipur — ancient & modern lead-zinc-silver), **Rampura-Agucha** (Bhilwara — world-class open-pit zinc deposit), Rajpura-Dariba (Rajsamand), Sindesar Khurd	• Hindustan Zinc Limited (HZL, Vedanta group). • Galena (lead ore) and Sphalerite (zinc ore). Silver is recovered as a by-product.
Mica (Ruby Mica / Muscovite)	**Andhra Pradesh** (1st in production), Rajasthan, Jharkhand	• **Andhra Pradesh**: Nellore Mica Belt (Gudur) • **Jharkhand**: **Koderma** (historically called the *"Mica Capital of the World"*), Giridih, Hazaribagh • **Rajasthan**: Bhilwara, Ajmer, Jaipur	• India was historic world leader in sheet mica (Muscovite / Ruby mica). • Essential electrical insulator resistant to high voltages.

11. Ocean Currents & Planetary Local Winds of the World

A. Major Ocean Currents (Warm vs Cold Master Matrix)

Rule of Thumb: Currents moving from Equator to Poles = Warm; Currents moving from Poles to Equator = Cold.

Ocean Basin	Warm Ocean Currents	Cold Ocean Currents	High-Yield Prelims Exam Takeaways
Atlantic Ocean	• **Gulf Stream** (powerful warm current) • **North Atlantic Drift** (keeps NW European ports ice-free year-round) • **Florida Current** • **Antilles Current** • **Brazil Current** (South Atlantic) • **Guinea Current** (West Africa)	• **Labrador Current** (flows from Arctic past Greenland) • **Canary Current** (North Africa coast) • **Benguela Current** (South-West Africa coast; creates **Namib & Kalahari Deserts**) • **Falkland / Malvinas Current** (Patagonia coast) • **East Greenland Current**	• **Grand Banks of Newfoundland**: Confluence of warm **Gulf Stream** and cold **Labrador Current** produces dense fog and creates world's richest cod-fishing ground. • **Sargasso Sea**: Clockwise North Atlantic gyre creates a calm, landless sea surrounded by currents, filled with *Sargassum* seaweed.
Pacific Ocean	• **Kuroshio (Japan Current)** (warm counterpart of Gulf Stream) • **North Pacific Drift** • **East Australian Current** (flows south along eastern Australia) • **Alaska Current** (keeps Alaskan coast ice-free) • **Tsushima Current** (Sea of Japan)	• **Oyashio (Kuril Current)** (cold current from Bering Strait) • **California Current** (West coast of USA) • **Peru / Humboldt Current** (flows along Chile & Peru; high upwelling, rich in anchovies; suppressed during **El Niño**) • **Okhotsk Current**	• Meeting of warm **Kuroshio** and cold **Oyashio** creates dense fog and prime fishing grounds off Japan. • **Atacama Desert** (driest non-polar desert) created by the desiccating effect of the cold **Humboldt/Peru Current**.
Indian Ocean	• **Mozambique Current** (East Africa) • **Agulhas Current** (South Africa) • **South-West Monsoon Drift** (summer)	• **West Australian Current** (flows along West Australia; creates **Great Sandy/Victoria Deserts**) • **Somali Current** (reverses direction seasonally; cold upwelling in summer)	• Indian Ocean currents completely **reverse their direction of flow twice a year** due to the seasonal reversal of the Monsoon winds.
Circumpolar	—	• **West Wind Drift (Antarctic Circumpolar Current)**	• **Largest ocean current by water volume**; circles Antarctica unbroken from West to East under roaring westerlies.

B. Famous Local Winds of the World (Hot vs Cold & Nicknames)

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Chinook	**Warm & Dry** (Katabatic)	Rocky Mountains (USA & Canada)	**"Snow Eater"** — descends eastern slopes of Rockies, melting winter snow rapidly and opening pastures for cattle.
Foehn (Föhn)	**Warm & Dry**	Northern slopes of the **Alps (Switzerland)**	Accelerates melting of alpine snow, aids grape ripening in Swiss valleys (*the climber's wind*).
Harmattan	**Warm, Dry & Dusty**	Sahara Desert blowing south-west to the **Gulf of Guinea (West Africa)**	**"The Doctor"** — provides welcome relief from the sultry, humid coastal tropical heat and kills pathogens.
Sirocco	**Hot, Dry & Dust-laden**	Sahara Desert blowing north across Mediterranean into **Southern Europe (Italy/Sicily)**	Causes **"Blood Rain"** (red Sahara dust particles falling with condensation rain in Italy and Greece).
Khamsin	**Hot & Dusty**	**Egypt**	Blows from the southern desert for approximately 50 days (*Khamsin* = fifty in Arabic).
Gibli	**Hot & Dry**	**Libya**	Sahara desert wind blowing north towards Libyan coast.
Brickfielder	**Hot & Dusty**	**Southern Australia**	Blows from the central Australian desert towards coastal cities (Melbourne/Sydney).

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Black Roller	**Hot & Dust-storm**	Great Plains of **North America / USA**	Intense summer dust squall darkening the sky.
Zonda	**Hot & Dry** (Foehn-type)	Eastern slopes of the **Andes (Argentina)**	Blows down Argentine pampas from the Andes.
Santa Ana	**Hot, Dry & Gusty**	Southern **California (USA)**	Blows from Mojave desert down coastal mountain canyons; notoriously triggers catastrophic wildfires.
Loo	**Scorching Hot & Dry**	Northern and North-Western Plains of **India (UP, Bihar, Punjab, Rajasthan)**	Blows during May–June afternoons; causes severe heatstrokes.
Nor'wester / Kalbaishakhi	**Hot-season thunderstorm**	**West Bengal, Odisha, Assam** (*Bordoisila*)	Violent afternoon squall in Baisakh (April–May); beneficial for early tea, jute, and aus paddy crops.
Mistral	**Extremely Cold & Dry**	Blows from Alps down the **Rhone Valley into Gulf of Lion (France & Spain)**	Winter/spring gale-force cold wind; trains often disrupted; houses in Provence face south without north windows.
Bora	**Extremely Cold, Gusty & Dry**	Blows from Dinaric Alps down towards the **Adriatic Sea (Italy, Croatia)**	Fierce northeastern catabatic katabatic wind reaching hurricane speeds in winter.
Blizzard	**Freezing Cold with Blinding Snow**	**Canada, Northern USA, Siberia, Polar regions**	Gale-force winds with sub-zero temperatures and zero visibility.
Buran / Purga	**Bitterly Cold & Fierce Blizzard**	**Siberia & Central Asian steppes (Russia)**	Buran brings icy blizzard conditions; Purga is tundra version.
Pampero	**Cold & Violent Wind**	**Pampas of Argentina and Uruguay**	Cold polar squall blowing from southwest across the pampas.

12. Structure of Atmosphere & Key Meteorological Phenomena

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
1. Troposphere	**0 to 18 km** (Equator: ~18 km; Poles: ~8 km)	**Decreases with altitude** (Lapse Rate: **6.5°C per km** or **1°C per 165 m**)	• Thickest at Equator due to strong vertical convectional currents. • Contains **~75% to 80% of total atmospheric mass** and **almost ALL atmospheric water vapour and dust particles**. • **ALL weather phenomena** (clouds, rainfall, lightning, thunder, cyclones, fog) occur EXCLUSIVELY in this layer. • Upper boundary is the **Tropopause** (temperature drops to -80°C over Equator and -45°C over Poles).
2. Stratosphere	**From Tropopause up to 50 km**	**Increases with altitude** (from -60°C up to 0°C at Stratopause)	• **Contains the Ozone Layer (Ozonosphere)** between **15 km and 35 km** (absorbs harmful solar ultraviolet UV-B and UV-C radiation). • Completely free from clouds and turbulent weather; ideal condition for flying **commercial supersonic jet aircraft**. • High-altitude **Subtropical Jet Streams** blow near the tropopause. • Rare iridescent **Nacreous Clouds (Mother-of-Pearl clouds)** form in polar winter stratosphere.

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
3. Mesosphere	**50 km to 80 km**	**Decreases rapidly with altitude** (reaches **-100°C at Mesopause**)	• **COLDEST LAYER OF THE ATMOSPHERE** (-100°C to -130°C). • **Meteors burn up** in this layer upon entering from space due to friction with thin gases. • Rare **Noctilucent Clouds** (night-shining polar clouds composed of ice crystals on meteor dust) appear here.
4. Thermosphere / Ionosphere	**80 km to 400+ km**	**Increases rapidly with altitude** (exceeds 1,500°C due to intense solar radiation absorption)	• Contains electrically charged ionized gas particles (**Ions**). • **Reflects terrestrial radio waves back to Earth**, enabling wireless global radio communication: - **D-layer (60–90 km)**: Reflects low-frequency radio waves, absorbs high-frequency; disappears at night. - **E-layer / Kennelly-Heaviside Layer (90–130 km)**: Reflects medium-frequency radio waves. - **F-layer / Appleton-Barnett Layer (140–400 km)**: Reflects high-frequency short radio waves. • **Auroras** (polar light displays caused by solar wind hitting charged atmospheric gases): - **Aurora Borealis** (Northern Lights, Arctic) - **Aurora Australis** (Southern Lights, Antarctic) • **International Space Station (ISS)** orbits in this layer (~400 km).
5. Exosphere	**>400 km to outer space** (~10,000 km)	Extremely high temperature, but imperceptible due to ultra-low density	• Outermost rarefied fringe of Earth's atmosphere merging into interplanetary space. • Dominated by light gases: **Hydrogen and Helium**. • Artificial communication and weather satellites orbit in higher exosphere.

13. Frontiers, Latitudinal/Longitudinal Framework & Territorial Limits of India

Locational / Border Indicator	Geographic Fact / Value	States / Locations Involved	Crucial Prelims Exam Triggers
Latitudinal Extent	**8°4′ N to 37°6′ N**	Mainland southern tip to northern tip	Total span = ~29°2′; Entirely in Northern & Eastern Hemispheres
Longitudinal Extent	**68°7′ E to 97°25′ E**	Westernmost point to easternmost point	Total span = ~29°18′; ~2 hours time lag between Arunachal & Gujarat
North–South Distance	**3,214 km**	Indira Col to Kanyakumari	Greater than East–West distance by 281 km
East–West Distance	**2,933 km**	Kibithu to Guhar Moti	
Total Land Frontier	**15,106.7 km**	16 States & 2 UTs share land borders	India shares land borders with 7 sovereign countries
Total Coastline	**7,516.6 km**	Mainland = 6,100 km; Island territories = 1,416.6 km	9 Coastal States + 4 Coastal UTs
Northernmost Point	**Indira Col** (37°6′ N)	Siachen Glacier (Ladakh)	Near trijunction of India, Pakistan, and China
Southernmost Point (Overall)	**Indira Point (Pygmalion / Parsons Point)** (6°45′ N)	Great Nicobar Island	Submerged partially during 2004 Tsunami; ~150 km from Sumatra (Indonesia)

Locational / Border Indicator	Geographic Fact / Value	States / Locations Involved	Crucial Prelims Exam Triggers
Southernmost Point (Mainland)	**Cape Comorin (Kanyakumari)** (8°4′ N)	Tamil Nadu	Confluence of Arabian Sea, Bay of Bengal, and Indian Ocean
Westernmost Point	**Guhar Moti / Sir Creek** (68°7′ E)	Kutch district, Gujarat	Sir Creek is a 96 km tidal estuary disputed between India & Pak
Easternmost Point	**Kibithu** (97°25′ E)	Anjaw district, Arunachal Pradesh	Lohit River enters India near Kibithu
Tropic of Cancer (23°30′ N)	Passes through **8 Indian States**	**Gujarat → Rajasthan → Madhya Pradesh → Chhattisgarh → Jharkhand → West Bengal → Tripura → Mizoram** (West to East)	• **Ranchi** is the closest state capital to Tropic of Cancer • **Mahi River** cuts the Tropic of Cancer TWICE! • Does NOT pass through Odisha, Bihar, or UP
Standard Meridian of India (82°30′ E)	**IST = GMT + 5 hours 30 mins**	Passes through **5 States**: **Uttar Pradesh** (Mirzapur / Shankargarh / Naini, Prayagraj) → **Madhya Pradesh → Chhattisgarh → Odisha → Andhra Pradesh**	• Tropic of Cancer and 82°30′ E meridian intersect in **Koriya / Surajpur district (Chhattisgarh)**

Land Frontiers by Length (Strict Descending Order)

1. **Bangladesh (4,096.7 km)** — **Longest border**; 5 States: West Bengal (longest border with BD), Assam, Meghalaya, Tripura (surrounded on 3 sides by BD; Zero Line), Mizoram.
2. **China (3,488 km)** — 1 UT (Ladakh) + 4 States: Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh; McMahon Line (1914).
3. **Pakistan (3,323 km)** — 2 UTs (J&K, Ladakh) + 3 States: Gujarat, Rajasthan (longest border with Pak), Punjab; Radcliffe Line (17 August 1947).
4. **Nepal (1,751 km)** — 5 States: Uttarakhand, Uttar Pradesh (longest border with Nepal), Bihar, West Bengal, Sikkim; Kali River forms western border.
5. **Myanmar (1,643 km)** — 4 States: Arunachal Pradesh (longest), Nagaland, Manipur, Mizoram; Purvanchal hills form watershed.
6. **Bhutan (699 km)** — 4 States: Assam (longest), West Bengal, Sikkim, Arunachal Pradesh.
7. **Afghanistan (106 km)** — **Shortest border**; Ladakh (POK - Wakhan corridor); Durand Line (1896).

Coastal States by Coastline Length (Descending)

- **Gujarat (1,214.7 km)** > **Andhra Pradesh (973.7 km)** > **Tamil Nadu (906.9 km)** > **Maharashtra (652.6 km)** > **Kerala (569.7 km)** > **Odisha (476.4 km)** > **Karnataka (280 km)** > **Goa (101 km — shortest among states)** > **West Bengal (157.5 km)**.
- **UTs**: **Andaman & Nicobar Islands (1,962 km — longest overall territory)** > Lakshadweep (132 km) > Puducherry (47.6 km) > Daman & Diu (42.5 km).

Maritime Channels & Straits Around India

- **8° Channel**: Separates **Minicoy (India)** from **Maldives**.
- **9° Channel**: Separates **Minicoy Island** from the **main Lakshadweep archipelago** (Kalpeni/Suheli).
- **10° Channel**: Separates **Little Andaman** from **Car Nicobar** (Andaman Islands from Nicobar Islands; ~150 km wide).
- **Duncan Passage**: Separates **South Andaman (Rutland Island)** from **Little Andaman**.
- **Sombrero Channel**: Separates **Car Nicobar** from **Central Nicobar (Nancowry)**.
- **Great / Grand Channel**: Separates **Great Nicobar (Indira Point)** from **Sumatra (Aceh, Indonesia)**.

- **Palk Strait & Adam's Bridge (Rama Setu)**: Separates **Tamil Nadu (Pamban Island / Dhanushkodi)** from **Sri Lanka (Mannar Island / Talaimannar)**; connects Palk Bay with Gulf of Mannar.
- **Coco Channel**: Separates **North Andaman (Landfall Island)** from **Coco Islands (Myanmar)**.

14. Peninsular & Himalayan Hill Sequences (North to South & West to East)

A. Trans-Himalayan & Himalayan Ranges (North to South Order)

1. **Karakoram Range** (K2 / Godwin-Austen 8,611 m; Siachen, Baltoro, Biafo, Hispar glaciers)
2. **Ladakh Range** (Khardung La pass; Leh lies between Ladakh and Zaskar ranges)
3. **Zaskar Range** (Kamet peak, Nanda Devi; Shipki La, Lipulekh passes)
4. **Great Himalayas (Himadri)** (Everest, Kanchenjunga, Makalu, Dhaulagiri, Nanga Parbat, Namcha Barwa)
5. **Lesser / Middle Himalayas (Himachal)** (Pir Panjal in J&K/HP, Dhauladhar in HP, Mussoorie & Nag Tibba in Uttarakhand, Mahabharat Lekh in Nepal)
6. **Shiwalik Range (Outer Himalayas)** (Youngest foothill range; flat-bottomed structural valleys called **Duns** in west e.g., Dehradun, and **Duars** in east).

B. Western Ghats (Sahyadri) Hill Sequence (Strict North to South)

1. **Northern Sahyadri**: **Kalsubai** (1,646 m — highest peak of Maharashtra / Northern Sahyadri), Salher, Mahabaleshwar (source of Krishna river), Harishchandra Range.
2. **Central Sahyadri**: **Mullayanagiri** (1,930 m — highest peak of Karnataka), **Kudremukh** (1,892 m), Baba Budan Hills (first coffee planted in India).
3. **Nilgiri Hills ("Blue Mountains")**: **Doda Betta** (2,637 m); junction meeting point of Western and Eastern Ghats; Nilgiri Tahr and Toda pastoral tribe.
4. *[Palghat / Palakkad Gap separates Nilgiris from Anaimalai Hills]*
5. **Anaimalai Hills ("Elephant Hills")**: **Anamudi** (2,695 m — **HIGHEST PEAK OF PENINSULAR / SOUTH INDIA**; Idukki, Kerala).
6. **Palani Hills**: Eastward spur from Anaimalai into Tamil Nadu; famous hill station **Kodaikanal** located around artificial star-shaped lake.
7. **Cardamom Hills (Yela Mala)**: Southernmost hill range of mainland India (Kerala-Tamil Nadu border); Periyar National Park.
8. *[Shenkottai Pass separates Cardamom Hills from Agasthyamalai]*
9. **Agasthyamalai Hills**: Southernmost biosphere reserve of Western Ghats (Kerala-TN border).

C. Eastern Ghats & Peninsular Discontinuous Hills (Strict North to South)

1. **Garhjat Hills**: Northern Odisha (Mayurbhanj / Keonjhar iron-ore belt).
2. **Nallamala Hills**: Andhra Pradesh (between Krishna and Pennar rivers; Srisailam dam and tiger reserve).
3. **Erramala Hills**: Kurnool district, Andhra Pradesh.
4. **Velikonda Range**: Eastern Andhra Pradesh (Nellore/Kadapa border).
5. **Palkonda Range**: Southern Andhra Pradesh (Tirupati sacred hills / Tirumala located on Seshachalam-Palkonda range).
6. **Nagari Hills**: Chittoor district, Andhra Pradesh.
7. **Javadi Hills**: Northern Tamil Nadu (Vellore / Tiruvannamalai; famous for blue-grey granites and sandalwood).
8. **Shevaroy Hills**: Salem district, Tamil Nadu; famous hill resort **Yercaud**; rich in bauxite deposits.
9. **Panchaimalai Hills**: Central Tamil Nadu (Tiruchirappalli).
10. **Sirumalai Hills**: Dindigul district, Tamil Nadu.
11. **Varushanad Hills**: Southern Tamil Nadu (source of Vaigai River).

D. Central Highlands & Transverse Order (North to South)

- **Vindhya Range** (Sandstone scarplands: Kaimur, Bhander, Bharner)
  ↓ *[Narmada River flows west in Rift Valley]*
- **Satpura Range** (Horst block mountain: **Rajpipla Hills** in west → **Mahadeo Hills / Dhupgarh 1,350 m** in middle → **Maikal Range / Amarkantak 1,065 m** in east)
  ↓ *[Tapi River flows west in Rift Valley south of Satpura]*
- **Ajanta Range & Satmala Hills** (Maharashtra; entirely within single state).

E. North-East Hills & Valleys

- **Arunachal Foothills (West to East)**: **Dafla Hills** → **Miri Hills** → **Abor Hills** → **Mishmi Hills** (Daphla-Miri-Abor-Mishmi named after local tribes).
- **Purvanchal Eastern Border Ranges (North to South)**: **Patkai Bum** (Arunachal-Myanmar border) → **Naga Hills** (Saramati Peak 3,826 m) → **Manipur Hills** → **Mizo Hills / Lushai Hills** (Phawngpui / Blue Mountain 2,157 m).
- **Meghalaya Plateau (West to East)**: **Garo Hills** (Nokrek Biosphere Reserve 1,412 m) → **Khasi Hills** (Shillong Peak 1,961 m; Mawsynram and Cherrapunji) → **Jaintia Hills** → **Mikir Hills** → **Barail Range** (connects Meghalaya plateau to Naga hills).

15. Major Lakes & Inland Wetlands of India Master Matrix

Lake Name	State / UT	Type / Origin	High-Yield Prelims Exam Triggers
Chilika Lake	**Odisha** (Puri, Khordha, Ganjam)	**Brackish coastal lagoon**	• **Largest brackish water lagoon in India** (fed by Daya River) • Designated **1st Indian Ramsar site (1981)** along with Keoladeo NP • Habitat of the endangered **Irrawaddy Dolphin** • Nalabana Island declared Bird Sanctuary
Vembanad Lake	**Kerala** (Kottayam, Alappuzha, Ernakulam)	**Brackish lagoon / backwater**	• **LONGEST LAKE IN INDIA** (96.5 km); largest in Kerala • Famous **Nehru Trophy Boat Race (Vallam Kali)** held on Punnamada Lake • Willingdon Island (Kochi Port) man-made in it • Kumarakom Bird Sanctuary on its eastern shore
Sasthamkotta Lake	**Kerala** (Kollam)	**Freshwater lake**	• Known as the **"Queen of Lakes"**; largest freshwater lake in Kerala • Contains unique larva *Cavaborus* which cleanses and purifies lake water
Ashtamudi Lake	**Kerala** (Kollam)	**Brackish estuarine lagoon**	• Eight-armed (*Ashta-mudi*) palm-shaped lake; gateway to Kerala backwaters; Ramsar site
Kolleru Lake	**Andhra Pradesh** (Eluru/Krishna)	**Freshwater lake**	• Located between **Krishna and Godavari deltas**; fed by Budameru and Tammileru streams • Atapaka Bird Sanctuary (pelican paradise); Ramsar site
Pulicat Lake	**Andhra Pradesh (96%) & Tamil Nadu (4%)**	**Brackish water coastal lagoon**	• **Second largest brackish lagoon in India** • Separated from Bay of Bengal by the barrier island of **Sriharikota** (ISRO Satish Dhawan Space Centre) • Famous annual **Flamingo Festival**

Lake Name	State / UT	Type / Origin	High-Yield Prelims Exam Triggers
Loktak Lake	**Manipur** (Moirang, Bishnupur)	**Freshwater lake**	• **Largest freshwater lake in North-East India** • Famous for heterogeneous mass of vegetation and soil called **Phumdis** • Hosts **Keibul Lamjao National Park** — **WORLD'S ONLY FLOATING NATIONAL PARK** • Exclusive natural habitat of endangered **Sangai (Brow-antlered / Dancing deer)**
Lonar Lake	**Maharashtra** (Buldhana district)	**Meteorite Impact Crater Lake**	• Created by hyper-velocity meteorite impact during Pleistocene epoch in Deccan basalt • Highly saline and alkaline (pH ~10.5); turns pink due to haloarchaea microbes • Notified National Geo-heritage Monument; Ramsar site (2020)
Wular Lake	**Jammu & Kashmir** (Bandipora)	**Tectonic Freshwater Lake**	• **LARGEST FRESHWATER LAKE IN INDIA** • Formed by tectonic activity; fed and drained by **Jhelum River** • **Tulbul Navigation Project** (disputed with Pakistan) located at its mouth
Dal Lake	**Jammu & Kashmir** (Srinagar)	Freshwater flood-plain lake	• "Jewel in the crown of Kashmir"; famous for **Shikaras** and floating gardens (*Radh*)
Sambhar Lake	**Rajasthan** (Jaipur, Nagaur, Ajmer)	**Inland Saline Playa Lake**	• **LARGEST INLAND SALINE LAKE IN INDIA**; represents an endorheic depression • Produces ~9% of India's total salt production; Ramsar site
Didwana, Pachpadra, Lunkaransar	**Rajasthan**	Inland saline playas	• Pachpadra produces high-grade sodium chloride (98% purity) used locally
Pushkar & Nakki Lake	**Rajasthan**	Pushkar: Sacred holy lake; Nakki: Mount Abu	• Nakki is highest lake in Rajasthan; legend says dug by gods' nails (*nakh*)
Pangong Tso	**Ladakh** (4,225 m elevation)	Endorheic saltwater lake	• **World's highest saltwater lake**; ~1/3 lies in India and ~2/3 in Tibet (China)
Tso Moriri	**Ladakh** (Rupshu valley)	High-altitude brackish lake	• Breeding ground for endangered Black-necked Crane and Bar-headed Goose; Ramsar site
Roopkund Lake	**Uttarakhand** (Chamoli, 5,029 m)	Glacial tarn lake	• Known as the **"Mystery Lake" / "Skeleton Lake"**; hundreds of human skeletons found
Bhimtal, Nainital, Naukuchiatal, Sattal	**Uttarakhand** (Kumaon)	Tectonic / fault freshwater lakes	• **Bhimtal** is the largest lake in Kumaon with a central island; Naukuchiatal has 9 corners
Suraj Tal & Chandra Tal	**Himachal Pradesh** (Lahaul & Spiti)	High-altitude glacial lakes	• Chandra Tal (source of Chandra river) and Suraj Tal (source of Bhaga river); join to form Chenab
Renuka Lake	**Himachal Pradesh** (Sirmaur)	Freshwater wetland lake	• Largest natural lake in Himachal Pradesh; shaped like a reclining woman; Ramsar site
Deepor Beel	**Assam** (Kamrup / Guwahati)	Freshwater oxbow lake	• Former channel of Brahmaputra; vital staging site for migratory waterbirds; Ramsar site
Rudrasagar Lake	**Tripura** (Melaghar)	Freshwater lowland lake	• Historic **Neermahal Palace** situated in the center of the lake; Ramsar site

16. Soils of India — ICAR Classification, Characteristics & Soil Health Master Card

A. ICAR 8 Soil Groups (% Share & Leading Characteristics)

1. **Alluvial Soil (~43.4% of total area — Largest)**: * **Genesis**: Deposited by Indus, Ganga, Brahmaputra river systems and coastal waves. * **Nutrient Profile**: **Rich in Potash and Lime**; severely **deficient in Nitrogen, Phosphorus, and Humus**. * **Classification**:
   - **Khadar**: New alluvium deposited by annual floods; fine loamy texture, highly fertile; light color.
   - **Bhangar**: Older alluvium situated above flood levels; clayey texture, less fertile; contains calcareous nodules (**Kankar**).
   - **Bhabar**: Narrow porous pebble-bed at foot of Shiwaliks where rivers disappear underground.
   - **Terai**: Marshy, waterlogged damp tract south of Bhabar with tall elephant grass; re-emerging streams.
   - **Crops**: Wheat, Rice, Sugarcane, Jute, Oilseeds, Maize, Pulses.
2. **Red & Yellow Soil (~18.6% — 2nd Largest Group)**: * **Genesis**: Weathering of ancient crystalline igneous (granites, gneisses) and metamorphic rocks in low-rainfall areas. * **Color Rationale**: Red color due to wide diffusion of **Iron (Ferric oxides)**; appears yellow when hydrated. * **Nutrient Profile**: **Deficient in Nitrogen, Phosphorus, Lime, and Humus**. * **Regions & Crops**: Tamil Nadu, Southern Karnataka, Telangana, Odisha, Chotanagpur, MP; ideal for millets, pulses, groundnut, tobacco.
3. **Black Soil / Regur / Tropical Chernozem (~15.2%)**: * **Genesis**: Weathering of basaltic lava rocks of the **Deccan Trap (Cretaceous lava flows)**. * **Distinctive Features**: Highly argillaceous (clayey >50%); swells and becomes sticky when wet; shrinks and develops wide deep cracks when dry (**"self-ploughing" capacity**); exceptional moisture-retention capability. * **Color**: Black color attributed to **Titaniferous Magnetite** and iron compounds. * **Nutrient Profile**: **Rich in Lime, Iron, Magnesia, Alumina, and Potash**; severely **deficient in Phosphorus, Nitrogen, and Organic Matter**. * **Regions & Crops**: Maharashtra, Gujarat, Malwa (MP), Northern Karnataka, Telangana; **Ideal for Cotton (Black Cotton Soil), Soyabean, Sugarcane, Citrus fruits**.
4. **Laterite Soil (~3.7%)**: * **Genesis**: Formed under conditions of **high temperature and alternating heavy wet and dry seasons with intense tropical rainfall**. * **Process**: Intense **leaching (desilication)** washes away silica and lime, leaving behind residual oxides of **Iron and Aluminium (Bauxite)**. * **Properties**: Highly acidic (pH 4.5–5.5); hardens like brick upon exposure to atmosphere (Latin *Later* = brick). * **Nutrient Profile**: Poor in Nitrogen, Phosphorus, Potash, Lime, and Humus. * **Regions & Crops**: Summits of Western Ghats (Kerala, Karnataka, Maharashtra), Eastern Ghats, Meghalaya plateau, Rajmahal Hills; **Premier soil for Plantation Crops: Tea, Coffee, Rubber, Cashewnut, Coconut**.
5. **Arid / Desert Soil (~4.4%)**: * **Properties**: Sandy to gravelly texture, saline in nature; low moisture and very low humus; phosphate normal but nitrogen deficient; with canal irrigation (Indira Gandhi Canal) produces bumper crops of wheat, barley, cotton.
6. **Saline & Alkaline Soils (Reh, Kallar, Usar, Rakar, Thur, Chopan) (~1.3%)**: * **Genesis**: Excessive canal irrigation in poorly drained arid/semi-arid regions causing high capillary action that pulls soluble sodium, calcium, and magnesium salts to topsoil. * **Reclamation**: Application of **Gypsum ($CaSO_4 \cdot 2H_2O$)** or Iron Pyrites to neutralize alkalinity; growing salt-tolerant green manure crops like **Dhaincha (*Sesbania*)**.
7. **Peaty and Marshy Soil (Kari) (~2%)**: * **Genesis**: Formed in humid regions with heavy rainfall and dense vegetation; high organic matter accumulation (40–50%). * **Properties**: Dark, heavy, highly acidic; found in **Kottayam and Alappuzha districts of Kerala (Kari lands)**, Sundarbans delta.
8. **Forest and Mountain Soil (~8.7%)**: * Immature loamy soil rich in humus from fallen leaf litter; acidic in higher alpine zones; terrace cultivation of apples, peaches, spices, tea.

17. Natural Vegetation & Forests of India (Champion & Seth / ISFR Key Indicators & Species)

A. Core Forest Statistics (India State of Forest Report - ISFR)

- **Total Forest & Tree Cover**: **~24.62%** of geographical area (Forest Cover: ~21.71%; Tree Cover: ~2.91%).
- **National Forest Policy (1988) Target**: **33%** of total geographical area (60% in mountainous/hilly regions).
- **State with Largest Forest Cover by Area**: **Madhya Pradesh** (1st) > Arunachal Pradesh (2nd) > Chhattisgarh (3rd) > Odisha (4th) > Maharashtra (5th).
- **State with Highest Forest Cover as % of Total Area**: **Mizoram** (84.5%) > Arunachal Pradesh (79.3%) > Meghalaya (76%) > Manipur (74.3%).
- **UT with Highest Forest Cover %**: **Lakshadweep** (90.3%) and **Andaman & Nicobar Islands** (81.7%).

B. Major Forest Types & Indicator Species (Match the Following Goldmine)

Forest Type	Annual Rainfall & Climate	Typical Regions	Dominant Commercial Tree Species (Prelims Match)
Tropical Wet Evergreen	**>250 cm**; Temp: 25–27°C; High humidity (>77%)	Western slopes of Western Ghats, NE Hills (Assam, Meghalaya), Andaman & Nicobar	• **Rosewood, Mahogany, Ebony, Aini, Cinchona (Quinine), Rubber, Wild Mango, Bamboo** • Dense multi-layered canopy; trees do not shed leaves simultaneously.
Tropical Semi-Evergreen	**200–250 cm**	Bordering wet evergreen belts in Western Ghats, Assam, Odisha	• **White Cedar, Hollock, Kadam, Kail, Champa**
Tropical Moist Deciduous	**100–200 cm**; distinct dry season	Eastern slopes of Western Ghats, Chotanagpur, foothills of Himalayas (Bhabar-Terai), Odisha, MP, UP	• **Teak (Sagan), Sal (Sakhu), Shisham, Sandalwood (Chandan), Mahua, Mulberry, Amla, Kusum, Semul, Bamboo** • Economically **most valuable timber forests of India**; shed leaves for 6–8 weeks in spring.
Tropical Dry Deciduous	**70–100 cm**; transitional between moist deciduous & thorn	Vast plains of UP, Bihar, MP, eastern Rajasthan, Deccan plateau	• **Tendu (Bidi wrapper), Palas / Dhak (*Flame of the Forest*), Amaltas, Bel, Khair, Axlewood (Dhaora)** • **MOST WIDESPREAD FOREST COVER IN INDIA (~41.8% of total forest area)**.
Tropical Thorn & Scrub	**<70 cm / <50 cm**	Rajasthan, SW Punjab, Haryana, Gujarat, rain-shadow leeward parts of Deccan	• **Acacia (Babool), Kikar, Ber, Wild Date Palm, Khair, Cacti, Euphorbia** • Deep roots, xerophytic thorns, fleshy stems to conserve water.
Montane Subtropical / Temperate	Altitude: **1,000 to 3,000 m**	Himalayas (J&K, HP, Uttarakhand, Sikkim) & Nilgiris/Anamalai (*Shola* forests)	• **1000–2000 m**: Broad-leaved **Oak and Chestnut** • **1500–3000 m**: Coniferous **Chir Pine** (resin & turpentine), **Deodar (*Cedrus deodara* — durable structural timber, western Himalayas), Blue Pine, Spruce, Silver Fir** • *Shola*: Stunted evergreen temperate grasslands/forests in Nilgiris and Anamalai.
Alpine & Sub-Alpine	Altitude: **>3,000 to 4,000 m**	High Himalayas above timber line up to snow line	• **Silver Fir, Juniper, Birch (Bhojpatra — ancient parchment bark), Rhododendron** • Alpine pastures called **Bugyals** in Uttarakhand and **Margs** in Kashmir (Gulmarg, Sonamarg).
Littoral / Mangrove / Tidal	Coastal saline estuaries, deltas, tidal creeks	Sundarbans (WB), Mahanadi delta (Bhitarkanika, Odisha), Godavari-Krishna deltas, Pichavaram (TN)	• **Sundari (*Heritiera fomes*), Goran, Kankra, Rhizophora, Sonneratia** • Halophytic adaptations: **Pneumatophores (breathing roots)**, stilt/prop roots, **vivipary** germination.

18. Shifting Cultivation (Jhum) Regional & Global Names Master Match-Sheet

A. Indian Regional Names for Shifting Cultivation

- **Jhum / Jhoom**: **North-Eastern India** (Assam, Meghalaya, Nagaland, Mizoram, Manipur, Arunachal Pradesh)
- **Podu / Dabi / Koman / Bhringa**: **Odisha** and Northern **Andhra Pradesh**
- **Bewar / Dahiya**: **Madhya Pradesh** (practiced by Baiga tribe in Bundelkhand/Baghelkhand)
- **Penda / Dipa**: **Bastar district (Chhattisgarh)** (practiced by Maria Gonds)
- **Valre / Waltre**: **South-Eastern Rajasthan** (Aravalli tribal belt)
- **Khil**: **Himalayan belt** (Himachal Pradesh, Uttarakhand)
- **Kuruwa / Khallu**: **Jharkhand** (Santhal Parganas)
- **Kumari**: **Western Ghats** (hilly tracts of Kerala and Karnataka)
- **Zabo**: **Nagaland** (indigenous forest-clearing + terraced pond water management)

B. International Names for Shifting Cultivation (Direct Exam Matches)

- **Milpa**: **Mexico and Central America**
- **Ladang**: **Indonesia and Malaysia**
- **Roca**: **Brazil** (Amazon basin)
- **Conuco**: **Venezuela** (Orinoco basin)
- **Taungya**: **Myanmar (Burma)**
- **Chena**: **Sri Lanka**
- **Ray**: **Vietnam and Laos**
- **Tamrai**: **Thailand**
- **Caingin**: **Philippines**
- **Masole**: **Congo Basin (Central Africa)**
- **Chitemene**: **Zambia, Zimbabwe and Malawi**

19. Agricultural Revolutions, Commodity Boards & Premier National Research Institutes

A. Agricultural Revolutions & Famous Pioneers

- **Green Revolution**: **Foodgrains (Wheat & Rice)** — Global Father: **Dr. Norman Borlaug** (Nobel Peace Prize 1970; semi-dwarf wheat at CIMMYT Mexico); Indian Father: **Dr. M.S. Swaminathan** (Bharat Ratna); Pantnagar University (1st agri university in India, 1960).
- **White Revolution (Operation Flood)**: **Milk & Dairy** — Father: **Dr. Verghese Kurien**; NDDB Anand, Gujarat (AMUL); launched in 1970.
- **Yellow Revolution**: **Oilseeds (Mustard & Sunflower)** — Pioneer: **Sam Pitroda** (Technology Mission on Oilseeds 1986).
- **Blue Revolution**: **Fish & Aquaculture** — Pioneers: **Dr. Hiralal Chaudhari** and **Dr. Arun Krishnan**.
- **Golden Revolution**: **Horticulture, Honey & Fruits** — Pioneer: **Nirpakh Tutej**.
- **Golden Fibre Revolution**: **Jute Production**.
- **Silver Revolution**: **Eggs & Poultry Production** — Pioneer: **Smt. Indira Gandhi**.
- **Silver Fibre Revolution**: **Cotton Production**.
- **Pink Revolution**: **Onion, Prawn / Shrimp & Pharmaceuticals** — Pioneer: **Durgesh Patel**.
- **Red Revolution**: **Meat and Tomato Production** — Pioneer: **Vishal Tewari**.
- **Grey Revolution**: **Chemical Fertilizers**.
- **Round Revolution**: **Potato Production**.
- **Brown Revolution**: **Leather, Cocoa & Non-Conventional Energy**.

- **Evergreen / Rainbow Revolution**: **Holistic sustainable agriculture** (coined by M.S. Swaminathan).

B. Commodity Boards Headquarters in India (Match Goldmine)

- **Tea Board of India**: **Kolkata** (West Bengal)
- **Coffee Board of India**: **Bengaluru** (Karnataka)
- **Rubber Board of India**: **Kottayam** (Kerala)
- **Spices Board of India**: **Kochi** (Kerala)
- **Tobacco Board of India**: **Guntur** (Andhra Pradesh)
- **National Jute Board**: **Kolkata** (West Bengal)
- **Central Silk Board**: **Bengaluru** (Karnataka)
- **Coir Board**: **Kochi** (Kerala)
- **National Fisheries Development Board (NFDB)**: **Hyderabad** (Telangana)

C. Premier National Research Institutes (Geography / Agriculture / Forestry)

- **Uttarakhand (UKPCS Top Triggers!)**:
- **Forest Research Institute (FRI)**: **Dehradun** (est. 1906, Greco-Roman architecture)
- **Indian Institute of Remote Sensing (IIRS)**: **Dehradun**
- **Wildlife Institute of India (WII)**: **Dehradun**
- **Wadia Institute of Himalayan Geology (WIHG)**: **Dehradun**
- **Indira Gandhi National Forest Academy (IGNFA)**: **Dehradun**
- **Govind Ballabh Pant Institute of Himalayan Environment & Development**: **Kosi-Katarmal, Almora**
- **Uttar Pradesh (UPPCS Top Triggers!)**:
- **Indian Institute of Sugarcane Research (IISR)**: **Lucknow**
- **National Sugar Institute (NSI)**: **Kanpur**
- **Indian Institute of Vegetable Research (IIVR)**: **Varanasi**
- **Indian Grassland and Fodder Research Institute (IGFRI)**: **Jhansi**
- **Central Institute for Research on Goats (CIRG)**: **Makhdoom, Mathura**
- **Central Institute for Subtropical Horticulture (CISH)**: **Rahmankhera, Lucknow**
- **Indian Institute of Pulses Research (IIPR)**: **Kanpur**
- **National Botanical Research Institute (NBRI)**: **Lucknow**
- **Central Drug Research Institute (CDRI)**: **Lucknow**
- **Central Institute of Medicinal and Aromatic Plants (CIMAP)**: **Lucknow**
- **All-India Premier Geographic & Agricultural Stations**:
- **Central Potato Research Institute (CPRI)**: **Shimla** (Himachal Pradesh)
- **National Rice Research Institute (NRRI / CRRI)**: **Cuttack** (Odisha)
- **Indian Agricultural Research Institute (IARI / Pusa)**: **New Delhi**
- **Central Arid Zone Research Institute (CAZRI)**: **Jodhpur** (Rajasthan)
- **National Dairy Research Institute (NDRI)**: **Karnal** (Haryana)
- **Central Soil Salinity Research Institute (CSSRI)**: **Karnal** (Haryana)
- **National Institute of Oceanography (NIO)**: **Dona Paula, Panaji** (Goa)
- **Indian National Centre for Ocean Information Services (INCOIS)**: **Hyderabad** (Telangana)
- **National Geophysical Research Institute (NGRI)**: **Hyderabad** (Telangana)
- **Central Plantation Crops Research Institute (CPCRI)**: **Kasaragod** (Kerala)
- **Central Tuber Crops Research Institute (CTCRI)**: **Thiruvananthapuram** (Kerala)
- **Central Tobacco Research Institute (CTRI)**: **Rajahmundry** (Andhra Pradesh)
- **Central Sheep and Wool Research Institute (CSWRI)**: **Avikanagar** (Rajasthan)

- **Central Institute for Research on Buffaloes (CIRB)**: **Hisar** (Haryana)

20. World Strategic Straits, Canals & Maritime Chokepoints Master Matrix

Strait / Canal	Water Bodies Connected	Landmasses Separated	High-Yield Prelims Exam Triggers
Strait of Malacca	**Andaman Sea (Indian Ocean) & South China Sea (Pacific Ocean)**	**Malay Peninsula & Sumatra (Indonesia)**	• **World's busiest shipping chokepoint**; shortest sea route between Persian Gulf/India and China/Japan
Strait of Hormuz	**Persian Gulf & Gulf of Oman (Arabian Sea)**	**Iran & Oman / UAE (Musandam Peninsula)**	• **World's most critical oil transit chokepoint** (~20% of global petroleum passes through it)
Bab-el-Mandeb	**Red Sea & Gulf of Aden (Indian Ocean)**	**Yemen (Arabia/Asia) & Djibouti / Eritrea (Horn of Africa)**	• Known as the **"Gate of Tears"**; connects Indian Ocean shipping to the Suez Canal
Suez Canal	**Mediterranean Sea & Red Sea**	**Sinai Peninsula & Mainland Egypt (Africa)**	• Opened in **1869**; designed by French engineer **Ferdinand de Lesseps** • Eliminates passage around Cape of Good Hope; Port Said (north) and Port Tewfik/Suez (south)
Strait of Gibraltar	**Atlantic Ocean & Mediterranean Sea**	**Spain (Europe) & Morocco (Africa)**	• Known as the **"Key to the Mediterranean"** / ancient **"Pillars of Hercules"**; Rock of Gibraltar (UK) on north
Bosporus (Bosphorus) Strait	**Black Sea & Sea of Marmara**	**European Istanbul & Asian Istanbul (Turkey)**	• World's narrowest strait used for international navigation; separates Europe from Asia
Dardanelles Strait	**Sea of Marmara & Aegean Sea (Mediterranean)**	**Gallipoli Peninsula (Europe) & Asian Turkey**	• Historic naval battle of Gallipoli (WWI); Turkish Straits system regulated by Montreux Convention
Bering Strait	**Chukchi Sea (Arctic Ocean) & Bering Sea (Pacific Ocean)**	**Russia (Siberia/Asia) & Alaska (USA/North America)**	• **International Date Line (IDL)** passes directly through this strait
Palk Strait	**Bay of Bengal & Palk Bay / Gulf of Mannar**	**Tamil Nadu (India) & Jaffna Peninsula (Sri Lanka)**	• Named after Robert Palk (Governor of Madras)
Strait of Dover	**English Channel & North Sea**	**England (Great Britain) & France (Mainland Europe)**	• Narrowest part of English Channel (34 km); Channel Tunnel (*Chunnel*) runs beneath it
Panama Canal	**Atlantic Ocean (Caribbean Sea) & Pacific Ocean**	**North America & South America (Isthmus of Panama)**	• Opened in **1914**; 82 km long; uses 3 sets of freshwater locks (Gatun, Pedro Miguel, Miraflores)
Kiel Canal	**North Sea & Baltic Sea**	**Jutland Peninsula (Germany)**	• Bypasses the Danish straits; world's busiest artificial waterway by vessel count
Sunda Strait	**Java Sea & Indian Ocean**	**Java and Sumatra (Indonesia)**	• **Krakatoa** volcanic island sits directly inside this strait
Lombok Strait	**Java Sea & Indian Ocean**	**Bali and Lombok (Indonesia)**	• Deeper alternative to Malacca for supertankers; marks the **Wallace Line** biogeographical boundary
Torres Strait	**Arafura Sea & Coral Sea (Pacific Ocean)**	**Cape York (Australia) & Papua New Guinea**	• Shallow tropical strait filled with coral reefs
Bass Strait	**Tasman Sea & Southern Ocean / Indian Ocean**	**Victoria (Mainland Australia) & Tasmania**	• King Island and Flinders Island lie in it
Cook Strait	**Tasman Sea & South Pacific Ocean**	**North Island & South Island (New Zealand)**	• Connects capital Wellington with South Island

Strait / Canal	Water Bodies Connected	Landmasses Separated	High-Yield Prelims Exam Triggers
Strait of Magellan	**Atlantic Ocean & Pacific Ocean**	**Mainland South America (Chile) & Tierra del Fuego**	• Natural passage discovered by Ferdinand Magellan (1520) avoiding turbulent Drake Passage
Drake Passage	**South Atlantic Ocean & South Pacific Ocean**	**Cape Horn (South America) & South Shetland Islands (Antarctica)**	• Roughest open water body on Earth; connects Atlantic to Southern Ocean
Florida Strait	**Gulf of Mexico & Atlantic Ocean**	**Florida Keys (USA) & Cuba**	• Conveys the powerful warm Gulf Stream northward

21. Major World Rivers, Drainage Features & Riverbank Cities Master Table

A. World Landmark Rivers & Unique Geographic Traits

River Name	Length & Basin	Source & Mouth	Landmark Exam Triggers & Geographic Superlatives
Nile River	**6,650 km** (Africa)	White Nile (Lake Victoria) + Blue Nile (Lake Tana) → **Mediterranean Sea**	• **LONGEST RIVER IN THE WORLD** • Confluence of White & Blue Nile at **Khartoum (Sudan)** • Forms classic **Arcuate Delta** in Egypt • **Aswan High Dam** creates **Lake Nasser** • Capitals on banks: **Cairo** (Egypt), **Khartoum** (Sudan)
Amazon River	**6,400 km** (South America)	Peruvian Andes (Nevado Mismi) → **Atlantic Ocean**	• **LARGEST RIVER IN THE WORLD BY WATER VOLUME & BASIN AREA** (~20% of world river discharge) • Flows across dense tropical rainforests (**Selvas** — "Lungs of the Earth") • Has **NO bridges** across its entire main stem • Spectacular tidal bore known as **Pororoca**
Yangtze River (Chang Jiang)	**6,300 km** (Asia / China)	Tibetan Plateau (Tangjula Mtns) → **East China Sea** (Shanghai)	• **LONGEST RIVER IN ASIA** (3rd longest in world); flows entirely in China • Hosts **Three Gorges Dam** (world's largest hydroelectric plant by capacity) • Cities: **Chongqing, Wuhan, Nanjing, Shanghai**
Mississippi–Missouri	**6,275 km** (North America)	Lake Itasca (Minnesota) → **Gulf of Mexico**	• Longest river system in North America; forms textbook **Bird's Foot Delta** • Cities: Minneapolis, St. Louis, Memphis, New Orleans
Danube River	**2,850 km** (Europe)	Black Forest (Germany) → **Black Sea**	• Flows through **10 countries** (more than any other river in the world) • Passes through **4 national capitals**: **Vienna** (Austria), **Bratislava** (Slovakia), **Budapest** (Hungary), **Belgrade** (Serbia)
Rhine River	**1,230 km** (Europe)	Swiss Alps (Lake Constance) → **North Sea**	• Flows through a **rift valley** between **Vosges Mountains (France)** and **Black Forest (Germany)** • **Busiest commercial inland waterway in Europe**; mouth at **Rotterdam Port** (Netherlands) • Basel, Strasbourg, Bonn, Cologne on its banks
Congo River (Zaire)	**4,700 km** (Central Africa)	Katanga Plateau → **Atlantic Ocean**	• **DEEPEST RIVER IN THE WORLD** (>220 m); 2nd largest discharge after Amazon • **CROSSES THE EQUATOR TWICE!** • Boyoma (Stanley) Falls and Livingstone Falls; **Kinshasa** and **Brazzaville** face each other across it
Limpopo River	**1,750 km** (Southern Africa)	Witwatersrand (South Africa) → **Indian Ocean** (Mozambique)	• **CROSSES THE TROPIC OF CAPRICORN TWICE!**

River Name	Length & Basin	Source & Mouth	Landmark Exam Triggers & Geographic Superlatives
Mahi River	**583 km** (India)	Vindhya Range (MP) → **Gulf of Khambhat**	• **CROSSES THE TROPIC OF CANCER TWICE!**
Volga River	**3,530 km** (Russia / Europe)	Valdai Hills → **Caspian Sea**	• **LONGEST RIVER IN EUROPE**; internal drainage into landlocked Caspian Sea; Volgograd (Stalingrad) on its bank
Amur River (Heilong Jiang)	**2,824 km** (Asia)	Confluence of Shilka & Argun → **Sea of Okhotsk**	• Forms the natural international border between **Russia and China** for >1,600 km
Mekong River	**4,350 km** (Southeast Asia)	Tibetan Plateau → **South China Sea**	• Known as the "Danube of the East"; flows through 6 countries: China, Myanmar, Laos, Thailand, Cambodia, Vietnam; **Vientiane** and **Phnom Penh** on banks
St. Lawrence River	**1,197 km** (North America)	Lake Ontario → **Gulf of St. Lawrence (Atlantic)**	• Natural outlet of the Great Lakes; forms part of USA–Canada international border; Montreal and Quebec City on its banks

B. Famous World Cities on Rivers (Direct Match List)

- **London**: **Thames** (UK)
- **Paris**: **Seine** (France)
- **Rome**: **Tiber** (Italy)
- **Berlin**: **Spree** (Germany)
- **Vienna, Budapest, Belgrade**: **Danube** (Austria, Hungary, Serbia)
- **Warsaw**: **Vistula** (Poland)
- **Kyiv (Kiev)**: **Dnieper** (Ukraine)
- **Cairo, Khartoum**: **Nile** (Egypt, Sudan)
- **Baghdad**: **Tigris** (Iraq)
- **Washington, D.C.**: **Potomac** (USA)
- **New York**: **Hudson** (USA)
- **New Orleans**: **Mississippi** (USA)
- **Buenos Aires**: **Río de la Plata** (Argentina)
- **Bangkok**: **Chao Phraya** (Thailand)
- **Yangon**: **Yangon River / Irrawaddy delta** (Myanmar)
- **Lahore**: **Ravi** (Pakistan)
- **Karachi**: **Indus** (Pakistan)

22. World Lakes, Inland Seas & Salinity Master Matrix

Lake Name	Location / Countries Bordering	Key Geographic Superlative	High-Yield Prelims Exam Triggers
Caspian Sea	Bordered by **5 Countries (mnemonic: TARIK)**: **T**urkmenistan, **A**zerbaijan, **R**ussia, **I**ran, **K**azakhstan	**LARGEST LAKE / INLAND WATER BODY IN THE WORLD**	• Surface area ~371,000 km^2; slightly saline (~1.2%); largest endorheic basin • Baku (capital of Azerbaijan) situated on its western shore • Fed by Volga, Ural, and Kura rivers; has NO natural outlet
Lake Superior	**USA & Canada border** (Minnesota, Wisconsin, Michigan, Ontario)	**LARGEST FRESHWATER LAKE IN THE WORLD BY SURFACE AREA**	• Surface area ~82,100 km^2; part of North American Great Lakes • Connected to Lake Huron by St. Mary's River and **Soo Locks**

Lake Name	Location / Countries Bordering	Key Geographic Superlative	High-Yield Prelims Exam Triggers
Lake Baikal	**Russia** (Siberia — Irkutsk & Buryatia)	**DEEPEST LAKE IN THE WORLD** (1,642 m)	• **Oldest freshwater lake in the world** (25–30 million years) • **Largest lake by volume of unfrozen freshwater (~20% of world's surface freshwater)** • Drained by only ONE river: the **Angara River** (tributary of Yenisey); UNESCO WHS
Lake Tanganyika	Shared by **Tanzania, DR Congo, Burundi, Zambia**	**LONGEST FRESHWATER LAKE IN THE WORLD** (673 km)	• 2nd deepest lake in the world (1,470 m); situated in East African Great Rift Valley
Lake Victoria	Shared by **Tanzania (49%), Uganda (45%), Kenya (6%)**	**LARGEST TROPICAL LAKE IN THE WORLD**	• 2nd largest freshwater lake by surface area (68,800 km^2) • **Equator passes directly through it** • Primary reservoir and source of the **White Nile River**
Lake Titicaca	**Andes Mountains** (Border of **Peru and Bolivia**)	**HIGHEST COMMERCIALLY NAVIGABLE LAKE IN THE WORLD**	• Elevation: **3,812 m** above sea level; high alpine lake • Indigenous Uros people live on artificial floating islands made of **Totora reeds**
Dead Sea	Bordered by **Jordan, Israel, and West Bank**	**LOWEST WATER BODY ON EARTH'S LAND SURFACE**	• Surface elevation: **~430 m below sea level** • Hypersaline terminal lake (~34% salinity — 10x oceans); high density allows humans to float effortlessly • Fed by Jordan River
Lake Van	**Eastern Turkey** (Anatolia)	**HIGHEST SALINITY AMONG MAJOR WATER BODIES (~330‰ / 33%)**	• Saline soda lake with no outlet; water does not freeze even at high elevation
Lake Assal	**Djibouti** (Afar Depression, Horn of Africa)	Hypersaline lake (~348‰ salinity)	• Lowest point in Africa (-155 m below sea level); 2nd highest salinity on Earth
Lake Maracaibo	**Venezuela** (South America)	Largest lake in South America	• Connected to Gulf of Venezuela by Tablazo Strait; rich petroleum basin; famous for **Catatumbo Lightning**
Aral Sea	Border of **Kazakhstan and Uzbekistan**	Catastrophic environmental shrinkage	• Formerly 4th largest lake; shrunken to <10% original size due to Soviet diversion of **Amu Darya and Syr Darya** rivers for cotton fields
Lake Balkhash	**Kazakhstan** (Central Asia)	Unique half-fresh, half-saline lake	• Western part is fresh water (fed by Ili River), while eastern part is saline water
Lake Chad	Shared by **Chad, Cameroon, Niger, Nigeria**	Endorheic Sahelian lake	• Crucial wetland in the Sahel; shrunken by >90% since 1960s

The Great Lakes of North America (Strict West-to-East Chronology)

- **Superior → Michigan → Huron → Erie → Ontario** (Mnemonic: **SMHEO** or **HOMES**).
- **Lake Michigan** is the **ONLY ONE of the 5 Great Lakes situated entirely within the United States** (the other four are shared between USA and Canada).
- **Niagara Falls** is located on the Niagara River between **Lake Erie and Lake Ontario**!
- **Welland Canal** bypasses Niagara Falls to connect Lake Erie and Lake Ontario for commercial shipping.

23. World Major Grasslands & Deserts Master Cheat-Sheet

A. Grasslands of the World (Tropical Savannas vs Temperate Steppes)

Grassland Name	Type (Temperate / Tropical)	Geographic Region & Countries	High-Yield Prelims Exam Triggers
Prairies	**Temperate** (Mid-latitude)	**North America (USA & Canada)**	• Known as the **"Granary of the World"** (massive commercial wheat belt) • Warm katabatic wind **Chinook** ("Snow Eater") melts winter snow on eastern slopes of Rockies.
Steppes	**Temperate** (Mid-latitude)	**Eurasia** (Ukraine, Southern Russia to Central Asia & Mongolia)	• Extensive temperate grassland plain; dark fertile **Chernozem** soils; premier wheat & grain farming.
Pampas	**Temperate** (Mid-latitude)	**South America (Argentina & Uruguay)**	• Fertile alluvial soils; rich in nutritious **Alfalfa** grass; extensive cattle ranching (**Estancias**) and cowboys (**Gauchos**); wheat cultivation.
Veld / Veldt	**Temperate** (Mid-latitude)	**South Africa** (High Veld & Bush Veld plateau)	• Renowned for rearing **Merino sheep** for high-grade wool and maize cultivation (Maize Triangle).
Downs	**Temperate** (Mid-latitude)	**Australia** (Murray–Darling River Basin)	• Rolling grassland plains; premier sheep-farming stations for wool and wheat growing.
Canterbury Plains	**Temperate** (Mid-latitude)	**New Zealand** (South Island)	• Rich pastoral plains for sheep rearing and dairy farming.
Puszta	**Temperate** (Mid-latitude)	**Hungary** (Danube River Basin)	• Traditional Hungarian horse-breeding and grain plain.
Savanna / Sudan Type	**Tropical**	**East & Central Africa** (Kenya, Tanzania, Sudan)	• Characterized by tall coarse grasses (elephant grass) and scattered umbrella-acacia / baobab trees • Known as **"Big Game Country"** (abundant herbivore & carnivore diversity); Serengeti; Masai pastoralists.
Llanos	**Tropical**	**Venezuela and Colombia** (Orinoco River Basin)	• Tropical savanna grassland with distinct wet and dry seasons; extensive cattle ranching.
Campos	**Tropical**	**Brazilian Highlands (Brazil)**	• Subtropical / tropical savanna plateau grasslands south of the Amazon forest.
Cerrado	**Tropical**	**Central Brazil** (Mato Grosso)	• Vast tropical savanna ecoregion; converted into world's largest soyabean and cattle production frontier.
Gran Chaco	**Tropical / Semi-arid**	**Paraguay, Bolivia, Northern Argentina**	• Scrub woodland and savanna; Quebracho tree (*axe-breaker*) source of tannin.

B. Major Deserts of the World & Origin Factors

Desert Name	Location & Continents	Type & Origin Factors	Unique Exam Superlative / Highlights
Sahara Desert	**North Africa** (11 countries)	**Subtropical Hot Desert**	• **LARGEST HOT DESERT ON EARTH** (~9.2 million km^2); covers ~30% of Africa; Ergs (sand seas), Regs (stone plains), Hamadas (rock plateaus).
Atacama Desert	**Northern Chile & Southern Peru**	**Coastal Hyper-Arid Desert**	• **DRIEST NON-POLAR DESERT ON EARTH** • Formed by rain-shadow of high Andes + cold desiccating **Humboldt (Peru) Current** • Vast deposits of **Sodium Nitrate (Caliche)** and world-class copper mines.
Namib Desert	**Namibia** (South-West Africa)	**Coastal Fog Desert**	• Formed by the cold **Benguela Current** offshore; ancient red sand dunes; home to ancient living fossil plant *Welwitschia mirabilis*.

Desert Name	Location & Continents	Type & Origin Factors	Unique Exam Superlative / Highlights
Kalahari Desert	**Botswana, Namibia, South Africa**	**Semi-Arid Sandy Basin**	• Inhabited by the indigenous **San (Bushmen)** hunter-gatherers; vast red sand dunes (*Kgalagadi*).
Australian Deserts	**Central & Western Australia** (Great Victoria, Great Sandy, Gibson, Simpson)	**Subtropical Hot / Trade Wind Desert**	• Known as the **"Red Centre" / "Outback"**; formed by subtropical high-pressure belt + cold **West Australian Current**; Great Victoria is largest.
Gobi Desert	**Southern Mongolia & Northern China**	**Cold Continental Rain-Shadow Desert**	• Located in the rain-shadow of the Himalayas; extreme winter cold (-40°C); home to indigenous two-humped **Bactrian camels**.
Taklamakan Desert	**Xinjiang, Northwest China** (Tarim Basin)	**Cold Desert**	• Bounded by Tian Shan and Kunlun mountains; known as the "Sea of Death" / "Place of No Return".
Rub' al Khali	**Southern Arabian Peninsula** (Saudi Arabia, Oman, UAE, Yemen)	**Continuous Sand Desert (Erg)**	• Known as the **"Empty Quarter"**; **LARGEST CONTIGUOUS SAND DESERT IN THE WORLD** (~650,000 km^2).
Patagonian Desert	**Southern Argentina** (South America)	**Cold Winter Rain-Shadow Desert**	• Rain-shadow of the Southern Andes; influenced by the cold **Falkland Current**; gravelly plateau.
Dasht-e Kavir & Dasht-e Lut	**Iran** (Central Plateau)	**Salt Deserts (Playa)**	• *Kavir* = great salt swamp; *Lut* recorded world's highest satellite land skin temperature (>70.7°C).
Mojave & Sonoran Deserts	**Southwestern USA (California, Arizona) & Mexico**	**Subtropical Hot Desert**	• Contains **Death Valley** (**Badwater Basin, -86 m** — lowest point in North America; hottest recorded air temp 56.7°C); giant **Saguaro cactus**.

24. World Volcanoes, Oceanic Trenches & Landmark Islands

A. Famous Volcanoes of the World (Direct Match Sheet)

Volcano Name	Country / Geographic Location	Type / Activity	High-Yield Prelims Exam Triggers
Mount Stromboli	**Lipari Islands / Sicily, Italy** (Tyrrhenian Sea)	Active Stratovolcano	• Known as the **"LIGHTHOUSE OF THE MEDITERRANEAN"** due to continuous glowing eruptions visible for miles.
Mount Etna	**Sicily, Italy**	Active Stratovolcano	• **Highest active volcano in Europe** (~3,357 m); continuously active; UNESCO World Heritage Site.
Mount Vesuvius	**Gulf of Naples, Italy**	Active / Dormant Stratovolcano	• Catastrophic Plinian eruption in **79 AD** completely buried and preserved the Roman cities of **Pompeii and Herculaneum**.
Mauna Loa & Kilauea	**Big Island of Hawaii, USA** (Pacific Ocean)	Active Shield Volcanoes	• **Mauna Loa** is the **world's largest subaerial shield volcano** by volume and area • **Kilauea** is among the world's most continuously active volcanoes; fluid basaltic lava flows.
Cotopaxi	**Andes Mountains, Ecuador** (South America)	Active Stratovolcano	• One of the world's highest active volcanoes (5,897 m); textbook symmetrical snow-capped cone.
Ojos del Salado	**Andes Mountains (Border of Chile & Argentina)**	Active Stratovolcano	• **HIGHEST ACTIVE VOLCANO ON EARTH (6,893 m)**; 2nd highest peak in Western Hemisphere after Mount Aconcagua.
Mount Kilimanjaro	**Tanzania** (East Africa)	Dormant Stratovolcano	• **HIGHEST PEAK IN AFRICA (5,895 m)**; freestanding volcanic mountain with 3 volcanic cones: Kibo, Mawenzi, Shira; permanent snowcap near Equator.

Volcano Name	Country / Geographic Location	Type / Activity	High-Yield Prelims Exam Triggers
Krakatoa (Krakatau)	**Sunda Strait (Between Java and Sumatra, Indonesia)**	Active Caldera Volcano	• Cataclysmic eruption in **1883** generated massive tsunamis and shockwaves heard 4,800 km away; modern growing cone is **Anak Krakatau**.
Mount Merapi & Mount Sinabung	**Java & Sumatra, Indonesia**	Highly Active Stratovolcanoes	• Merapi (*Mountain of Fire*) is the most active volcano in Indonesia; deadly pyroclastic density currents.
Mount Pinatubo & Mayon	**Luzon, Philippines**	Active Stratovolcanoes	• Pinatubo's massive **1991 eruption** injected huge SO2 aerosol clouds causing global temporary cooling of ~0.5°C • **Mayon** is celebrated for having the **world's most perfectly symmetrical volcanic cone**.
Mount Fuji (Fujisan)	**Honshu Island, Japan**	Dormant Stratovolcano	• Highest peak in Japan (3,776 m); sacred mountain; last erupted in 1707 (Hoei eruption).
Mount Erebus	**Ross Island, Antarctica**	Active Stratovolcano	• **SOUTHERNMOST ACTIVE VOLCANO ON EARTH**; features a long-lived persistent phonolitic lava lake.
Barren Island	**Andaman Sea, India** (~135 km NE of Port Blair)	**Active Volcano**	• **THE ONLY CONFIRMED ACTIVE VOLCANO IN SOUTH ASIA / INDIA**; erupted in 1991, 2005, and continues sporadic activity.
Narcondam Island	**Andaman Sea, India**	**Dormant Volcanic Island**	• Extinct/dormant andesite volcano; exclusive habitat of the endemic, endangered Narcondam Hornbill.

B. Deepest Oceanic Trenches of the World

- **Mariana Trench (Challenger Deep)**: **10,994 m** (Western Pacific Ocean near Guam) — **DEEPEST POINT ON PLANET EARTH**.
- **Tonga Trench (Horizon Deep)**: **10,882 m** (Southwest Pacific Ocean) — 2nd deepest trench on Earth.
- **Kuril–Kamchatka Trench**: **10,542 m** (Northwest Pacific Ocean).
- **Philippine Trench (Galathea Deep)**: **10,540 m** (Western Pacific Ocean).
- **Puerto Rico Trench (Milwaukee Deep)**: **8,376 m** (Atlantic Ocean) — **DEEPEST POINT IN THE ATLANTIC OCEAN**.
- **Java / Sunda Trench**: **7,450 m** (Eastern Indian Ocean off Sumatra/Java) — **DEEPEST POINT IN THE INDIAN OCEAN**.
- **Factorian Deep (South Sandwich Trench)**: **7,434 m** (Southern Ocean) — **DEEPEST POINT IN THE SOUTHERN OCEAN**.
- **Molloy Deep**: **5,550 m** (Fram Strait, Arctic Ocean) — **DEEPEST POINT IN THE ARCTIC OCEAN**.

25. World Major Industrial Regions & Specialised Manufacturing / Mining Centres

Region / City	Country	Primary Manufacturing / Mineral Specialization	High-Yield Prelims Exam Nickname / Triggers
Ruhr Valley & Saar Basin	**Germany**	**Coal mining, Iron and Steel, Heavy Engineering**	• Known as the **"Ruhrgebiet" / Industrial Heart of Europe**; Duisburg (world's largest inland port).
Pittsburgh	**Pennsylvania, USA**	**Iron and Steel Industry**	• Celebrated as the **"STEEL CITY OF THE WORLD"**; confluence of Allegheny and Monongahela rivers.
Detroit	**Michigan, USA**	**Automobile Manufacturing**	• Celebrated as the **"AUTOMOBILE CAPITAL OF THE WORLD"** (Headquarters of Ford, GM, Chrysler).

Region / City	Country	Primary Manufacturing / Mineral Specialization	High-Yield Prelims Exam Nickname / Triggers
Birmingham	**West Midlands, England, UK**	**Iron, Steel, Heavy Metal Hardware**	• Historic heart of the Industrial Revolution; central to the English **"Black Country"**.
Manchester & Lancashire	**North-West England, UK**	**Cotton Textile Manufacturing**	• World's premier historic cotton textile spinning cluster due to humid oceanic climate.
Katanga (Shaba) & Copperbelt	**DR Congo & Zambia** (Central Africa)	**Copper and Cobalt Extraction**	• World's richest mineral belt for battery-grade **Cobalt and high-grade Copper**.
Witwatersrand ("The Rand")	**Johannesburg, South Africa**	**Gold Mining**	• **World's largest gold-bearing reef formation**; produced ~40% of all gold mined in human history.
Kimberley	**Northern Cape, South Africa**	**Diamond Mining**	• Historic diamond capital; the **"Big Hole"**; origin of the geological term **Kimberlite pipe**.
Baku	**Azerbaijan** (Caspian Sea)	**Petroleum Extraction & Refining**	• Historic birthplace of modern commercial oil extraction; Nobel Brothers and Rothschild oil legacy.
Krivoy Rog & Donbass (Donets Basin)	**Ukraine**	**Iron Ore & Coking Coal**	• Krivoy Rog = massive iron-ore basin; Donbass = premier historic coking coalfield.
Magnitogorsk & Kuznetsk (Kuzbass)	**Russia** (Urals & Siberia)	**Iron & Steel Combine**	• Magnitogorsk built around Magnetic Mountain (iron ore); Kuznetsk is a giant Siberian coal basin.
Upper Silesia	**Poland**	**Coal Mining, Zinc & Lead, Heavy Industry**	• Industrial core of Central Europe around Katowice.
Chuquicamata	**Antofagasta, Atacama Desert, Chile**	**Copper Mining**	• **World's largest open-pit copper mine by volume** and second deepest open-pit mine on Earth.
Potosí (Cerro Rico)	**Bolivia** (Andes Mountains)	**Silver Mining**	• Historic "Rich Mountain" that bankrolled the Spanish Empire with vast silver wealth for centuries.
Kalgoorlie & Coolgardie	**Western Australia**	**Gold Mining**	• Historic Australian desert gold rush of 1890s; famous **"Golden Mile"** open-pit Super Pit gold mine.
Sudbury Basin	**Ontario, Canada**	**Nickel, Copper, Platinum Group Metals**	• Ancient 1.85-billion-year-old meteorite impact basin; premier global source of nickel.

Famous Indian Industrial Nicknames & City Specializations

- **Detroit of India**: **Chennai** (Tamil Nadu — automobile manufacturing hub of South Asia).
- **Manchester of India**: **Ahmedabad** (Gujarat — premier cotton textile hub).
- **Manchester of North India**: **Kanpur** (Uttar Pradesh — historic cotton mills and leather industry).
- **Manchester of South India**: **Coimbatore** (Tamil Nadu — cotton spinning mills).
- **Cottonopolis of India**: **Mumbai** (Maharashtra — historic concentration of textile mills).
- **Silicon Valley of India / Electronic City**: **Bengaluru** (Karnataka — premier IT & aerospace cluster).
- **Pittsburgh of India / Steel City**: **Jamshedpur / Tatanagar** (Jharkhand — TISCO established 1907 at Sakchi, confluence of Subarnarekha and Kharkai rivers).
- **Brass City (*Peetal Nagari*)**: **Moradabad** (Uttar Pradesh — world-renowned brass handicraft industry).
- **Glass City / Bangle City (*Suhag Nagari*)**: **Firozabad** (Uttar Pradesh — glass blowing and bangles).
- **Leather City of the World**: **Kanpur** (Uttar Pradesh).
- **Perfume Capital of India (*Itra Nagari*)**: **Kannauj** (Uttar Pradesh).
- **Weaver City / City of Handlooms**: **Panipat** (Haryana).
- **City of Locks**: **Aligarh** (Uttar Pradesh).

- **Sports Goods Hub**: **Meerut** (Uttar Pradesh) and **Jalandhar** (Punjab).

Topic 1 — Indian Physical Geography: Mountains & Hills

Consolidated — 50 Must-Score Facts

1. India has **six** physiographic divisions: Northern Mountains, Northern Plains, Peninsular Plateau, Indian Desert (Thar), Coastal Plains, and Islands. The Thar is a **Pleistocene–recent** sand sheet, not a Tertiary desert.
2. Himalayan belts north to south are **Trans-Himalaya → Himadri (Greater) → Himachal (Lesser) → Shiwalik**. Himadri is crystalline and **fossil-less**; Himachal carries **marine fossils**; Shiwalik is outermost and holds **human remains** / unconsolidated sediments.
3. Among usual age options, **Himadri is the youngest** Himalayan belt and **Aravalli is the oldest** fold system of India. **Guru Shikhar (1722 m)** on Mount Abu is Rajasthan's highest peak.
4. The Kashmir Valley sits between **Pir Panjal (south)** and **Himadri (north)**. **Karewas** are lacustrine terraces famous for saffron. A **dun** is a longitudinal valley between Himachal and Shiwalik (e.g. Dehra Dun) — not a karewa.
5. Central India hills west to east run **Satpura → Mahadeo → Maikal → Chhotanagpur**. **Vindhya** lies **north** of the Narmada; **Satpura** lies **south** of the Narmada.
6. State–peak pairs: **Tamil Nadu–Doddabetta**, **Rajasthan–Guru Shikhar**, **Nagaland–Saramati**, **Madhya Pradesh–Dhupgarh**, **Kerala–Anaimudi**, **Uttarakhand–Nanda Devi**.
7. **Lipulekh, Niti and Mana** are all in **Uttarakhand** (Lipulekh ≠ Ladakh; Mana ≠ Himachal). **Nathu La** = Sikkim; **Shipki La** = Himachal; **Zoji La** = J&K / Ladakh axis.
8. Tirupati's Venkateswara temple stands on the **Tirumala / Mallamalla Hills** of the **Eastern Ghats** (Andhra Pradesh) — not the Shevaroy Hills of Tamil Nadu.
9. **Intertrappean beds** between Deccan lava flows hold **land and freshwater** fossils, not marine biota. Trap thickness teaching figures: upper ~**450 m**, middle ~**1200 m**, lower ~**150 m**.
10. India is the **seventh**-largest country by area (~**3.28 million km²**, ~**2.4%** of world land). The Tropic of Cancer cuts the **middle** of India — the country is **not** wholly tropical.
11. The **Atal Tunnel** runs under **Rohtang** in the **Pir Panjal** (Himachal). Safe tag: longest highway tunnel **above 10,000 ft** — not an unqualified "world's longest." **Sela Tunnel** (2024) is the Arunachal Tezpur–Tawang all-weather link.
12. Tropic of Cancer (**23°30′ N**) crosses **eight states only** W→E: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram — **not Uttar Pradesh** and **not Ladakh**.
13. India's Standard Meridian is **82°30′ E** near **Mirzapur (UP)** and passes through **five states**: UP, Madhya Pradesh, Chhattisgarh, Odisha, Andhra Pradesh (**not** Telangana / Maharashtra). **IST = GMT + 5 h 30 min** for the whole country.
14. **Namcha Barwa** is in **Tibet**, not India. **K2** stands in the **Karakoram**, not on Himadri. **Kanchenjunga** is the highest peak **fully in India**.
15. The **Konkan** coast is a coast of **submergence**. The **Malabar** and **Coromandel** coasts are coasts of **emergence**.
16. The **Marwar Plateau** lies **east** of the Aravalli; the **Marwar Plain / Thar** lies **west** of the Aravalli. **Marusthali** is the sandy desert core; **Bagar** is the semi-arid eastern fringe.
17. **Palghat (Palakkad) Gap** is a **rift** and the widest break in the Western Ghats, linking Kerala with Tamil Nadu. **Anaimudi (2695 m)** is South India's highest peak and the hub of Anamalai, Palani and Cardamom.
18. The snowline is **lower in the western Himalaya** than in the east. **Siachen (~76 km)** is India's longest glacier (Karakoram / Nubra). **Zemu** (Sikkim) feeds the **Teesta**. **Sonapani** is the largest glacier of the Pir Panjal.
19. Western Ghats are a **continuous** wall from the Tapi gap toward Kanyakumari; Eastern Ghats are **discontinuous** and lower. **Nilgiri** is the junction of both Ghats with the southern hills.
20. **Purvanchal Hills** are the NE fold hills beyond the Dihang (Patkai, Naga, Mizo, Mishmi) — **not** the eastern Uttar Pradesh plain also called Purvanchal.
21. **Garo, Khasi and Jaintia** hills form the **Meghalaya Plateau** and are geologically **peninsular**, not Himalayan folds. Mawsynram and Cherrapunji sit on the **Khasi Hills**.

22. Extreme points: north **Indira Col** (Siachen / Ladakh); southernmost **territory Indira Point** (Great Nicobar); southernmost **mainland Kanyakumari**; east **Kibithu** (Arunachal); west **Guhar Moti** (Gujarat).
23. Total coastline ≈ **7,516.6 km** (mainland ~5,422.6 + islands ~2,094). Longest mainland state coast: **Gujarat**, then Andhra Pradesh, then Tamil Nadu. **Telangana is not** a coastal state. Mean Sea Level is measured from the **Chennai** coast.
24. The **10° Channel** separates Andaman from Nicobar; the **9° Channel** separates Minicoy from the rest of Lakshadweep; the **8° Channel** lies between Minicoy and the Maldives. Andaman–Nicobar are largely **volcanic / tectonic**; Lakshadweep is **coral**.
25. The Himalayan arc is about **2400 km** from the Indus gorge to the Dihang gorge. Western syntaxial bend ≈ **Nanga Parbat**; eastern bend ≈ **Namcha Barwa** (Tibet).
26. **Duars** are the West Bengal–Assam foothills opening toward Bhutan — not the same as UP **Terai**. **Bhabar** = pebble fans where streams sink; **Terai** = marshy belt where streams re-emerge.
27. Uttar Pradesh's highest point is **Amsot (~941 m)** on the Kaimur / Sonbhadra fringe — a **Vindhyan** peak, not Himalayan. Bhabar–Terai foothills appear in Pilibhit, Lakhimpur Kheri, Bahraich and Shravasti.
28. The core UP plain is the **Ganga–Yamuna Doab**. South-west Bundelkhand is granite–gneiss and drought-prone. Eastern UP "Purvanchal" is a **plain region**, not the NE Purvanchal Hills.
29. **Annamalai** and **Sirumalai** are **Tamil Nadu peninsular** hills — never park them in a Himalayan match list. **Shevaroy** is also Tamil Nadu, not Tirupati's hill.
30. The **Deccan Plateau** is the southern tableland of the old peninsular block; **Central Highlands** (Malwa, Bundelkhand, Baghelkhand) are the northern part. The plateau slopes **high west → low east**.
31. The **Aravalli** runs ~**800 km** from Palanpur (Gujarat) toward Delhi as a **relict Archaean** fold belt and helps cast the rain shadow that feeds the Thar.
32. The **Northern Plains** are Indus–Ganga–Brahmaputra alluvium with the ladder **Bhabar → Terai → Bhangar → Khadar**. Eduard Suess (1893) framed the Indo-Gangetic depression as a **fore-deep** later filled by fluvial alluvium.
33. **Artesian wells** of the northern plains / Uttarakhand teaching set sit in the marshy **Terai**, not in Bhabar or Shiwalik.
34. Holland's Indian rock systems run **Archaean → Purana → Dravidian → Aryan**. Age ladder often asked: Archaean → Dharwar (metalliferous) → Cuddapah → Vindhyan → Gondwana (~**98%** of Indian coal) → Deccan Traps → Tertiary → Quaternary.
35. **Kuttanad (Kerala)** lies about **1.2–3.0 m below sea level** — India's lowest inhabited tract and FAO GIAHS "Rice Bowl of Kerala."
36. **Luni** is the **only large river** of the Indian Desert; the arid core often receives **less than 150 mm** rainfall annually.
37. Himalayan thrusts south to north: **MFT** (plain / Shiwalik) → **MBT** (Shiwalik / Lesser) → **MCT** (Lesser / Greater) → **STDS** → **Indus–Tsangpo Suture**.
38. **Kashmir Valley** and **Imphal Basin** are classic **lacustrine** plains. Mizoram's unconsolidated sediments are often tagged a **molasse basin**.
39. **Jindhagada Peak (~1690 m)** in Araku Valley (Andhra Pradesh) is the usual highest peak of the **Eastern Ghats** (ahead of Arma Konda / Deomali).
40. Tropic of Cancer and IST meridian (**23°30′ N / 82°30′ E**) intersect in **Koria (Korea) district, Chhattisgarh**, on the Baghelkhand Plateau.
41. Gujarat–Arunachal longitude span gives ~**2 hours** local-time lag (~**118 min**). At noon IST, longitude **120° E** reads **14:30**. At 10:00 IST, Shillong (~**92° E**) is about **10:38**.
42. Land frontier length ascending: Afghanistan < Bhutan < Myanmar < Nepal < Pakistan < China < **Bangladesh (longest)**. Total land frontier ≈ **15,106.7 km**.
43. Glacier length tags: **Siachen** longest in India; among other Karakoram names **Hispar** and **Biafo** are next-tier long glaciers. **Bara Shigri** is a major Himachal glacier.
44. Island peaks: **Saddle Peak (732 m, North Andaman)** highest in Andaman; **Mount Thuillier (642 m, Great Nicobar)** highest in Nicobar. **Barren Island** is India's only confirmed **active** volcano (Andaman).

45. **Kangra** is a strike / longitudinal valley; **Kulu** is a transverse valley (Himachal). Do not swap the two valley types.
46. Karnataka hill–plain tags: **Malnad** = forested hill tract; **Maidan** = rolling granite plain.
47. Coastal erosion on open shores is driven mainly by **waves**. **Raman Sagar Mountain** (~1505 m undersea, Arabian Sea SW of Mumbai) is a named submarine relief fact.
48. **Chotanagpur** and **Rajmahal** sit on the eastern peninsular highland belt; Rajmahal is linked with Santhal Pargana teaching maps.
49. **Cardamom Hills** continue south of Anamalai toward Kerala–Tamil Nadu; they are peninsular, not Himalayan.
50. **Sela Tunnel (2024)** and **Z-Morh / Sonamarg Tunnel (2025)** are current all-weather Himalayan highway tags — pair them with Atal / Rohtang and Zoji La axes, not with Eastern Ghat passes.

📋 Must-score facts — belts, peaks, passes

Himalayan belts (N → S)

Trans-Himalaya → Himadri (Greater) → Himachal (Lesser) → Shiwalik

Item	Lock
Youngest belt (usual options)	**Himadri** (हिमाद्रि)
Oldest fold system (usual options)	**Aravalli** (अरावली)
Kashmir Valley	Pir Panjal **south**; Himadri **north**

Central India hills (W → E)

Satpura → Mahadeo → Maikal → Chhotanagpur · **Vindhya** lies **north** of Narmada

State ↔ peak

State	Peak
Tamil Nadu	**Doddabetta**
Rajasthan	**Guru Shikhar** (Aravalli)
Nagaland	**Saramati**

Passes / temple

Item	Lock
Lipulekh / Niti / Mana	All in **Uttarakhand** (Lipulekh ≠ Ladakh)
Venkateswara / Tirupati	**Tirumala / Mallamalla** — Eastern Ghats (AP)
Intertrappean beds	Land / freshwater fossils (not marine)
India area rank	**7th** (~3.28 million km^2; ~2.4% world land)

Topic 2 — Climate of India

Consolidated — 48 Must-Score Facts

1. India's climate is **tropical monsoon**, not wholly tropical, because the north is subtropical. The Himalaya is a winter **wall** against Central Asian cold and a summer **orographic** barrier on southern slopes.
2. Monsoon comes from Arabic **mausim** (season). About **75–90%** of India's rain falls with the south-west monsoon (**June–September**). Normal Kerala onset is around **1 June**.
3. IMD seasons: Cold (**Dec–Feb**, Western Disturbances in the NW), Hot (**Mar–May**, Loo / Nor'westers / mango–blossom showers), SW Monsoon (**Jun–Sep**), Retreating Monsoon (**Oct–Nov**, October heat + Bay / Andaman cyclones).
4. SW monsoon starts when the **ITCZ** shifts north and a thermal low forms over NW India. SE trades cross the equator near **40–60° E**; the **Mascarene High** and **Somali Jet** feed the two branches.
5. Classical **thermal** monsoon theory = land–sea differential heating. **Flohn's dynamic** concept = seasonal migration of planetary pressure / wind belts with the ITCZ.
6. The **Arabian Sea branch** soaks Western Ghats windward slopes and leaves the leeward Deccan dry. The **Bay of Bengal branch** waters the north-east and the Ganga plain.
7. The Tamil Nadu SE coast stays **dry in the SW monsoon** (parallel to the Bay branch + Arabian-Sea rain shadow). Its main rain is the **north-east monsoon** later.
8. Typical onset ladder: Kerala ~**1 June** → Mumbai / Kolkata ~**10 June** → Delhi ~**29 June** → Rajasthan last ~**mid-July**. Burst is sudden; withdrawal is gradual NW → SE.
9. **Active** monsoon = trough on the **Ganga plain** (plains wet). **Break** monsoon = trough shifts onto the **Himalaya** (central India dry). Break ≠ retreating monsoon.
10. Northern-plains rainfall generally **declines east → west**. The rainy season is **shorter south → north**. Rajasthan stays dry partly because the Aravalli runs **parallel** to the Arabian Sea branch.
11. IMD counts a **rainy day** from **≥ 2.5 mm** in 24 hours. India can still face seasonal water stress because most rain is packed into four monsoon months.
12. The **north-east monsoon (Oct–Dec)** mainly serves Tamil Nadu, south Andhra, the SE Karnataka–Kerala fringe and Puducherry — **not** all-India winter rain.
13. Retreating-monsoon months are mainly **October–November** (October heat + Bay / Andaman cyclone risk). Do not call this the Western Disturbance season.
14. **Western Disturbances** are Mediterranean **extra-tropical** systems bringing NW winter rain that **decreases west → east**. Core belt: Punjab, Haryana, Himachal, J&K, Uttarakhand, western UP, northern Rajasthan.
15. Winter anticyclonic high over northern India comes from cold, dense, **subsiding continental air** — it is **not** caused by Western Disturbance rain.
16. Mid-latitude jets are mainly **westerly**. Winter **Subtropical Westerly Jet** south of the Himalaya steers Western Disturbances after Tibet splits the jet. The northern branch does **not** drive India's winter rain.
17. The **Tropical Easterly Jet** is a summer **easterly** around ~**14° N** and supports the SW monsoon. The **Somali Jet** is the cross-equatorial monsoon feed.
18. **El Niño** = warm eastern Pacific / Peru phase; usually weakens the Indian monsoon and **cuts** plankton (less upwelling). **La Niña** = cool phase; usually strengthens the monsoon.
19. The **Indian Ocean Dipole** compares west vs east **Indian Ocean**, not the Pacific. **+IOD** helps the Indian monsoon; **−IOD** hurts it.
20. Köppen India map: **Am** = Kerala / Konkan / NE; **Aw** = Deccan; **As** = Tamil Nadu dry-summer Coromandel; **BWh** = Thar; **Cwg** = Ganga plains / north Bihar / most of the UP plain.
21. January **18°C** isotherm ≈ tropical–subtropical divide. **Diurnal range** is highest in the Rajasthan desert. **Leh** is among the driest class stations; **Mawsynram / Cherrapunji** are the wettest class.

22. Troposphere is heated mainly by **long-wave terrestrial** radiation. Normal lapse ≈ **6.5 °C / 1000 m**. Relative humidity **falls as temperature rises**. Ozone UV shield peaks in the **stratosphere**.
23. Circulation cells: **Hadley (0–30°)**, **Ferrel (30–60°)**, **Polar (60–90°)**. **Doldrums** = rainy ITCZ calm; **horse latitudes** = dry subtropical **high** near **30°**.
24. Rain types: **orographic**, **convectional**, **cyclonic**. An **isohyet** is a line of equal rainfall.
25. **Loo** = hot dry May–June wind of the Indo-Gangetic plains. **Kal Baisakhi / Nor'westers** = violent April–May thunderstorms of eastern / north-eastern India.
26. **Mango showers** = pre-monsoon rains of Kerala / Karnataka (and coastal Tamil Nadu). **Blossom showers** help coffee in Tamil Nadu / Kerala.
27. **Sea breeze** = day, sea → land; **land breeze** = night, land → sea. **Anabatic** = day upslope; **katabatic** = night downslope. **Foehn / Chinook** = warm dry lee winds, not cold katabatic drainage.
28. Tropical cyclones form more often over the **Bay of Bengal** than the Arabian Sea. SST needs ~**26–27 °C**. **Eye** calm; **eyewall** fiercest. They do **not** form on the equator.
29. Regional cyclone names: **Hurricane** (USA / Atlantic), **Typhoon** (NW Pacific), **Baguio** (Philippines), **Willy-willy** (Australia), **Cyclone** (North Indian Ocean). Willy-willy ≠ Australian hot wind **Brickfielder**.
30. Tornado intensity scale = **Fujita**; hurricane scale = **Saffir–Simpson**. Cumulonimbus = thunder / intense rain.
31. Thornthwaite's key line: **vegetation is the true index of climate** — that is **not** Köppen's letter-code system.
32. Absolute humidity = **mass of vapour**; relative humidity = **% of saturation**. RH can fall when temperature rises even if vapour mass is unchanged.
33. Traditional six ritus order: **Vasanta** → **Grishma** → **Varsha** → **Sharad** → **Hemant** → **Shishir**.
34. Spatial rainfall teaching ladder often runs Kochi > Kolkata > Patna > Delhi. Rajasthan driest class station ≈ **Jaisalmer**; wettest class ≈ **Mawsynram / Cherrapunji**.
35. Heavy Pushkar-hills rain can drain via the **Luni** and flood low **Balotra** (Barmer) in the Pachpadra depression.
36. In Uttar Pradesh, May–June **Loo** hits the plains; winter rain / fog track **Western Disturbances**, and **western UP is wetter than eastern UP** in that winter belt.
37. UP's SW-monsoon rain often arrives with **Bay of Bengal depressions**. Active spells need the trough on the plain. Pre-monsoon lightning and winter **inversion smog** (west UP / NCR fringe) are key hazards.
38. Coromandel / Chennai coast is **NE-monsoon dependent**, not a SW-monsoon rain belt. Retreating-monsoon October heat ≠ WD winter rain.
39. Pressure belts shift with the seasons; ITCZ follows the overhead sun into the Indian landmass in northern summer.
40. Jet-stream speeds in teaching notes often reach **300–500 km/h**. "All jets are easterly" is false — main jets are westerly; TEJ is the summer easterly exception.
41. **Albedo** = reflecting power (fresh snow highest). Atmosphere heat budget teaching still parks nuclear / geothermal outside "stored solar" energy.
42. Local foreign winds often paired in distractors: **bora** (cold Adriatic), **mistral** (cold Rhône), **harmattan** (West Africa dust), **khamsin** (Egypt heat) — none is India's Loo.
43. El Niño tendency in a given year is **not** a guaranteed all-India drought; IOD phase can offset or reinforce the Pacific signal.
44. **Mission Mausam** (MoES / IMD) is the institutional monsoon / severe-weather forecast upgrade tag used in current-affairs stems.
45. Desert Development Programme teaching start is **1977–78** in arid / semi-arid blocks (Rajasthan–Gujarat–Haryana core, later expanded).
46. Temperature **inversion** = temperature rises with height in a short layer; calm winter nights trap pollution in west-UP / NCR basins.
47. Condensation releases **latent heat**, which helps fuel deep convection and monsoon thunderstorms.
48. India's mean annual precipitation is high in national totals, yet seasonal concentration + runoff + evaporation create the classic "wet country, dry months" water-stress paradox.

📋 Must-score facts — seasons, monsoon, onset

Frame

Item	Lock
Climate type	**Tropical monsoon** (not wholly tropical)
Etymology	Arabic **mausim** (season)
SW monsoon share	~**75–90%** of annual rain

IMD seasons

Season	Months	Tag
Cold	Dec–Feb	Western Disturbances (NW)
Hot	Mar–May	Pre-monsoon / loo
SW monsoon	Jun–Sep	Main rains
Retreating / NE	Oct–Dec	TN coast rain

Onset / branch tags

Place	Typical onset
Kerala	~**1 June**
Mumbai / Kolkata	~**10 June**
Delhi	~**29 June**
Branch	Lock
--------	------
Arabian Sea	Windward Western Ghats wet; leeward Deccan dry
Tamil Nadu SE coast	**Dry in SW monsoon**; rain in **NE monsoon** (उत्तर-पूर्व मानसून)
Active vs break	Trough on Ganga plain vs shift to Himalaya foothills
Plains rain	Declines **east** → **west**

Topic 3 — Drainage System

Consolidated — 50 Must-Score Facts

1. About **77%** of India's drainage **area** faces the **Bay of Bengal** and about **23%** faces the **Arabian Sea**; inland basins cover ~**8%** of area. Over **90% of river water** still goes to the Bay of Bengal.
2. Water-share ladder: **Brahmaputra ~40%**, **Ganga ~25%**, **Godavari ~6%**. Brahmaputra leads volume; **Ganga** leads basin area **inside India**.
3. Western Ghats are the main peninsular divide. From **Amarkantak**, drainage is radial — **Narmada** west and **Son** toward the Ganga.
4. **Ganga** = largest basin in India; **Godavari** = largest **peninsular** basin. Among Tapi / Narmada / Mahanadi / Cauvery alone, **Mahanadi** is the largest basin.
5. Himalayan rivers are mostly **perennial** (melt + rain) and often **antecedent**. Peninsular rivers are mostly **rain-fed**, older and graded.
6. Drainage types: **consequent** follows original slope; **subsequent** follows weak belts later; **antecedent** (Indus, Sutlej, Brahmaputra) cuts rising Himalaya; **superimposed** (Chambal) inherits an older course.
7. **Trans-Himalayan** rivers rise north of the Great Himalaya in Tibet and cut through — Indus, **Sutlej** and Brahmaputra (Tsangpo). Among Jhelum / Sutlej / Ganga / Ravi, the trans-Himalayan pick is **Sutlej**.
8. Indus reaches the Arabian Sea in Pakistan. Punjab rivers E→W: **Sutlej–Beas–Ravi–Chenab–Jhelum**. Indus Treaty: India gets Ravi, Beas, Sutlej; Pakistan gets Indus, Jhelum, Chenab.
9. Dam–river pairs: **Baglihar = Chenab**, **Pandoh = Beas** (≠ Ravi), **Srisailam = Krishna** (≠ Tungabhadra), **Tulbul = Jhelum / Wular**.
10. The river is named **Ganga** only at **Devprayag** (Alaknanda + Bhagirathi). Length ~**2525 km**. In Bangladesh it is the **Padma**; Brahmaputra is the **Jamuna**; both join the Meghna system to the Bay.
11. Ganga joins: Yamuna at **Prayagraj**, Ramganga at **Kannauj**, Ghaghara at **Chhapra**, Gandak at **Sonpur**, Son at **Patna**, Kosi at **Kursela**.
12. Left-bank Himalayan Ganga tributes include Ramganga, Ghaghara, Gandak, Kosi and Mahananda; **Gomti** is a **plains-origin** left-bank river. Right-bank facts: Yamuna, Son, Damodar.
13. Yamuna right-bank tributes W→E: **Chambal–Sind–Betwa–Ken**. Chambal joins near Etawah; Betwa at **Hamirpur**; Ken at **Banda**. Betwa does **not** meet at Prayagraj.
14. **Jonk** joins the **Mahanadi**, not the Ganga. **Gandak** does **not** flow through Uttar Pradesh in the usual state list. **Punpun** meets near Fatuha. Chambal ravines start near **Mhow**.
15. Panch Prayag upstream → downstream: **Vishnu** (Dhauli) → **Nanda** (Nandakini) → **Karn** (Pindar) → **Rudra** (Mandakini) → **Dev** (Bhagirathi) = Ganga. Badrinath is on the **Alaknanda**; Kedarnath is on the **Mandakini**.
16. The eastern **Dhauliganga** joins the **Kali**, not the Alaknanda — do not confuse it with Vishnuprayag's Dhauli.
17. Brahmaputra rises as Angsi / Tsangpo, makes a **U-turn near Namcha Barwa**, and enters India as Siang / Dihang. **Majuli** is the classic river island. North/right tributes include Subansiri, Manas, Sankosh, Teesta; south/left include Burhi Dihing and Dhansiri.
18. **Teesta** was captured from the Ganga system into the Brahmaputra. **Barak** becomes Meghna and reaches the Bay — it is **not** inland drainage.
19. Peninsular length ladder: **Godavari (~1465 km) > Krishna (~1400) > Narmada (~1312) > Mahanadi (~851) > Tapi (~724)**. Godavari (Dakshin / Vriddha Ganga) rises near **Trimbak**.
20. Origins: Yamuna–Yamunotri; Krishna–Mahabaleshwar; Godavari–Nashik / Trimbak; Mahanadi–Sihawa; Narmada–Amarkantak; Tapi–**Multai (Satpura)** not Western Ghats; Kaveri–Brahmagiri.
21. **Pranhita** → Godavari; **Hemavati** → Kaveri; **Malaprabha** → Krishna. Hyderabad stands on the **Musi**, not the Krishna stem. Ludhiana is on the **Sutlej**, not the Ravi.
22. West-flowing set: Narmada, Tapi, Mahi, Sabarmati, **Periyar**. Narmada and Tapi form **estuaries** in rift valleys; most east-flowing peninsular rivers build **deltas**.

23. **Luni** is inland drainage: upper course fresh, lower course **saline**. Classic fault-valley tag = **Damodar**; Narmada and Tapi are also rift valleys.
24. Delta types: **arcuate** = Sundarbans; **bird's-foot** = Mississippi (not Ganga). **Trellis** = folded structure (e.g. Singhbhum); **radial** = Amarkantak; **dendritic** = homogeneous slope (Ganga / Godavari / Krishna / Kaveri class).
25. National Waterways: **NW-1** Ganga–Hooghly; **NW-2** Brahmaputra; **NW-3** Kerala; **NW-4** Krishna–Godavari; **NW-5** Brahmani–Mahanadi. National Waterways Act (2016) lists **106** NWs.
26. **Ken–Betwa** is the **only** National Perspective Plan link under implementation (Bundelkhand MP–UP; Panna TR forest link).
27. UP anchors: Sangam at **Prayagraj**; **Gomti** = Lucknow's river; **Ghaghara** = Ayodhya's river; core plain = Ganga–Yamuna **Doab**.
28. National Chambal Sanctuary spans **UP + MP + Rajasthan** — not Haryana.
29. Kaveri island sequence: **Srirangapatna** → **Shivasamudram** → **Srirangam**. Bhubaneswar is **not** a Mahanadi-bank city in the usual trap set.
30. East-flowing peninsular rivers N→S often run **Subarnarekha** → **Mahanadi** → **Godavari** → **Krishna** → **Pennar** → **Cauvery** → **Vaigai**.
31. Indus is longest among the four Himalayan systems overall, but most of its course lies in Pakistan. **Ganga** is the longest river **in India**.
32. Concordant drainage follows structure; **discordant** drainage cuts across it — Himalayan antecedents are the classic discordant case.
33. **Kosi** = Sorrow of Bihar; **Damodar** = Sorrow of Bengal. Do not swap the titles.
34. Indian **trellis** ground = folded **Singhbhum** (Chotanagpur). **Rectangular** streams follow joints on **Vindhyan** rocks. **Sharavati** shows a **parallel** Western Ghats pattern.
35. **Centripetal** streams drain into **Loktak** and into **Sambhar** (Mendha, Rupangarh).
36. Gangetic alluvium depth teaching: Oldham ~**4000–6000 m**; Glennie ~**2000 m**.
37. **Son** rises from the Amarkantak belt and has the **largest catchment among major Ganga tributes** in the usual comparison set; it joins the Ganga near Patna.
38. The **Kaimur Range** divides the Son system (south) from the Tons / Yamuna system (north).
39. In Nepal, Gandak = **Gandaki / Narayani**. **Triveni Canal** draws Gandak water in West Champaran (Bihar). Gandak joins Ganga at Sonpur and does **not** traverse UP.
40. In Bihar the Ganga passes through **12 districts**; longest district stretches are often tagged **Patna** then **Bhagalpur**. Punpun joins at **Fatuha**.
41. Kosi is formed by seven Himalayan streams (**Arun** principal) and is **not** a simple glacier-origin stem. **Kanwar Lake** (Begusarai) is an oxbow of the **Burhi Gandak**, not the Gandak.
42. **Nagavali** and **Vamsadhara** rise from the Eastern Ghats; **Brahmani** and **Subarnarekha** rise from the Chotanagpur / Ranchi plateau.
43. **Chandra** and **Bhaga** unite at **Tandi** (Lahaul) to form the **Chenab (Chandrabhaga)**. Punjab doabs: **Bist** (Beas–Sutlej), **Bari** (Beas–Ravi), **Rechna** (Ravi–Chenab), **Chaj** (Chenab–Jhelum).
44. **Gandikota canyon** (Kadapa, AP) was carved by the **Pennar**. In S.P. Chatterjee's division, the Mahanadi Basin sits under the **Northern Deccan Plateau**.
45. City–river anchors: **Kanpur** largest city on the Ganga; **Uttarkashi** on Bhagirathi; **Leh** on the **right bank** of the Indus; **Pahalgam** on the **Lidder**; **Hampi** on the **Tungabhadra**; **Pandharpur** on the Bhima / Chandrabhaga; **Govindghat** on Alaknanda / Lakshman Ganga (≠ Mandakini).
46. Rectangular / joint-controlled streams on Vindhyan rocks differ from dendritic plains patterns — pattern stems often park both traps together.
47. **Periyar** is a major west-flowing Kerala river; do not list it with east-coast delta builders like Godavari / Krishna.
48. River rejuvenation / knickpoints matter for waterfall chapters, but drainage stems still ask rejuvenated Himalayan / peninsular course evidence via gorges and terraces.
49. Inland drainage classic = **Luni** (Thar) plus centripetal playa catchments such as **Sambhar** — not Barak / Meghna.

50. Namami Gange / GAP are **cleaning programmes**, not river-origin facts; do not use them to answer confluence or source stems.

📋 Must-score facts — basins, types, dam–river

Area / water share

Item	Lock
Bay of Bengal drainage area	~**77%**
Arabian Sea drainage area	~**23%**
Water volume	Brahmaputra ~**40%** · Ganga ~**25%** · Godavari ~**6%**
Largest basin in India	**Ganga**
Largest peninsular basin	**Godavari**
Main peninsular divide	**Western Ghats** (पश्चिमी घाट)
Amarkantak radial	**Narmada** west · **Son** east

Drainage types

Type	Lock
Consequent	Follows original slope
Subsequent	Follows weak belts later
Antecedent	Cuts rising land (Indus, Sutlej, Brahmaputra)
Himalayan vs peninsular	Mostly perennial vs mostly rain-fed / seasonal

Dam ↔ river

Dam	River
Baglihar	**Chenab** (चिनाब)
Pandoh	**Beas** (ब्यास) (≠ Ravi)
Srisailam	**Krishna** (कृष्णा) (≠ Tungabhadra)

Master Table — Panch Prayag of Uttarakhand (पंच प्रयाग)

(Tested heavily in match-the-pair: UPPCS 2023 Q50, UKPCS)

Prayag (प्रयाग)	Headstream 1	Headstream 2 (Tributary)	District / Location	Key Shrine / Geographical Landmark
1. Vishnuprayag (विष्णुप्रयाग)	**Alaknanda** (अलकनंदा)	**Dhauliganga** (धौलीगंगा - Western)	Chamoli, Uttarakhand	Nearest prayag to Badrinath & Joshimath; origin from Satopanth glacier
2. Nandaprayag (नंदप्रयाग)	**Alaknanda**	**Nandakini** (नंदाकिनी)	Chamoli, Uttarakhand	Confluence named after King Nanda; Nandakini rises from Nanda Ghunti
3. Karnaprayag (कर्णप्रयाग)	**Alaknanda**	**Pindar** (पिंडर)	Chamoli, Uttarakhand	Pindar rises from Pindari glacier (Kumaon/Garhwal border)
4. Rudraprayag (रुद्रप्रयाग)	**Alaknanda**	**Mandakini** (मंदाकिनी / Kali Ganga)	Rudraprayag, Uttarakhand	Mandakini flows down from **Kedarnath** (Chorabari glacier)
5. Devprayag (देवप्रयाग)	**Alaknanda**	**Bhagirathi** (भागीरथी)	Tehri Garhwal, Uttarakhand	**The river is officially named GANGA (गंगा) from this confluence onward**; Bhagirathi rises from Gaumukh (Gangotri glacier)

Master Table — Tributaries of Major Indian River Systems (बाएँ एवं दाएँ तट की सहायक नदियाँ)

(High-yield for list-matching and tributary identification questions)

River System	Origin (उद्गम)	Left-Bank Tributaries (बायाँ तट)	Right-Bank Tributaries (दायाँ तट)	Drainage Mouth / Key Notes
Ganga (2,525 km)	Devprayag (Bhagirathi + Alaknanda)	• **Ramganga** (meets at Kannauj) • **Gomti** (origin Gomat Taal/Phulhar Jheel, Pilibhit; meets near Saidpur) • **Ghaghara / Saryu** (meets at Chhapra) • **Gandak** (meets at Sonpur) • **Burhi Gandak** • **Kosi** ("Sorrow of Bihar"; meets at Kursela) • **Mahananda** (easternmost left tributary)	• **Yamuna** (largest tributary; meets at Prayagraj Sangam) • **Tamas / Tons** (meets near Sirsa) • **Son** (radial from Amarkantak; meets at Patna/Danapur) • **Punpun** (meets at Fatuha)	Enters Bangladesh as **Padma**; combines with **Jamuna** (Brahmaputra) and **Meghna** to form Sundarbans delta
Yamuna (1,376 km)	Yamunotri Glacier (Bandarpunch)	• Tons (largest Himalayan tributary) • Hindon (meets near Noida/Greater Noida) • Rind, Sasur Khaderi	• **Chambal** (meets near Muradganj/Etawah) • **Sindh** (meets in Jalaun) • **Betwa** (meets at Hamirpur) • **Ken** (meets at Chilla, Banda)	Parallel to Ganga; forms fertile Ganga-Yamuna Doab
Indus (Sindhu, 2,880 km)	Bokhar Chu (near Lake Mansarovar, Tibet)	• **Zanskar** • Suru, Soan • **Panjnad** (combined flow of Jhelum, Chenab, Ravi, Beas, Sutlej)	• **Shyok** (with Nubra) • **Gilgit** • Hunza • **Kabul** (with Swat) • Kurram, Gomal	Trans-Himalayan; enters Arabian Sea south of Karachi
Brahmaputra (2,900 km)	Angsi / Chemayungdung Glacier (Tibet as *Yarlung Tsangpo*)	• **Lohit** • **Dibang** (Sikang) • **Burhi Dihing** • **Dhansiri** (South) • Kopili	• **Subansiri** (gold-bearing; largest tributary) • Kameng (Jia Bhoreli) • **Manas** • Sankosh • **Teesta** (captured from Ganga system in 1787 flood)	Syntaxial U-turn at Namcha Barwa; world's largest river island **Majuli** (Assam)
Godavari (1,465 km) *"Dakshin Ganga" / "Vriddha Ganga"*	Trimbakeshwar (Nashik, Maharashtra)	• **Dharna** • **Penganga** • **Wainganga** • **Wardha** (combined: **Pranhita**) • **Indravati** (Chitrakote Falls) • **Sabari** & Sileru	• **Pravara** • Mula • **Manjra / Manjira** (longest right-bank tributary, flows through Nizamabad) • Maner	**Largest Peninsular river basin**; forms lobate delta in Andhra Pradesh

River System	Origin (उद्गम)	Left-Bank Tributaries (बायाँ तट)	Right-Bank Tributaries (दायाँ तट)	Drainage Mouth / Key Notes
Krishna (1,400 km)	Near Mahabaleshwar (Western Ghats, Maharashtra)	• **Bhima** (longest tributary, 861 km) • Dindi • Peddavagu • **Musi** (Hyderabad sits on its banks) • Paleru, Munneru	• **Venna** • **Koyna** (Shivsagar reservoir) • Panchganga • Dudhganga • **Ghataprabha** (Gokak falls) • **Malaprabha** • **Tungabhadra** (Tunga + Bhadra; largest basin tributary)	Second largest east-flowing peninsular river; major dams: Almatti, Srisailam, Nagarjuna Sagar
Cauvery / Kaveri (800 km) *"Ganga of the South"*	Talakaveri, Brahmagiri Range (Kodagu, Karnataka)	• **Harangi** • **Hemavati** • **Shimsha** • **Arkavathi**	• **Lakshmantirtha** • **Kabini** • Suvarnavathi • **Bhavani** • **Noyyal** • **Amaravati**	Perennial due to dual monsoons (SW monsoon in upper basin + NE monsoon in lower basin); islands: Srirangapatna, Shivanasamudra, Srirangam
Mahanadi (851 km)	Sihawa highlands (Dhamtari, Chhattisgarh)	• **Seonath** (longest tributary, 383 km) • **Hasdeo** • **Mand** • **Ib**	• **Jonk** (UPPCS repeated trap: Jonk joins Mahanadi, NOT Ganga!) • **Ong** • **Tel**	Hirakud Dam (longest earthen dam); drains Chhattisgarh and Odisha into Bay of Bengal
Narmada (1,312 km)	Amarkantak Plateau (Anuppur, MP)	• **Sher** • **Shakkar** • **Dudhi** • **Tawa** (longest tributary) • Ganjal, Kundi	• **Hiran** • Barna • Choral • Man, Uri, Orsang	Flows through **rift valley** between Vindhyas (north) & Satpuras (south); Dhuandhar falls; Sardar Sarovar Dam; forms **estuary** (no delta)
Tapi / Tapti (724 km)	Multai (Betul district, Satpura Range, MP)	• **Purna** (main tributary; drains Vidarbha) • **Girna** • Panjhra • Bori, Waghur	• Suki • Gomai • Arunavati	Parallel rift valley south of Satpuras; Ukai & Kakrapar dams; forms estuary at Surat

Master Table — Major Mountain Passes of India (भारत के प्रमुख पर्वतीय दर्रे)

(High-frequency match-the-pair: Pass ↔ State / Mountain Range)

Pass Name (दर्रा)	State / Union Territory	Mountain Range	Strategic / Historical Connection
Zoji La (ज़ोजिला)	Ladakh (UT)	Great Himalayas	Connects **Srinagar to Leh**; Zojila Tunnel under construction
Karakoram Pass	Ladakh (UT)	Karakoram Range	Highest pass in India (~5,540 m); ancient silk route to Tarim Basin / Yarkand
Bara-lacha La (बारालाचा)	Himachal Pradesh	Zanskar Range	Connects **Mandi/Manali to Leh**; source of Bhaga river (Chenab)
Shipki La (शिपकी ला)	Himachal Pradesh	Zanskar Range	**Sutlej river enters India from Tibet** through this gorge; India-China border trade post
Rohtang Pass (रोहतांग)	Himachal Pradesh	Pir Panjal Range	Connects Kullu Valley with Lahaul and Spiti; bypassed by **Atal Tunnel** (9.02 km)

Pass Name (दर्रा)	State / Union Territory	Mountain Range	Strategic / Historical Connection
Mana Pass (माना दर्रा) / *Dungri-La*	Uttarakhand	Greater Himalayas	Located in Nanda Devi Biosphere; connects Uttarakhand with Tibet; near Badrinath
Niti Pass (नीति दर्रा)	Uttarakhand	Greater Himalayas	Ancient trade route between Uttarakhand (Chamoli) and Tibet
Lipulekh Pass (लिपुलेख)	Uttarakhand (Pithoragarh)	Kumaon Himalayas	**Tri-junction of India, Nepal, and Tibet**; pilgrimage route for **Kailash-Mansarovar Yatra**
Nathu La (नाथुला)	Sikkim	Dongkya Range	Ancient Silk Route to Lhasa (Chumbi Valley, Tibet); reopened for border trade in 2006
Jelep La (जेलेप ला)	Sikkim	Greater Himalayas	Passes through Chumbi Valley connecting Sikkim/Kalimpong with Lhasa (Tibet)
Bomdi La (बोमडिला)	Arunachal Pradesh	Greater Himalayas	Connects Arunachal Pradesh (Tawang) with Lhasa (Tibet)
Diphu Pass (दिफू दर्रा)	Arunachal Pradesh	Mishmi Hills	Tri-junction border of India, China, and Myanmar
Thal Ghat (थाल घाट) / *Kasara Ghat*	Maharashtra	Western Ghats (Sahyadri)	Links **Mumbai to Nashik & Kolkata** (NH-3 & Central Railway)
Bhor Ghat (भोर घाट)	Maharashtra	Western Ghats (Sahyadri)	Links **Mumbai to Pune & Chennai** (Mumbai-Pune Expressway & rail line)
Palghat Gap (पालघाट)	Kerala & Tamil Nadu	Between Nilgiris & Anaimalai	Major gap connecting **Palakkad (Kerala) with Coimbatore (Tamil Nadu)**
Shencottah Gap (शेनकोट्टा)	Kerala & Tamil Nadu	Between Cardamom & Agasthyamalai	Connects **Kollam (Kerala) with Madurai (Tamil Nadu)**

Topic 4 — Lakes, Waterfalls & Water Resources

Consolidated — 48 Must-Score Facts

1. Classify lakes first by origin: lagoon, inland saline, freshwater, crater, glacial, oxbow or artificial. Origin pairs: **Wular** tectonic; **Chilika** lagoon; **Lonar** meteorite crater; **Sambhar** wind / playa; **Kabartal** oxbow.
2. **Chilika** is the largest **east-coast lagoon** (Odisha). If a stem says "largest saline" and omits "inland," Chilika is often the intended key.
3. **Sambhar** is the largest **inland** saline lake (Rajasthan). Sister Rajasthan saline names include Didwana, Kuchaman, Sargol and Khatu.
4. **Wular** is the largest-freshwater volume fact (J&K, on the Jhelum, tectonic). **Dal** is the tourism lake. **Roopkund** is Uttarakhand's Mystery / Skeleton Lake.
5. **Loktak** (Manipur) holds **phumdis**, **Keibul Lamjao** and the Sangai. **Kabartal** is Bihar's oxbow Ramsar lake.
6. **Pulicat** is the second major brackish lagoon (AP–TN, Sriharikota bar). **Kolleru** is mainly **freshwater**, not a lagoon king.
7. **Vembanad** is Kerala's largest lake and India's **longest**; it is a **west-coast kayal / Ramsar**, not India's overall largest lake. **Kayal** = Kerala coastal lagoon / backwater.
8. **Phulhar** (Pilibhit) links to the Gomti source. **Kodaikanal** lake is artificial. Periyar Lake is **not** a coastal lagoon.
9. World lake traps: **Onega** and **Ladoga** are in **Russia** (not Canada). Michigan is USA-only; Superior is USA + Canada. Size ladder often asked: Superior > Victoria > Michigan >> Chilika.
10. Not-UP Ramsar traps: **Rudrasagar** = Tripura; **Sultanpur** = Haryana; **Surinsar–Mansar** = J&K; **Hokera** = J&K (≠ Punjab); **Sasthamkotta** = Kerala; **Renuka** = Himachal.
11. India has **100 Ramsar** sites; Uttar Pradesh has **13**. The **100th** national site is **Surha Tal / JP Narayan Bird Sanctuary, Ballia** (5 Jun 2026).
12. India has >**17%** of world people, ~**4%** of water and ~**2.45%** of land. Precipitation ~**4000 BCM**; available ~**1869 BCM**; utilisable ~**1122 BCM** (surface ~690 + groundwater ~433).
13. The Ganga–Brahmaputra–Barak belt is ~**one-third** of area but ~**60%** of surface water. Irrigation takes ~**89%** of surface use and ~**92%** of groundwater use.
14. Groundwater now covers >**60%** of irrigated area. Canals dominate the Indo-Gangetic plains; **tanks** matter on the peninsula. Among major states, **UP** leads groundwater resources available for irrigation.
15. Quality tags: **arsenic** in the Ganga belt; **fluoride** in Rajasthan / parts of Maharashtra; **usar** from over-irrigation in the NW; Yamuna's Delhi–Etawah stretch is the classic dirty stretch.
16. National Water Policy **2012** puts **drinking first**. **PMKSY** = har khet + more crop per drop (approved **1 July 2015**). **Atal Jal** covers seven states including **UP**. Ken–Betwa is the live interlinking project.
17. Irrigation by CCA: **minor** ≤ **2000 ha** (~**62%** of potential); **medium** 2000–10,000 ha; **major** > **10,000 ha**. Wells / tubewells — not canals alone — dominate national source share.
18. Source-share teaching (2018–19 class): tubewells ~**48.5%**; wells + tubewells ~**64%**; canals ~**23%**; tanks ~**2.3%**.
19. **Kunchikal** (Varahi) = usual **highest** waterfall key; **Nohkalikai** = tallest **plunge**; **Jog** (Sharavati) = famous for **width**, not height.
20. **Shivanasamudra** (Cauvery) = volume king. **Kapildhara** is on the **Narmada**, not Godavari. Sharavati is a **west-flowing** Karnataka river, not a Cauvery tributary.
21. Fall pairs: **Dudhsagar–Goa (Mandovi)**; Barkana–Karnataka; Khandadhar–Odisha; **Hundru–Subarnarekha**; **Lodh / Budha Ghagh–Burha** (Jharkhand's highest ≠ Kanchi); **Dhuandhar–Narmada** (Bhedaghat); **Chitrakote–Indravati**.
22. **Chachai–Bihad** is a **wrong** pair. **Bhalkund** is on the **Bina**, not the Narmada. Other pairs: Amritdhara–Hasdeo; Chulia–Chambal; Duduma–Machhkund; Gokak–Ghataprabha.

23. **Bhakra–Nangal** = Sutlej + **Gobind Sagar** (PB–HR–RJ). **Tehri** = tallest rock-fill on **Bhagirathi** (+ Bhilangana). **Hirakud** (Mahanadi, Odisha) = longest-earthen narrative (**1948–57**).
24. **Indira Sagar** (Narmada, **MP**) = volume king (Harsud submerged). **Sardar Sarovar** = Narmada in **Gujarat** (beneficiaries GJ–RJ–MH–MP). **Ban Sagar** = **Son**, not Narmada. **Idukki** = arch dam on **Periyar**.
25. **Nagarjuna Sagar** and **Srisailam** are both on the **Krishna**. **Tulbul** = Jhelum / Wular. **Baglihar / Dulhasti** = Chenab. **Pandoh / Pong** = Beas.
26. Ramganga project tags: length ~**825.8 m**, power ~**198 MW**, Garhwal location; irrigation is **not** a neat **6 lakh ha**.
27. **Indira Gandhi Canal** = India's **longest** (Harike Barrage; Sutlej–Beas–Ravi waters → western Rajasthan). **Gang Canal (1927**, Ganga Singh) is among the oldest developed RJ systems. Upper Ganga Canal offtake = **Bhimgoda / Haridwar**; Lower Ganga Canal = **Narora**.
28. A **dam stores** water; a **barrage diverts** it. **Farakka** (1975) diverts ~**40,000 cusec** toward the Hooghly — it is not a storage dam.
29. **DVC (1948)** is the first multipurpose valley project of independent India; Damodar = **Sorrow of Bengal**. Tilaiya / Maithon on **Barakar**; Panchet on Damodar.
30. Oldest hydro = **Sidrapong (Darjeeling, 1897)**; second = **Shivasamudram (Cauvery, 1902)**. Cauvery dispute parties = **TN–KA–KL–Puducherry**.
31. Chambal chain = **Gandhi Sagar (MP) + Rana Pratap Sagar + Jawahar Sagar (RJ)**. **Telugu Ganga** carries Krishna water to **Chennai**. Nehru called dams the **"temples of modern India."**
32. **Narmada Bachao Andolan / Medha Patkar** opposed **Sardar Sarovar** height. Narmada plan scale ≈ **30 major / 135 medium / 3000 small**.
33. Multipurpose logic = flood control + irrigation + power (+ navigation / recreation). Koyna is linked to reservoir-induced seismicity (**1967**).
34. India–Bhutan hydro names: **Chukha**, Tala, Kurichhu, **Mangdechhu / Mungdechhu**. Eastern UP / Bihar cannot tap the Himalaya immediately to their north because those ranges lie in **Nepal**.
35. **Garland Canal** idea = Captain **Dinshaw J. Dastur**. First river-linking thought often tagged to **M. Visvesvaraya**; later **K.L. Rao**. **Hariyali** (27 Jan **2003**) = watershed / rural water storage.
36. Oxbow = cut-off meander on a floodplain; lagoon = barred coastal water body. Chilika is a **lagoon**, not an oxbow. Lonar is a **meteorite crater** in Maharashtra basalt — not a volcanic caldera and not tectonic like Wular.
37. West-coast kayals (Vembanad) ≠ inland playas (Sambhar) ≠ tectonic freshwater basins (Wular). **Yaya Tso** is a Ladakh biodiversity heritage site.
38. Pant Sagar is linked with **Rihand / Sonbhadra** — do not swap it with Gobind Sagar (Bhakra) or the Chambal chain.
39. **Jawai Project** is in **Rajasthan** (Luni tributary) — not Tamil Nadu.
40. Jal Jeevan Mission is a rural tap-water **supply** overlay — not a dam–river multipurpose fact.
41. Peninsula irrigation classic = **tanks / ponds** on hard rock + seasonal rivers; Indo-Gangetic irrigation mixes canals and tubewells.
42. **Pant / Gobind / Chambal** reservoir names are high-swap traps: Gobind = Bhakra–Sutlej; Pant = Rihand; Chambal = Gandhi / RPS / Jawahar Sagar.
43. **Idukki vs Mullaperiyar**: both on the **Periyar** belt; Idukki is the arch hydro tag — do not put Idukki on Cauvery.
44. Waterfall formation teaching: youth stage / knickpoint / hard–soft rock contact — height crowns (Kunchikal / Nohkalikai) still decide most match stems.
45. **Palani** fall teaching pair is Himachal in the usual match set — do not park it with South-Indian Cardamom hills without reading the stem.
46. Surface vs groundwater: utilisable total ~**1122 BCM** splits roughly surface **690** + groundwater **433** in the standard teaching figures.
47. Interlinking live project = **Ken–Betwa** only under NPP implementation; do not invent multiple "operational" national links.

48. Ramsar count stems need both national (**100**) and UP (**13**) numbers plus the Ballia 100th-site name — count alone is not enough if the stem asks the latest site.

📋 Must-score facts — lakes, lagoons, Ramsar traps

Lake ↔ type / state

Lake	Lock
Chilika	Largest **east-coast lagoon** (Odisha)
Sambhar	Largest **inland saline** (Rajasthan)
Wular	Largest freshwater volume (J&K; Jhelum; tectonic)
Loktak	Phumdis; Keibul Lamjao; Sangai (Manipur)
Pulicat	2nd major brackish lagoon (AP–TN)
Kolleru	Mainly **freshwater** (AP)
Vembanad	Kerala's largest; India's **longest**; west-coast kayal
Kabartal	Bihar oxbow Ramsar
Phulhar	Pilibhit; Gomti source link
Kodaikanal	**Artificial**

Not in UP (common traps)

Rudrasagar (Tripura) · Sultanpur (Haryana) · Surinsar–Mansar (J&K)

World lake traps

Lake	Lock
Onega / Ladoga	**Russia** (≠ Canada)
Michigan	Wholly **USA**
Superior	USA + Canada

Topic 5 — Soils

Consolidated — 48 Must-Score Facts

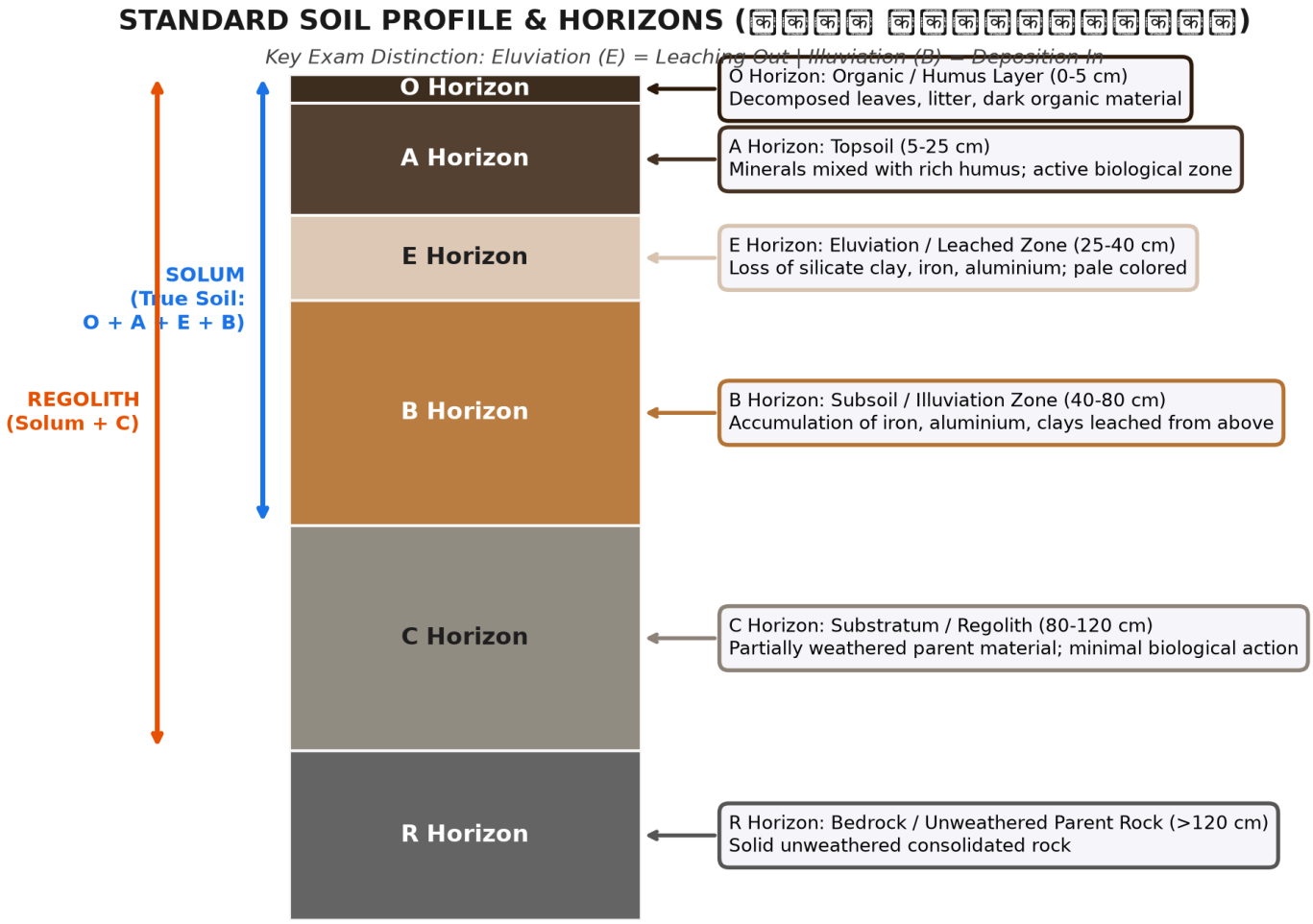
1. Ancient Indian names: **Urvara** = fertile and **Usara** = sterile. Jenny's five factors are Climate, Living organisms, Relief, Parent rock, and Time (**CLORPT**).
2. **Edaphic** means a **soil** factor — not climate, biotic, or topography. Residual / sedentary soils form on bedrock (regur, red). Transported / azonal soils are the alluvium of the plains.
3. World zonal pairs: **Podzol** = cold temperate woods; **Chernozem** = temperate steppe; **Laterite** = hot humid; **Terra Rossa** = limestone/dolomite weathering; **Sierozem** = mid-latitude desert. **Terra Roxa** = Brazil's basaltic coffee soil (not Indian regur).
4. Capillary action is strongest in **clay** (Clay > silt/loam > sand). Halophytes live on saline soils. Contour bunding belongs on **hill slopes**, not desert margins or flood plains.
5. Profile downward is **O → A (topsoil) → E → B → C → R**. **Eluviation** washes material **out** of A/E; **illuviation** deposits it **into B**. Young khadar has weak horizons; mature residual soils show clear horizons.
6. The usual eight-type teaching set is Alluvial, Black / Regur, Red & Yellow, Laterite, Arid / Desert, Saline / Usara, Peaty / Marshy, and Forest / Mountain.
7. Parent-material pairs: crystalline rock → red; Cuddapah–Vindhyan → calcareous / clay; Gondwana → immature poor; Deccan Trap → regur; extra-peninsular sediment → alluvium. Climate can override rock — granite becomes laterite if wet, or sandy if arid.
8. **Pedocal** soils accumulate lime in arid tracts such as Rajasthan. Himalayan soils develop better on gentler **north-facing** slopes; many south faces are too steep for deep profiles.
9. **Alluvial** is India's **largest** soil group (about **33.5%–40%**, covering ~11 lakh sq km). It is rich in potash and lime but **poor in phosphorus, nitrogen, and humus** (NCERT). Uttar Pradesh is about **90%** alluvial.
10. **Khadar** is new, fine, and fertile. **Bangar / Bhangar** is older, coarser, and has **more kankar**. **Bhur** is sandy ridges of the western Gangetic belt. **Rarh** is older West Bengal / Jharkhand fringe upland.
11. **Bhabar** (about 8–16 km) is pebble fans where streams **disappear**. **Tarai** (about 15–30 km) lies south, where streams **re-emerge**; it is moist and nitrogen-rich but phosphorus-poor.
12. **Black / regur** sits mainly on Deccan and Rajmahal Trap basalt; Tamil Nadu black may sit on gneiss / schist. Clay is often **≥62%**, montmorillonite is the key clay, and cotton is the crop. Black soil is **not** Himalayan.
13. Black soil area is about **16.6% / ~5.46 lakh km²**. It holds moisture well, sticks when wet, and cracks when dry (**self-ploughing**). It lacks nitrogen, phosphorus, and humus.
14. **Red / yellow** soils form on ancient crystalline and metamorphic rocks. Colour comes from **ferric oxides**; hydrated form looks yellow. Coverage is about **8.7 lakh sq km**.
15. **Laterite** was named by **F. Buchanan (1807)** from Latin *later* (brick). Hot-wet leaching leaves soils **rich in Fe and Al oxides** but poor in N, lime, P, and humus. Most widespread in **Kerala (Malabar)**, then Maharashtra.
16. Laterite crops classically include **cashew, tapioca, tea, and coffee** on well-drained slopes — not default cotton (cotton = black / regur).
17. **Desert / arid** soils cover about **29 million ha**. Western Rajasthan soils are often alkaline/saline with a calcareous **kankar** base and high calcium.
18. **Saline / usara** soils carry sodium, potassium, and magnesium salts. **Gujarat** has the largest saline soil area; **Uttar Pradesh** has the largest alkaline / sodic wasteland area (~2,129 sq km).
19. **Peaty / kari** soils of Kerala (Kottayam–Alappuzha / Kuttanad), plus Sundarbans and north Bihar pockets, grow paddy after rains. Their black colour is **not** Deccan cotton regur.
20. **Forest / mountain** soils cover about **8.67%**. Steep Himalayan slopes often lack deep humus despite forest cover because of erosion and slow accumulation.
21. **Karewa** soils are Kashmir **lacustrine** deposits with moraines, famous for **saffron (zafran)**, almond, and walnut — not ordinary Gangetic alluvium.

22. Texture: sand drains, clay holds, loam is best for crops. Clay particles are smaller than **0.002 mm**; silt is **0.002–0.06 mm**; sand is about **0.06–2.0 mm**.
23. Among copper, iron, manganese, and zinc, **zinc (Zn)** is the most deficient micronutrient across Indian soils.
24. Water-erosion ladder: **Splash (raindrop) → Sheet → Rill → Gully → Ravine / stream channel**. **Chambal** ravines (badlands) hit Morena, Bhind, Gwalior, and Shivpuri hardest.
25. Farmland often erodes fastest under row crops such as **sorghum (jowar)**; cover crops such as **clover** keep erosion lowest among common options.
26. Crop rotation, mixed cropping, and multiple cropping replenish fertility. **Seed modification** raises yield / pest resistance but does **not** raise soil fertility by itself.
27. Agricultural water tricks: reduced / zero tillage, gypsum on clay–salt hardpans, and residue / mulch to cut evaporation.
28. Usar / sodic soils have **pH > 7** and need **gypsum ($CaSO_4 \cdot 2H_2O$)**, pyrites, or green manure — **not lime**. Tolerant paddy lines include Narendra Usar Paddy–1 and –2.
29. Optimal crop pH is about **6.0–7.0**. Acid soils of heavy-rain tracts are treated with **agricultural lime ($CaCO_3$ / CaO)**. Tea prefers **slightly acidic, calcium-poor**, well-drained slopes.
30. Sand content of Indo-Gangetic alluvium **decreases from west to east**. Minimum water retention = sandy alluvium; maximum = **clay**.
31. Standard loam is about **40% sand, 40% silt/clay mix teaching**, and **20%** silt in many coaching ratios — loam = balanced texture for crops.
32. Legumes (pea, soybean, alfalfa, black gram / urd) fix atmospheric nitrogen via root nodules and restore fertility.
33. In laterite profiles the **upper horizons are more acidic** than lower ones; excess iron oxides also cut fertility.
34. **Matasi** of Chhattisgarh is red–yellow soil covering about **55–60%** of the state and is prime for paddy; **Kanhar** is the lower black clay — do not swap.
35. Soil leaching is most severe in **tropical rainforest / high-rainfall monsoon** tracts (Western Ghats, Meghalaya).
36. West Bengal has about **60%** net sown area; Andhra Pradesh is near **40%** — do not invent laterite as Andhra's sole soil.
37. **Mitti Bachao** began at **Hoshangabad (Narmadapuram), Madhya Pradesh**. **World Soil Day** is **5 December** (not 5 June).
38. UP local names: Bundelkhand black = **Mar / Kabar**; Bundelkhand red = **Parua / Rakar**; Vindhyan = **Bhonta / Monta**; alkaline canal tracts = **Usar / Reh / Kallar / Thur**.
39. Saline soils = high soluble salts; sodic / alkali = high exchangeable sodium and high pH. **Gypsum** treats sodic; **lime** treats acid — never swap the remedies.
40. Wind erosion control uses **shelter belts / windbreaks**; gully control uses **check dams / bunds**. Contour ploughing and terracing cut slope wash.
41. Soil Health Card launched **19 Feb 2015** at **Suratgarh (Rajasthan)** with slogan *Swasth Dharaa, Khet Haraa*. Card reports **12 parameters**. By Jul 2025, **>25 crore** cards had been issued; the scheme now sits under RKVY as **Soil Health and Fertility**.
42. Acid soils concentrate in the North-East, Western Ghats, Kerala, and other heavy-leaching belts; arid pedocals concentrate lime in the west.
43. **Doab** belts of the plains are classically alluvial; Bundelkhand mixes red and black; the Tarai is a moist rice–sugarcane fringe south of Bhabar.
44. Sheet erosion removes a thin uniform layer; rill cuts small channels; gully is the deep stage that creates ravines — do not start the ladder at gully alone.
45. **ICAR / teaching problem-soil set** pairs saline, alkaline / sodic, and acidic soils as reclaimable classes with distinct chemistry and remedies.
46. Alluvial soils of the plains support rice–wheat systems where irrigation exists; black soils favour cotton–jowar–soy; red soils need fertiliser for millets and pulses.

47. Laterite can harden into a brick-like crust when exposed; the name itself points to that brick-making use in humid tropics.
48. Conservation package for slopes: contour bunding + terracing + cover crops + check dams; for dunes / arid wind: shelter belts and mulching — match method to agent (water vs wind).

📋 Must-score facts — profile, types, doab (दोआब) belts

Profile / terms



Item	Lock
Profile (down)	**O → A → E → B → C → R**
Eluviation	Washes **out** of A/E
Illuviation	Deposits **into** B
Edaphic	**Soil** factor
Urvara / Usara	Fertile / sterile (ancient names)
Pedocal	Lime accumulation in arid tracts

Soil type tags

Soil	Lock
Alluvial	Largest (~**40%**); potash-rich; **poor in P & N**
Khadar	New, fine, fertile

Soil	Lock
Bangar	Older; more **kankar**
Bhur	Sandy doab ridges
Bhabar	Pebble fans; streams **disappear** (~8–16 km)
Tarai	South of Bhabar; streams reappear (~15–30 km)
Black / regur	Deccan Trap; cotton soil
Laterite	High rain leaching; brick-like
Red & yellow	Crystalline parent

Topic 6 — Agriculture

Consolidated — 50 Must-Score Facts

1. **Net sown area (NSA)** is land sown at least once a year. **Gross cropped area (GCA)** counts every sowing. **Cropping intensity** = **GCA / NSA × 100** (about **111%** in 1950–51 to about **156%** now). About **86%** of holdings are small or marginal, and about half of NSA is still rainfed.
2. Groundwater covers about **64%** of irrigated area (**2018–19**: tubewell ~**48.5%**; canals ~**23%**; tanks ~**2.3%**). Canals suit clayey plains; tube wells dominate north-west and western Uttar Pradesh; tanks suit the peninsula; drip suits horticulture. Minor projects (CCA ≤ **2,000 ha**) create ~**62%** of irrigation potential.
3. **Kharif** (June–October) includes rice, maize, jowar, bajra, ragi, cotton, jute, groundnut, soy, and tur. **Rabi** (October–March) includes wheat, barley, gram, mustard, peas, and linseed, helped by Western Disturbances. **Zaid** (March–June) covers melons, cucumber, fodder, and vegetables. Cane is long-duration; tea, coffee, and rubber are perennial.
4. Rice needs hot-wet conditions (often above **20–27°C** and above **100 cm** rain) on clayey alluvium. Methods include transplant, broadcast, drill, **DSR**, and **AWD**. Aus / Aman / Boro are eastern season names. **Azolla** is a biofertiliser. Golden rice carries **Vitamin A**.
5. West Bengal often leads rice **volume**; Punjab leads **yield**. India is usually the world's **second** rice producer after **China**. A 2024–25 Union claim of first rank is a separate headline, not the usual textbook rank.
6. Wheat needs cool growing weather and bright ripening, about **50–75 cm** rain, and loam soils. It is a **rabi** crop with two belts: Ganga–Satluj plains and Deccan black soil. Assam–Wheat is a wrong pair. India is usually world **number two**.
7. Millet leaders: **Jowar = Maharashtra**, **Bajra = Rajasthan**, **Ragi = Karnataka**. Maize is food plus feed and mostly kharif. International Year of Millets 2023 marketed millets as **Shree Anna**.
8. Pulses fix nitrogen through **Rhizobium**. Gram is rabi; tur, moong, and urad are kharif. India is the world's top pulse producer. **Rajma** is the major pulse with poor symbiotic fixation. Cobalt is the micronutrient linked to the Rhizobium system.
9. Oilseed leaders: **Groundnut = Gujarat**, **Mustard = Rajasthan** (rabi), **Soybean = Madhya Pradesh** (kharif). The **Yellow Revolution** is oilseeds — not the Golden Revolution. Groundnut **pegging** needs gypsum for pod fill.
10. Cotton (**white gold**) needs above about **21°C**, **50–100 cm** rain, about **210 frost-free days**, and preferably black soil. It is **kharif**, lasts about **6–8 months**, is about two-thirds rainfed, hates waterlogging, and has three zones (north alluvial, central black, south mixed). India grows all four species; bulk is *hirsutum* / Bt; crop is mostly **medium** staple.
11. Ahmedabad lies in the cotton belt, but India's traditional largest textile mill centre is **Mumbai**, not Ahmedabad. Cotton belongs to the **Malvaceae** family; India pioneered hybrid cotton.
12. Jute (**golden fibre**) needs about **25–35°C**, above **150 cm** rain, alluvial soil, and standing water for **retting**. White (*C. capsularis*) and tossa (*C. olitorius*) are the two fibres; **mesta** is the drier-area ally. State leader is **West Bengal**. Uttar Pradesh–Jute is wrong.
13. Sugarcane needs about **21–27°C** and **75–150 cm** rain or irrigation. **Uttar Pradesh leads quantity**; **Maharashtra leads productivity** and cooperatives. Price policy is **FRP** (plus SAP), not cereal MSP. Ratoon means a crop from stubble. Adsali cane in Maharashtra takes **16–18 months**.
14. Tea is a plantation crop needing about **20–30°C**, **150–300 cm** rain, and slopes — Assam, West Bengal, and Nilgiri. Coffee needs shade and is strongest in **Karnataka**, then Kerala and Tamil Nadu.
15. Rubber needs about **25–35°C** and above **200 cm** rain; the correct state pair is **Kerala**. Pepper and cardamom also concentrate in **Kerala**. World citrus is a Mediterranean specialty.
16. Potato leadership is with **Uttar Pradesh**. CIP-SARC is at **Agra (Singna)**, not Aligarh. **Sultana, Gulabi and Kali Champa** are **guava** varieties — not grapes.

17. Planning Commission agro-climatic zones = **15**; NARP zones about **127**; agro-ecological regions about **20**. The Trans-Gangetic belt is the classic Green Revolution wheat–rice zone.
18. **CACP recommends** MSP; the **Cabinet decides**. Mandated MSP crops are **22**. Cane uses FRP, not the cereal MSP schedule. Budget 2018–19 set the **≥1.5×** cost target for MSP.
19. Green Revolution = HYV seed + water + fertiliser (Lerma Rojo, Sonora 64, IR-8) in Punjab–Haryana–western Uttar Pradesh. Costs include groundwater stress, monoculture, and the stubble window. **BGREI** targets eastern rice systems. Norman Borlaug's Nobel is for **Peace**.
20. **M.S. Swaminathan** is the Green / Evergreen face. **Verghese Kurien** is White Revolution / Operation Flood / NDDB / Amul — not milk for Swaminathan.
21. Revolution colours: **Golden** = horticulture + honey; **Grey** = fertiliser; **Yellow** = oilseed; **Black** = petroleum; **Silver** = egg; **Pink** = onion / meat / prawn; **Rainbow** = integrated; **Evergreen** = sustainable agriculture; **Blue** = fish (**Hiralal Chaudhuri**).
22. Institutions: **ICAR** Delhi; **IARI** Pusa; seed chain **Breeder → Foundation → Certified** (bag tags: golden yellow / white / blue); **KVK** at district level. First agri university = **Pantnagar (1960)**. **Agmark** = Act **1937**. FAO grain moisture ≤ **14%**.
23. **Jhum** is north-east shifting cultivation (also podu / bewar / dahiya / kumari / waltre). Organic farming bans synthetics; **Sikkim** is the first fully organic state (2016, NPOP / APEDA). Precision farming is GIS / GPS site-specific management.
24. India usually ranks **first** in milk, pulses, and jute, and often **second** in rice, wheat, and cane.
25. Red rot of cane is a **fungus**; citrus canker is a **bacterium**. Command Area Development improves an existing command — it is not a new dam.
26. Green Revolution costs include monoculture and groundwater stress in Punjab–Haryana. Bio-decomposer sprays are the free fungal stubble-management tool in recent papers.
27. Amartya Sen's food thesis is **entitlements**, not Swaminathan's breeding story.
28. Lift irrigation matters where southern canal layouts are irregular. Drip and sprinkler raise water-use efficiency in horticulture and dry tracts.
29. Cotton is **kharif**; wheat is **rabi** — never swap those seasons. Cane is long-duration and straddles seasons.
30. World coffee order often taught for 2016 is Brazil > Vietnam > Colombia > Indonesia. Do not put Gujarat–Tea or Assam–Wheat in a correct-pair list.
31. Mixed farming = crops + livestock. Double cropping = two crops in one year. Parallel cropping classic = **wheat + mustard**. Contract farming pioneer = **Punjab**.
32. Green manure N% fact: **cowpea** often highest among common options; sunhemp often max kg N/ha. Fertigation avoids rock / super phosphate. Conservation agri = min till + residue + rotation.
33. Cane points: Sugar Bowl **UP**; breeding **Coimbatore (SBI, 1912)**; first mill **Pratappur 1903**; SSI = WWF–ICRISAT **2009**. Rice Bowl of India = **Krishna–Godavari delta**.
34. Board HQ: Coffee **Bengaluru**, Tea **Kolkata**, Rubber **Kottayam**, Tobacco **Guntur**, Spices **Kochi**. History of Indian Agriculture = **M.S. Randhawa**.
35. **Norin-10** is the principal wheat dwarfing gene. **Sonora-64** was developed through induced mutation at IARI. **Macaroni wheat** (*T. durum*) suits rainfed tracts. **Triticale** is a wheat × rye cross.
36. **Crown Root Initiation (CRI)** at about **20–25 days** after sowing is the most critical irrigation stage for wheat.
37. **SRI** (Madagascar, 1980s) uses alternate wetting and drying and cuts seed rate, methane, and electricity use versus continuous flooding.
38. Eastern rice seasons: **Aman** (winter; sow Jun–Jul, harvest Nov–Dec); **Aus / Kar** (autumn); **Boro / Dalua** (summer; sow Nov–Dec, harvest Mar–Apr).
39. Sustainable Sugarcane Initiative uses fewer seed canes, bud chips, wide spacing, and drip fertigation (WWF–ICRISAT, 2009).
40. Sericulture: India alone produces all four commercial silks — **Mulberry** (Karnataka lead), **Tasar** (Jharkhand), **Muga** (Assam, endemic golden silk), and **Eri** (Assam lead).

41. Livestock flash: India usually leads world **milk** volume. 2019 Cattle Census order often taught as West Bengal #1, UP #2, MP #3 for cattle; UP leads buffaloes; Telangana leads sheep. **Jamunapari** is a high-yield goat; **Kharai** camels of Kutch swim to mangrove graze.
42. Horticulture / plantation boards keep HQ matches straight — Coffee Bengaluru, Tea Kolkata, Rubber Kottayam, Tobacco Guntur, Spices Kochi.
43. Animal husbandry flash pairs White Revolution / Operation Flood with **Verghese Kurien** and **NDDB Anand**.
44. Khandwa–Khargone (Nimar, MP) is a classic *White Gold* cotton region. Cotton fibres are seed hairs.
45. Safflower oil is high in linoleic acid in coaching nutrition notes; do not confuse safflower with mustard as the Rajasthan leader.
46. Balanced NPK teaching ratios for pulses often use **1:2:2** or **1:2:3** — pulses need less N because of fixation.
47. Evergreen Revolution (Swaminathan) means sustainable productivity with ecology — not a second chemical GR wave.
48. PM-AASHA (PSS via NAFED/NCCF), e-NAM, PM-KISAN, and PMFBY are the living scheme set around price support, markets, income, and crop insurance.
49. Shree Anna branding after International Year of Millets 2023 pushes millets for diversification away from rice–wheat monoculture.
50. Zaid crops fill the short summer window with irrigation; they are neither classic kharif nor rabi staples.

📋 Must-score facts — seasons, crop–state, irrigation

Land-use terms

Term	Lock
NSA	Land sown ≥ once a year
GCA	Counts every sowing
Cropping intensity	(GCA/NSA) × 100
Irrigation share	Groundwater ~**64%** (tubewell heavy)

Season crops

Season	Lock
Kharif (Jun–Oct)	Rice, maize, millets, cotton, jute, groundnut, soy, tur
Rabi (Oct–Mar)	Wheat, barley, gram, mustard, potato

Crop ↔ leader / need

Crop	Lock
Rice	WB volume often; Punjab yield; hot-wet; clayey alluvium
Wheat	Cool grow / bright ripen; ~50–75 cm; **rabi**
Jowar / Bajra / Ragi	**MH** / **RJ** / **KA**
Groundnut / Mustard / Soy	**GJ** / **RJ** (rabi) / **MP** (kharif)
Cotton	~21°C+; 50–100 cm; ~210 frost-free days; black soil preferred
Pulses	Rhizobium N-fix; India top producer

Topic 7 — Natural Vegetation & Biodiversity Geography

Consolidated — 48 Must-Score Facts

1. "Vegetation is the true index of climate" is **Thornthwaite's** line, not Köppen's letter codes. Controls are climate, soil, relief, biotic factors, and time.
2. India is a **mega-biodiversity** country. The four hotspots touching India are **Himalaya**, **Western Ghats–Sri Lanka**, **Indo-Burma**, and **Sundaland** (**Nicobar**, not Andaman). Do not split Himalaya into east and west as if those were the four.
3. Rain ladder: tropical evergreen **>200 cm**; moist deciduous **100–200 cm** (largest area, about **37%**); dry deciduous **70–100 cm** (about **28%**); thorn below about **70–75 cm**. Semi-evergreen is the mix zone. Some notes put wet evergreen above **250 cm**.
4. Champion–Seth teaching heads group into five major classes and about **16** sub-types (moist tropical, dry tropical, subtropical montane, temperate montane, alpine).
5. **Dry evergreen** of the Tamil Nadu Coromandel grows on about **100 cm** of **north-east monsoon** with jamun–neem–tamarind. It is not Western Ghat wet rainforest.
6. **Chir pine** belts the western Himalaya at about **1000–2000 m**. Alpine moist belts carry rhododendron–birch; dry alpine juniper sits above about **3500 m**. **Shola** forests are South Indian hill patches.
7. Species diversity **decreases from equator to poles**. Richness is not the same as evenness. Alpha is within one ecosystem, beta between ecosystems, gamma across a region.
8. India has two realms (Palearctic Himalaya + Indomalayan rest), five biomes, **ten** biogeographic zones, and about **25** provinces.
9. Evergreen forests of the Western Ghats, north-east, and Andamans are multilayered with rosewood, mahogany, and ebony and do **not** shed all leaves in one dry season. Moist deciduous forests (teak, sal, shisham, sandalwood) shed in the dry season and cover the **largest** area.
10. Thorn / scrub of Rajasthan, Gujarat, and rain-shadow tracts holds acacia, babool, ber, and khair (xerophytes). Littoral / mangrove forests of deltas hold **sundari** and **pneumatophores** and buffer storms.
11. World vegetation match: **Maquis** = Mediterranean; **Fynbos** = South Africa; **Chaparral** = California; **Matorral** = Chile. Epiphytes peak in equatorial forest; acacia in Sahara; baobab in savanna; cedars in Mediterranean.
12. Latest Forest Cover (ISFR 2023 teaching benchmark) stands at about **21.76%** of India's TGA (Forest + Tree Cover ≈ **25.17%**). Top state by forest **area** is **Madhya Pradesh**. **Forest cover ≠ Recorded Forest Area**. National Forest Policy 1988 aims at about **33%**.
13. Uttar Pradesh forest cover is about **6.24%** of geographical area (Forest + Tree Cover ≈ **9.96%**). Districts above 20% include **Chandauli** and **Shravasti**, not Bahraich. Ballia has near-nil forest yet holds **Surha Tal** Ramsar. Uttarakhand forest cover is about **45.44%** (legal RFA ~**71%**).
14. National Parks are stricter than Wildlife Sanctuaries. Biosphere Reserves have **core + buffer + transition**. India has **18** biosphere reserves; **Nilgiri (1986)** was the first.
15. **In-situ** conservation = National Park, Sanctuary, Biosphere Reserve, sacred grove. **Ex-situ** = zoo, botanical garden, seed bank. A botanical garden is **not** in-situ.
16. Chronology: Wildlife Protection Act **1972** → Project Tiger **1 April 1973** → Project Elephant **1992** → Biological Diversity Act **2002**.
17. Park–state pairs: **Simlipal = Odisha** (not Madhya Pradesh); **Nokrek = Meghalaya** (not Manipur); **Nanda Devi = Uttarakhand** (not Uttar Pradesh); **Agasthyamalai = Kerala and Tamil Nadu**; Manas = Assam; Keoladeo = Rajasthan; Kanha = Madhya Pradesh.
18. **Dachigam** protects the **Hangul** (Kashmir stag). **Keibul Lamjao** sits on **Loktak** (Manipur) for the Sangai.
19. Uttar Pradesh's **only National Park** is **Dudhwa** (Lakhimpur Kheri). Tiger Reserves are Dudhwa, Pilibhit, Amangarh, and Ranipur — do not invent extra National Parks.

20. Dudhwa Tiger Reserve = Dudhwa National Park + Kishanpur + **Katarniaghat** (1975). **Chandraprabha** Wildlife Sanctuary is in **Chandauli**.
21. Ramsar traps **not** in Uttar Pradesh: Rudrasagar (Tripura), Sultanpur (Haryana), Surinsar–Mansar (J&K). **Surha Tal / JP Narayan BS, Ballia** is India's **100th Ramsar** (5 Jun 2026) and UP's 13th.
22. **MISHTI** restores mangroves and belongs to Budget **2023–24**, not 2019. Mangrove breathing roots are **pneumatophores**. Sundarbans are the sundari mangrove fact.
23. Coral reefs fact to Andaman & Nicobar, Lakshadweep, Gulf of Mannar, and Gulf of Kachchh. Coral bleaching links to about **2°C** warming of sea water.
24. **Operation Olivia** is the Coast Guard Olive Ridley programme in Odisha. Nesting season is about **November–May**, not August.
25. Montane vegetation is an altitude elevator: tropical foothills → oak / deodar / chir → alpine → bugyal → snow. Plains rainfall ladders do not replace this altitude logic.
26. Keystone species collapse an ecosystem if lost; flagship species are public mascots — do not treat them as the same idea.
27. India has about **50,000** rice strains and about **1,000** mango varieties in the usual diversity teaching line.
28. Tea and rubber estates are **plantations**, not climax evergreen forest. Moist and dry deciduous belong to one monsoon-forest family split by rainfall.
29. Tropical evergreen is also called **Selvas**. Typical figures: rain **>200 cm**, mean annual temperature above about **22°C**.
30. **Butea monosperma** is the **flame of the forest** (**Dhak / Palash**) and the **state flower of Uttar Pradesh**.
31. **Katha** is obtained from **khair** wood. **Khejri** is the multipurpose "king of the desert" tree used in social forestry.
32. Tree-line is **lower** in the **Western** Himalaya (about **3300–3600 m**) than in the Eastern / Central Himalaya (about **3600–3800 m**).
33. **Southern** Himalayan slopes hold **denser** vegetation than northern slopes because they get **more rainfall and sunlight**.
34. **Deodar** is abundant in the **western** Himalayan temperate belt at about **1500–2500 m**. Among deodar, chir, sal, and teak, deodar grows at the **highest** elevation.
35. Classic mangrove match: **Achra–Ratnagiri = Maharashtra**; **Coondapur = Karnataka**; **Pichavaram = Tamil Nadu**; **Vembanad = Kerala**. **Silent Valley (Kerala)** = tropical evergreen.
36. **Mahogany** is **not** a Himalayan species. Himalayan set = juniper, silver fir, spruce, deodar, birch.
37. **Jim Corbett** (Uttarakhand) is India's **first** National Park (1936 as Hailey NP). **Kaziranga** (Assam) is the one-horned rhinoceros flagship. **Gir** (Gujarat) is the Asiatic lion home.
38. **Bhitarkanika (Odisha)** is the usual second Indian mangrove block after Sundarbans. West-coast mangroves (MH–Goa–KA) are often scrubby and degraded compared with east-coast deltas.
39. Several Indian biospheres (Nilgiri, Nanda Devi, Sundarbans, Gulf of Mannar, and others) also sit on UNESCO's **World Network of Biosphere Reserves**.
40. Halophytes tolerate salt; mangroves are the main Indian halophyte trees and often grow **pneumatophores** for oxygen in waterlogged mud.
41. Acclimatization is a short physiological adjustment; adaptation is a longer evolutionary trait — do not swap the two words in stems.
42. ISFR 2017 frozen PYQ figure for forest cover is **21.54%** — always read the **report year** in the stem before using a later percentage.
43. FSI (Dehradun) under MoEFCC publishes the biennial **India State of Forest Report**. Very dense / moderately dense / open forest classes are cover categories, not legal RFA classes.
44. Sacred groves are **in-situ** community conservation patches — group them with NP / WLS / BR, not with zoos.
45. Project Tiger followed WLPA 1972; open-ended recall of the launch year is **1973** even when some MCQ options omit it.

46. Sundarbans hold Royal Bengal Tiger and crocodile habitat inside **littoral / tidal** forest — not dry deciduous or alpine.
47. Coral needs clear, warm, shallow tropical seas and symbiotic **zooxanthellae**; bleaching is heat stress expulsion of those algae, not cooling.
48. Conservation International hotspot logic is high **endemism** plus heavy **habitat loss** — do not equate the four Indian hotspots with the **18** biosphere reserves.

📋 Must-score facts — rain ladder, hotspots, zones

Rain ↔ forest

Forest	Rain / tag
Tropical evergreen	**>200 cm**
Moist deciduous	**100–200 cm** (largest area ~**37%**)
Dry deciduous	Lower rain; teak–sal belts
Dry evergreen (TN Coromandel)	~**100 cm NE monsoon** (उत्तर-पूर्व मानसून)
Thorn / scrub	Rajasthan, Gujarat, rain-shadow; acacia–babool

Biodiversity tags

Item	Lock
"Vegetation is true index of climate"	**Thornthwaite**
Hotspots touching India	Himalaya · WG–Sri Lanka · Indo-Burma · Sundaland
Chir pine belt	W. Himalaya ~**1000–2000 m**
Realms	Palearctic (Himalaya) + Indomalayan
Biogeographic zones	About **10** zones

Topic 8 — Minerals, Energy & Industry

Consolidated — 48 Must-Score Facts

1. Most of India's mineral wealth lies **east of the Mangalore–Kanpur line**. The northern plains are almost empty of major ores. **Chota Nagpur** is the richest mineral belt (India's Ruhr). **Dharwar** rocks are the metallic storehouse; **Gondwana** holds nearly all coal.
2. India is poor in many non-ferrous metals but rich in **bauxite**. Mica's classic dielectric belt is **Koderma–Gaya–Hazaribagh** in Jharkhand (plus Nellore and Ajmer belts).
3. As per the Ministry of Mines Annual Report **2021–22**, the leading iron-ore states are **Odisha and Chhattisgarh**. Jharkhand and Karnataka do not always lead.
4. Iron-ore mine–state pairs: **Gurumahisani (Odisha)**, **Bailadila (Chhattisgarh)**, **Noamundi (Jharkhand)**, **Kudremukh (Karnataka)**. Mount Newman (Australia) is famous for **iron**, not manganese.
5. The four main iron belts are Odisha–Jharkhand, Durg–Bastar–Chandrapur, Ballari–Chitradurga, and Maharashtra–Goa. Bailadila is India's largest mechanised hematite mine; ore often exports via Visakhapatnam.
6. Manganese leaders include **Odisha, Maharashtra and Madhya Pradesh**. Postmasburg (South Africa) is a manganese centre. **Bhilwara** is mica and zinc (Rampura-Agucha), not manganese. **Chitradurga** is iron, not copper.
7. Bauxite feeds aluminium and often sits with **laterite**. Odisha leads (Panchpatmali). **MALCO** is at **Mettur, Tamil Nadu**. **HINDALCO** is at **Renukoot, Uttar Pradesh**. BALCO is at Korba; NALCO at Damanjodi/Angul.
8. Copper centres are **Khetri (Rajasthan)**, Singhbhum (Jharkhand) and Malanjkhand (Madhya Pradesh). **Panna (Madhya Pradesh)** is diamond. **Sukinda (Odisha)** is chromite. **Zawar** and **Rampura-Agucha** (Bhilwara) are zinc–lead.
9. Uranium's classic mine is **Jaduguda, Jharkhand**. Thorium and monazite sands line the **Kerala–Tamil Nadu** beaches. Lithium finds at **Reasi, Jammu & Kashmir** matter, but India still imports much Li, Ni and Co.
10. About **97%** of Indian coal is **Gondwana** coal in the Damodar–Son–Mahanadi–Godavari valleys. Tertiary coal is mainly in the North-East. Rank rises **peat (<40% C) → lignite (40–55%) → bituminous (60–80%) → anthracite (80–95%)**. Anthracite is scarce (J&K); lignite = **Neyveli**; ~**80%** of deposits are bituminous **non-coking**.
11. Coalfield centres: **Raniganj (West Bengal)**, **Jharia (Jharkhand, coking)**, Ramgarh (Jharkhand), **Talcher (Odisha)**. **Maolong** coal is in **Meghalaya**, not Arunachal. **Neyveli** is lignite in Tamil Nadu.
12. Valley–field pairs: Talcher–Mahanadi, Karanpura–Damodar, Singrauli–Son, Singareni–Godavari. **Singrauli** is the oldest large NTPC hub and sits on the **Madhya Pradesh–Uttar Pradesh** border, not northern Chhattisgarh.
13. World coal traps: Ruhr is **Germany**; Donetsk is **Ukraine**; Appalachian is USA; Kuzbass is Russia. **Pegu Yoma (Myanmar)** is mineral oil, not tin. **Kashagan** oil is in **Kazakhstan**, not Kuwait.
14. **Methane** is the main gas in natural gas and **CNG**. **LPG** is propane–butane. The **Black Revolution** means petroleum, not oilseeds (Yellow).
15. Oil–gas hubs include Digboi/Naharkatiya (Assam), Mumbai High, Ankleshwar (Gujarat), Barmer (Rajasthan) and the Krishna–Godavari / Cauvery basins. Digboi is among India's oldest refineries; **Jamnagar** is the largest refining–petrochemical complex.
16. Refinery–state pairs: Nayara (Gujarat), Manali (Tamil Nadu), Numaligarh (Assam), Tatipaka (Andhra Pradesh), **Mathura (Uttar Pradesh)**, Barauni (Bihar), Panipat (Haryana), Jamnagar (Gujarat), Paradip (Odisha), Bina (Madhya Pradesh), Haldia (West Bengal).
17. Conventional energy means coal, oil, gas and nuclear. Non-conventional means solar, wind, tidal, biogas and geothermal. Nuclear and geothermal energy are **not** stored solar energy.
18. Nuclear plant centres: **Tarapur (Maharashtra)** — first commercial, 1969; Kakrapar (Gujarat); Kudankulam (Tamil Nadu); Kaiga (Karnataka); Rawatbhata (Rajasthan); Kalpakkam (Tamil Nadu); **Narora (Uttar Pradesh)** — only UP plant. **Shivanasamudra** is **hydel**, not nuclear.

19. Solar points: electricity from **PV cells**; **Diu** was the first UT to go fully solar-powered; **Cochin** was the first solar airport. **International Solar Alliance** HQ is at **Gurugram**. **OSOWOG** is an ISA flagship idea.
20. Wind leaders include Tamil Nadu (**Muppandal**), Gujarat (Kutch) and Rajasthan (Jaisalmer). Tidal potential is higher in the **Gulf of Khambhat** than in Kutch.
21. The first cotton-mill cluster grew in **Maharashtra–Gujarat** on cotton, humidity and Mumbai port. **Mumbai** was traditionally the largest mill centre; Ahmedabad is in the belt but was not traditionally the largest. Coimbatore is a Tamil Nadu spinning hub; Kanpur is an Uttar Pradesh mill city.
22. Steel plant centres: **Bhilai (Chhattisgarh)**, **Rourkela (Odisha)**, **Bokaro (Jharkhand)**, **Durgapur (West Bengal)**, Vizag (Andhra Pradesh), Salem (Tamil Nadu), Jamshedpur (Jharkhand). Chotanagpur is the mineral-heavy industrial heartland.
23. Agency headquarters: ONGC Delhi, MECL Nagpur, NMDC Hyderabad, NALCO Bhubaneswar.
24. The **Delhi–Mumbai Industrial Corridor (DMIC)** and the **Eastern and Western Dedicated Freight Corridors** are the main corridor schemes. SEZs are export-oriented zones.
25. In Uttar Pradesh, **Sonbhadra** is the energy capital. **Mathura** has the major refinery. **Narora** is the nuclear plant. Obra, Anpara and Rihand are thermal hubs.
26. Shankargarh silica feeds the **Firozabad** glass industry. The **Chunar–Churk–Dalla** belt is cement. **HINDALCO Renukoot** is aluminium. Sonrai (Lalitpur) has a copper occurrence.
27. Uttar Pradesh city industries: Kanpur textile and leather, Moradabad brass, Aligarh padlocks and hardware, Firozabad glass, Meerut sports goods, Noida–Ghaziabad NCR engineering.
28. Uttar Pradesh coal is mostly **non-coking** Singrauli–Sonbhadra coal — not Jharia-type coking coal.
29. Hematite is the most **used** Indian iron ore; magnetite is higher grade (~**70%** Fe) and concentrates in the **southern** sector (Karnataka). About **79%** of hematite resources sit in the **eastern** sector.
30. Gold: **Kolar** (Karnataka) is the historic closed field; **Hutti** (Karnataka) is the main working goldfield. Mine output is tiny versus jewellery demand.
31. Non-metallic set includes mica, limestone, gypsum, graphite, diamond, salt and **magnesite** ($MgCO_3$ refractory — not manganese metal). Graphite and rare earths sit on the Union **critical minerals** list (24 minerals, 2023).
32. Coking coal makes metallurgical coke; **Jharia (Jharkhand)** is the main coking field. Non-coking bituminous coal feeds thermal power — about four-fifths of production.
33. Coal nationalisation of coking and non-coking grades ran through **1971–73**. Tertiary NE coals are younger and often lower rank than Damodar Gondwana coals.
34. Power mix teaching line: **thermal** dominates generation share; hydro is strong in Himalayan / Western Ghat pockets; nuclear sits where cooling water and security allow.
35. Industrial region tags: Chotanagpur = mineral-heavy heartland; Mumbai–Pune = cotton / engineering; Ahmedabad–Vadodara = textile / petrochem; Hugli = jute / engineering; Coimbatore = spinning.
36. Western DFC = **Dadri–JNPT**; Eastern DFC = **Ludhiana–Dankuni**. DMIC is the industrial-corridor spine linking Delhi–Mumbai manufacturing nodes.
37. ISA was launched by India and France; HQ **Gurugram**; member-count traps (not "38 only") matter — use living CA figures, not frozen wrong options.
38. Limestone is the flux and cement raw material that ties steel and cement belts together in Chotanagpur and the Vindhyan / Satpura fringes.
39. Salt production concentrates on Gujarat coasts and inland Rajasthan / Sambhar-type lakes in the usual map. Fuller's earth and silica sands feed ceramics and glass.
40. Atomic minerals: uranium = Jaduguda class; thorium / monazite = Kerala–TN beach sands. Do not park thorium at Jaduguda.
41. Magnesite vs manganese: magnesite is a **refractory carbonate**; manganese is a **ferro-alloy metal** — Bhilwara is not a manganese capital.
42. Guntur is **not** an iron-ore mining centre; do not list it with Ballari / Bailadila. Tobacco Board HQ at Guntur is the correct Guntur industry tag.

43. First commercial nuclear plant is **Tarapur (1969)**; Narora is the **only** Uttar Pradesh nuclear station (Bulandshahr).
44. Critical-minerals mission and block auctions after 2023 keep lithium, nickel, cobalt, REE and graphite in the living CA set with Reasi Li as the frozen 2023 PYQ anchor.
45. Mumbai High is offshore west-coast oil; Assam is the classic onshore east oil province; Barmer (Rajasthan) is the western onshore oil growth story.
46. Private port / refining flash: **Mundra** is the largest privately owned port; Jamnagar refining scale outranks Digboi's age prestige.
47. Aluminium chain: bauxite → alumina → smelter power need — Odisha bauxite + captive power explains NALCO / BALCO geography.
48. Energy "not from the Sun" pair remains **nuclear fission** and **geothermal** — wind, biomass and hydro are solar-linked in the usual stem.

📋 Must-score facts — ore belts, mine–state, energy

Frame

Item	Lock
Mineral wealth map	Mostly **east of Mangalore–Kanpur line**
Non-ferrous	Poor many metals; rich in **bauxite**
Mica belt	**Koderma–Gaya–Hazaribagh**

Iron / Mn / bauxite

Item	Lock
Iron leaders (2021–22 teaching)	**Odisha** (ओडिशा), Chhattisgarh
Gurumahisani	Odisha
Bailadila	Chhattisgarh
Noamundi	Jharkhand
Manganese leaders	Odisha, MH, MP
Bauxite	Feeds aluminium; laterite link

Energy / industry tags

Item	Lock
Coal belts	Damodar, Mahanadi, Godavari, Son
Petroleum	Mumbai High; Assam; Gujarat
Atomic minerals	Beach sands (monazite) South India

Topic 9 — Transport & Communication

Consolidated — 48 Must-Score Facts

1. Transport moves people and goods; communication moves information. About **95%** of India's trade by **volume** moves by sea.
2. India's first passenger train ran **Mumbai–Thane in 1853**. Indian Railways is organised into about **18 zones** (older books still say 16), including South Coast Railway at Visakhapatnam.
3. Zone–HQ pairs: **Northern Railway — New Delhi**; **North Eastern Railway — Gorakhpur**; **South Eastern Railway — Garden Reach, Kolkata** (not Cuttack); **East Central Railway — Hajipur**; **North Central Railway — Prayagraj**; **South Western Railway — Hubballi**; **East Coast Railway — Bhubaneswar**.
4. The **Western Dedicated Freight Corridor** runs **Dadri–JNPT**. The **Eastern Dedicated Freight Corridor** runs **Ludhiana–Dankuni**.
5. **Amrit Bharat** is a **station modernisation** programme, not a new-lines scheme. About **1,275** stations were identified all-India; in **February 2023**, **149 Uttar Pradesh stations** were listed.
6. The new **Pamban** bridge is an **RVNL** project and India's first **vertical-lift** railway sea bridge. Hubballi holds both the **South Western Railway HQ** and the **August 2020** railway museum.
7. National Highways are only about **2%** of road length but carry about **40%** of traffic. Rural roads are most of the length (~**80%**) and are linked to **PMGSY**.
8. The **Golden Quadrilateral** joins Delhi–Mumbai–Chennai–Kolkata (about **5,846 km**; NHDP **2001**). North–South is Srinagar–Kanyakumari; East–West is Porbandar–**Silchar**. They meet at **Jhansi**.
9. **Bharatmala** is a **highway** programme. **Sagarmala** is **port-led**. Do not swap the two. **PMGSY (25 Dec 2000)** = rural all-weather roads; **PMBJP** = highway BOT corridors.
10. The **Atal Tunnel** runs under **Rohtang** in the **Pir Panjal** (Himachal / BRO). A bare claim that it is the "world's longest highway tunnel" is not the safe line.
11. The built **Purvanchal Expressway** alignment is Lucknow–Azamgarh–Mau–Ghazipur–Ballia. It **passes Mau**; Basti is off the main spine.
12. India has **12 major ports** plus many non-major ports. **IWAI** dates from **1986**. The National Waterways Act **2016** notified **111 National Waterways**.
13. National Waterway pairs: **NW-1** = Ganga, Haldia–Prayagraj; **NW-2** = Brahmaputra, Dhubri–Sadiya; **NW-3** = Kerala West Coast Canal; **NW-4** = Godavari–Krishna / Kakinada–Puducherry; **NW-5** = Brahmani–Mahanadi–East Coast Canal.
14. A July 2020 containership link from Kolkata toward Agartala used the **Chattogram** route.
15. Port–state pairs: **Paradip (Odisha)**, **Tuticorin / VOC (Tamil Nadu)**, **Kakinada (Andhra Pradesh)**, **Alappuzha (Kerala)**. **JNPT** is the container flagship; **Kolkata–Haldia** is riverine.
16. A **harbour** is sheltered water; a **port** is harbour plus cargo facilities. **Igarka** is in **Russia**, not China.
17. **Kushinagar** became an international airport on **24 June 2020**. **Cochin** was India's first solar-powered airport. **Jewar** is Noida International Airport.
18. FLY91's inaugural flight (18 March 2024) was from **Manohar International Airport** (Goa Mopa).
19. The Motihari–**Amlekhganj** (Nepal) petroleum pipeline was inaugurated in **September 2019**.
20. **PM Gati Shakti** was launched on **13 October 2021** with the "seven engines" model. The **National Logistics Policy** (September 2022) complements it. The **Blue Economy** means ocean-based growth.
21. Optical-fibre communication uses **total internal reflection (TIR)**, not simple refraction.
22. Uttar Pradesh has **no major seaport**. Its rail HQ fact is **NER Gorakhpur**; its waterway fact is **NW-1** on the Ganga.
23. NHAI was set up in **1995**. Expressways are access-controlled highways, distinct from ordinary National Highways.
24. West-coast major ports: Kandla (Deendayal), Mumbai, JNPT, Mormugao, New Mangalore, Cochin. East-coast majors: Kolkata–Haldia, Paradip, Vizag, Chennai, Ennore (Kamarajar), Tuticorin.

25. **Incredible India** is the MoT tourism brand; **ITDC** dates from **1966**. Match hill stations carefully: Chakrata–UK, Haflong–Assam, Kalimpong–WB, Kufri–HP.
26. **City of Lakes = Udaipur** (not Jaisalmer). **Pink City = Jaipur**. **Sabarimala = Kerala**. UNESCO tourist set includes Champaner–Pavagarh, CST Mumbai, Mamallapuram and Konark.
27. **Srisailam = Nallamala**; Omkareshwar and Pushkar are **not** on Satmala / Mahadeo in the classic trap. **Biligiriranga** holds the standing Ranganatha note in Karnataka.
28. Railway track gauges: **broad (1676 mm / 1.676 m)**, metre (**1 m**), and narrow (**762 / 610 mm**). Most Indian main-line traffic is now broad gauge.
29. **Konkan Railway** main line runs **Roha (Maharashtra)–Mangaluru (Karnataka)** through Maharashtra–Goa–Karnataka — it does **not** enter Kerala as the Konkan Railway corporation main line.
30. **Deendayal (Kandla)** is a tidal / gulf port on the west; **Ennore / Kamarajar** is a major east-coast port near Chennai (first major port registered as a company). **Mundra** is the largest privately owned port.
31. **HBJ / HVJ** (Hazira–Bijaipur–Jagdishpur) is the classic gas-pipeline spine in coaching notes.
32. Bharatmala Phase-I progress (MoRTH, 31 Oct 2024): about **18,714 km** constructed; **Rajasthan** led constructed length (~**2,241 km**).
33. LNG terminals teaching set: Dahej, Hazira, Kochi, Dabhol, Ennore — **not Kandla**.
34. Metro Railway Kolkata entered the zone list as a separate unit in older zone-count stories; always read whether the stem wants **16** or **18** zones.
35. Road hierarchy: National Highways (NHAI / MoRTH) → State Highways → District / rural roads (PMGSY). Do not claim NH are most of India's road **length**.
36. Communication basics: postal / telecom move information; transport moves goods and people — never treat the two words as synonyms.
37. Natural harbours need sheltered deep water; artificial harbours need breakwaters and dredging. Diamond Harbour is also a tidal-harbour teaching name.
38. Sagarmala is port-led coastal economic development; Bharatmala is road-corridor development — year-of-launch stems still test the **name swap** first.
39. Pilgrimage geography: Sabarimala (Kerala), Srisailam (Nallamala), Omkareshwar (Narmada / Madhya Pradesh framing), Pushkar (Rajasthan) — keep range matches tight.
40. Among density traps, some UKPCS keys pick **USA** for highest railway density among Canada / USA / Russia / India options — India is the common distractor.
41. Nhava Sheva (**JNPT**) is western; Visakhapatnam, Chennai and Tuticorin are eastern — coast-side stems settle on one western port.
42. Inland waterways need draft, terminals and navigable stretches; NW-1 on the Ganga is the UP waterway fact even though UP has no major seaport.
43. Jewar (Noida International) is UP's new greenfield international hub; Kushinagar's 2020 upgrade is the Buddhist-circuit international airport fact.
44. Amrit Bharat's **149** figure is UP **stations** for modernisation (Feb 2023), not 149 new railway lines.
45. Gati Shakti year is **2021**; NLP is **2022** — do not park both in 2022.
46. First passenger railway fact stays **1853 Mumbai–Thane**; do not swap it with later freight or metro inaugurations.
47. Tourism brand **Incredible India** is Ministry of Tourism; ITDC (1966) is the public hospitality / tourism corporation — different tags.
48. Optical fibre = **TIR**; microwave / radio links are not explained by TIR — keep the physics tag with fibre only.

📋 Must-score facts — rail, road, ports, waterways

Rail / freight

Item	Lock
First passenger train	Mumbai–Thane **1853**
Zones	~**18** (older books: 16)
Northern / NER / SER HQ	New Delhi / **Gorakhpur** (गोरखपुर) / Garden Reach, Kolkata
Western DFC	**Dadri–JNPT**
Eastern DFC	**Ludhiana–Dankuni**
Amrit Bharat	Station modernisation (≠ new lines); ~**149** UP stations
New Pamban	RVNL; first **vertical-lift** railway sea bridge

Road / schemes

Item	Lock
Golden Quadrilateral	Delhi–Mumbai–Chennai–Kolkata (~5846 km)
NS / EW	Srinagar–Kanyakumari / Porbandar–Silchar
Bharatmala vs Sagarmala	Highways vs **port-led**
Bharatmala top constructed state (31 Oct 2024)	**Rajasthan** (~2,241 km)
PMGSY	Rural all-weather (**2000**)
Atal Tunnel	Under **Rohtang** (Pir Panjal; BRO)
Gati Shakti	**13 Oct 2021**; NLP **Sep 2022**

Ports / waterways / air

Item	Lock
Sea trade volume	~**95%** of India's trade volume
Major ports	**12** + many non-major
NW-1 / NW-2 / NW-3	Ganga Haldia–Prayagraj / Brahmaputra / Kerala W. Coast Canal
Paradip / Tuticorin / Kakinada	Odisha / TN / AP
Harbour vs port	Sheltered water vs harbour + cargo facilities
Kushinagar intl	**24 Jun 2020**; Jewar = Noida greenfield
UP	**No major seaport**; rail HQ NER Gorakhpur; **NW-1**

Topic 10 — Tribes & Institutions

Consolidated — 45 Must-Score Facts

1. Scheduled Tribe lists are **state-wise** under **Article 342**. Parliament amends the lists by law. A community listed in one State need not be listed in another.
2. STs are about **8.6%** of India's population (Census 2011). Among the largest groups are **Bhil, Gond and Santhal**. **Madhya Pradesh** often leads in absolute ST population.
3. **Lokur Committee** criteria for ST listing include primitive traits, distinctive culture, geographical isolation, shyness of contact, and economic backwardness.
4. There are **75 Particularly Vulnerable Tribal Groups (PVTGs)** (Dhebar PTG renamed PVTG in **2006**). **Odisha** has the most PVTG communities. **PM-JANMAN** is the main outreach mission.
5. PVTG criteria stress pre-agricultural technology, stagnant or declining population, extremely low literacy, and a subsistence economy.
6. The **Fifth Schedule** covers Scheduled Areas in many States. The **Sixth Schedule** covers autonomous councils in **Assam, Meghalaya, Tripura and Mizoram** only.
7. State–tribe pairs: **Angami / Rengma = Nagaland**; **Toda / Paliyan = Tamil Nadu**; **Birhor = Jharkhand**; **Khasi = Meghalaya** (not Arunachal); **Yanadi = Andhra Pradesh**; **Chenchu = Andhra / Telangana**.
8. **Kuki** is a North-East tribe and is **not** a Uttar Pradesh Scheduled Tribe. **Bodo = Assam**; **Lepcha = Sikkim**; **Meena = Rajasthan**; **Warli = Maharashtra**; **Apatani = Arunachal**.
9. **Khasi and Garo** of Meghalaya are classic **matrilineal** societies.
10. Uttar Pradesh has **15** notified ST entries. The classic five are **Tharu, Bhotia, Buksa, Jaunsari and Raji**. UP PVTGs are **Buksa and Raji**.
11. In Uttar Pradesh, **Tharu** live in the **Terai**; **Buksa** on the Bijnor fringe; **Saharya** in **Lalitpur**; **Agariya** are linked to **iron smelting** in the Sonbhadra–Mirzapur belt.
12. After the **2003** Act, many UP ST notifications are **district-limited**. Two clusters matter: Terai north versus Vindhyan / Sonbhadra south.
13. Central Indian heartland tribes include **Gond** (with Maria/Muria), **Bhil**, Baiga, Kol, Korku, Sahariya, Halba, Warli and Meena.
14. Jharkhand core tribes include **Santhal, Munda, Oraon**, Ho, Kharia, Birhor, Bhumij, Asur and **Birjia**. Birjia is **not** an Andaman tribe.
15. Santhali uses the **Ol Chiki** script. **Birsa Munda** and the **Hul of 1855** are classic pairs. **Janjatiya Gaurav Diwas** is **15 November** (Birsa's birth anniversary).
16. Andaman **Negrito** natives are Great Andamanese, **Onge**, **Jarawa** and **Sentinelese** (North Sentinel Island). Nicobar **Mongoloid** groups are Nicobarese and **Shompen** (Great Nicobar PVTG).
17. World habitat pairs: **Bushman = Kalahari**; **Eskimo = northern Canada**; **Pygmy = Congo**; **Masai = East Africa** (not West Africa); **Ainu = Japan**; **Punan = Borneo**; Lapps/Sami = Sweden–Finland.
18. Also: **Semang = Malaysia** (not Indonesia); Bedouin = Arabia; **Kirghiz = Central Asia** (not Sudan); **Maori = New Zealand**; **Zulu = South Africa**; Bantu ≠ Sahara.
19. **Transhumance** = seasonal valley–mountain herding. Gypsies' original home in the usual teaching line = **India**. Cro-Magnon is the most recent among classic fossil-human options.
20. **Koryak** live in north-east Siberia, not Alaska. **Rengma** is a Nagaland tribe, not an Andaman island tribe.
21. **IMD** headquarters is New Delhi (**1875**). **ICAR** headquarters is New Delhi (**1929**). **IARI** is at **Pusa, New Delhi** — it is not the apex ICAR council itself.
22. **FRI** and **ICFRE** are at **Dehradun**. **WII** is also Dehradun but is not the FRI answer. **NDRI** is at **Karnal**. **IVRI** is at **Izatnagar, Bareilly (UP)**.

23. **NDDB** is at **Anand** and ran **Operation Flood** — it is not NDRI. **IIHR** is at Bengaluru. **CISH** is at Lucknow. **CSAUAT** is at Kanpur. **Pantnagar** is the first State Agricultural University tradition.
24. The **Forest Rights Act** is **2006**. It recognises individual and community forest rights. **Adi Karmayogi** (June 2025) is a **Ministry of Tribal Affairs** programme.
25. Main tribal belts are the North-Eastern hills, the central Indian plateau, western Bhil country, and the Andaman–Nicobar isolates.
26. Krishi Vigyan Kendras (KVKs) are the frontline **extension** arms under the ICAR system.
27. **PESA, 1996** extends panchayat provisions to Fifth Schedule areas with special gram-sabha powers over land, minor forest produce and local resources.
28. **TRIFED** markets tribal products; **Van Dhan Vikas** clusters support value-addition of minor forest produce under MoTA / TRIFED framing.
29. **Eklavya Model Residential Schools (EMRS)** are the flagship residential schooling network for ST students under the Ministry of Tribal Affairs.
30. Odisha PVTG names often asked include **Dongria Kondh, Bonda, Juang and Lodha**. Do not park Birjia among Andaman Negrito groups.
31. **Sentinelese** remain largely isolated on North Sentinel Island; **Jarawa** inhabit the Andaman trunk-road belt — do not treat them as Nicobar Mongoloid groups.
32. Gond country spans **Madhya Pradesh–Chhattisgarh–Maharashtra–Odisha** belts; Bhil country is strongest in **Rajasthan–Gujarat–MP–Maharashtra**.
33. **Article 275(1)** grants provide special Central assistance for Scheduled Tribes / Scheduled Areas welfare schemes.
34. Sixth Schedule autonomous councils (e.g. Bodo / BTC framing) sit only in the four NE States named above — Fifth Schedule States do not get Sixth Schedule councils by default.
35. **Keria / Kharia** map to **Odisha** (Kharia also Jharkhand). Do not park them as western-desert tribes.
36. Institute traps: **NDRI = Karnal dairy research**; **NDDB = Anand cooperative board**; **IVRI = Izatnagar veterinary**; **FRI = Dehradun forestry**.
37. World habitat extras: **Masai = East Africa**; **Ainu = Japan**; **Punan = Borneo**; Lapps / Sami = Sweden–Finland; **Maori = New Zealand**; **Zulu = South Africa**.
38. **CMFRI** is classically linked to Kochi fisheries research; do not confuse it with FRI Dehradun or IARI Pusa.
39. Primitive traits + isolation + distinctive culture remain the Lokur spine; economic backwardness alone does **not** make a community ST.
40. UP ST desk: **15** entries, classic five, PVTGs **Buksa + Raji**, Tharu–Terai / Saharya–Lalitpur / Agariya–iron — keep district-limited notifications after 2003 in mind.
41. Andaman Negrito set = Great Andamanese, Onge, Jarawa, Sentinelese; Nicobar Mongoloid set = Nicobarese + **Shompen** — **Rengma** is never in either island set.
42. IMD's **150-year** anniversary note still keeps headquarters at **New Delhi**; do not move IMD HQ to Pune or Lucknow in institute stems.
43. **Ministry of Tribal Affairs** (not Social Justice / AYUSH) is the nodal ministry for PM-JANMAN and Adi Karmayogi framing.
44. Santhal **Hul (1855)** and Birsa's anti-colonial resistance are separate Jharkhand identity anchors — Janjatiya Gaurav Diwas marks Birsa's birth, not the Hul date.
45. Matriliny fact: Khasi–Garo of Meghalaya; do not extend that tag blindly to all North-East tribes.

📋 Must-score facts — tribe↔state, PVTG, institutes

Frame

Item	Lock
ST lists	State-wise Art. **342**
ST share (2011)	~**8.6%**; largest groups **Bhil, Gond, Santhal**
PVTGs	**75**; most in **Odisha**; PM-JANMAN outreach
5th vs 6th Schedule (6वीं अनुसूची)	Many States' Scheduled Areas vs Assam–Meghalaya–Tripura–Mizoram councils

Tribe ↔ state

Tribe	Lock
Angami / Rengma	**Nagaland**
Toda / Paliyan	**Tamil Nadu**
Birhor	**Jharkhand**
Khasi / Garo	**Meghalaya** (मेघालय) (matrilineal)
Kuki	NE (**not** UP ST)
UP classic five	Tharu, Bhotia, Buksa, Jaunsari, Raji (**15** notified)
UP PVTGs	**Buksa + Raji**
Tharu / Buksa / Saharya	Terai / Bijnor fringe / **Lalitpur**
Jharkhand core	Santhal, Munda, Oraon, Ho…
Andaman Negrito	Great Andamanese, Onge, Jarawa, Sentinelese

World habitats / institutes

Pair	Lock
Bushman / Eskimo / Pygmy / Masai	Kalahari / N. Canada / Congo / **East Africa**
IMD / ICAR HQ	**New Delhi** (1875 / 1929)
IARI	**Pusa, New Delhi**
FRI / ICFRE / WII	**Dehradun** (WII ≠ FRI answer)
NDRI / IVRI	**Karnal** / Izatnagar teaching
CISH	**Lucknow** (लखनऊ); CSAUAT **Kanpur** (कानपुर)
FRA	**2006**
Janjatiya Gaurav Diwas	**15 Nov** (Birsa birth)

Topic 11 — Population Geography

Consolidated — 45 Must-Score Facts

1. India's first non-synchronous census was in **1872** (some papers key **1871**). The first **synchronous** all-India census was in **1881**.
2. **1921** is the **Great Divide** year of Indian census history. **2011** was the **15th** census and the **7th** after Independence.
3. Census 2011 India population was about **121.09 crore**. Decadal growth **2001–11** was **17.64%** (often rounded **17.7%**).
4. Arithmetic density is **population / total area**. India's 2011 arithmetic density is **382** persons per km^2.
5. Physiological density is **population / net sown area**. Agricultural density is **agricultural population / net sown area**. Do not swap these three density types.
6. Among States, **Bihar** has the highest density (**1106**). **Arunachal Pradesh** has the lowest State density (**17**). Uttar Pradesh is **829**. Delhi UT is very dense but is not a "lowest density State" answer.
7. Core 2011 figures: sex ratio **943**, child sex ratio **919**, literacy **74.04%** (age **7+**), urban share **31.16%**. SC ~**16.6%**; ST ~**8.6%**.
8. **Uttar Pradesh** is the most populous State and has the largest rural population. **Sikkim** is the least populous State. **Nagaland** showed negative growth in 2001–11.
9. Highest State sex ratio is **Kerala**. Among States in the usual 2011 set, lowest sex ratio is **Haryana**. Lowest child sex ratio (rural + urban) is also **Haryana**.
10. Literacy is highest in **Kerala** and lowest among States in **Bihar**. In Uttar Pradesh, **Shrawasti** has the lowest female literacy among districts.
11. Absolute growth is $\mathbf{P_2 - P_1}$. Growth rate is the percentage change. **Natural growth = CBR − CDR**. Induced change comes from **migration**.
12. Replacement-level **TFR is 2.1** children per woman, not "per thousand". NFHS-4 reported **2.2**; NFHS-5 reports about **2.0**.
13. The National Population Policy **2000** aimed at population stability by **2045**. World Population Day is **11 July**. The World Population Report is associated with **UNFPA**.
14. The demographic dividend window is the large share of working ages **15–59**, not 60+ or 0–6. Dependency compares young plus aged with workers.
15. Urbanisation acceleration in the classic curve is linked to the **second stage** of demographic transition. Among religions, **Jains** are the most urbanised.
16. Demographic Transition Theory is linked to **Thompson** (with **Notestein**). Optimum population is linked to **Edwin Cannan**. Social mal-adjustment is linked to **Henry George**.
17. Malthus argued population grows **geometrically** while food grows **arithmetically**. Positive checks raise deaths; preventive checks lower births. **Karl Marx** criticised Malthus; **Ester Boserup** stressed agricultural intensification.
18. The largest internal migration stream is **rural → rural**. Female migration is often for marriage; male migration is often for work. **Immigration** = in; **emigration** = out.
19. Push factors drive people from the origin; pull factors attract them to the destination. Out-migration sources are often Uttar Pradesh–Bihar; destinations are often Maharashtra–Delhi–Gujarat.
20. A **census town** needs population **≥5,000**, density **≥400/km²**, and **≥75%** of **male main workers** in non-agriculture.
21. Census 2011 listed **53** million-plus urban agglomerations. Kanpur crossed the million mark in **1971**; Lucknow in **1981**. 2011 UP UA order is **Kanpur > Lucknow > Ghaziabad > Agra**.
22. Uttar Pradesh district facts (2011): **Prayagraj** most populous; **Ghaziabad** densest; **Jaunpur** among higher sex-ratio districts; **Shrawasti** lowest female literacy.

23. Uttar Pradesh holds about **16.5%** of India's population. Rural share ~**77.7%** vs India **68.84%**. Urban share ~**22.3%** vs India **31.16%**. State sex ratio **912** vs India **943**. Literacy ~**67.7%** vs India **74.04%**.
24. Crowded belts are the Ganga plain and coasts. Sparse belts are the Himalaya, North-East hills, Thar and dry interior pockets.
25. Keep **Census 2011** numbers until **Census 2027** results replace them. UN estimates that India became most populous around **2023** do not rewrite the 2011 tables.
26. A broad-base population pyramid signals high fertility. Ageing shows earlier in southern States than in the high-fertility northern belt.
27. Occupational structure is taught as primary / secondary / tertiary. Primary still takes a large share of India's workforce, with a slow shift toward secondary and tertiary.
28. **CBR** and **CDR** are expressed **per thousand** of population; **TFR** is children **per woman**. Do not mix the units.
29. Demographic Transition stages: high birth–high death → falling death (boom) → falling birth → low birth–low death.
30. Age structure teaching bands are often **0–14**, **15–59** (workers) and **60+**.
31. **Lorenz curve** is used for **income inequality**, not literacy or sex ratio.
32. **Social capillarity** is linked to **Arsène Dumont** — aspiration and smaller families — distinct from Henry George's social mal-adjustment tag.
33. Literacy in Census 2011 is counted for age **7+**. Do not swap with school-enrolment rates or 0–6 literacy claims.
34. Physiological density rises when net sown area is scarce relative to population — useful for comparing pressure on farmland across States.
35. Million-city / UA stems use the **2011** Mumbai > Delhi > Kolkata order; the **2001** Mumbai > Kolkata > Delhi order is a trap.
36. Sex ratio improved from **933 (2001)** to **943 (2011)**; child sex ratio worsened from **927** to **919** — keep the opposite directions.
37. Highest urban % among States in the usual set is **Goa**; lowest urban % among large teaching States often points to **Himachal Pradesh**.
38. SC absolute leader is **Uttar Pradesh**; SC proportion leader is often **Punjab**. ST absolute leader is **Madhya Pradesh**; ST proportion leaders include **Lakshadweep / Mizoram**.
39. Natural increase can be high even when migration is low; induced change can reshape city growth without a high CBR–CDR gap.
40. World Population Day (**11 July**) and UNFPA's World Population Report are the usual international anchors — not IMF or WHO as the report house.
41. Replacement fertility **2.1** is the long-run stationarity bench; NFHS-5 ~**2.0** means India is near/below replacement in the survey, while Census 2011 still frames older stems.
42. Dependency ratio rises when the share of children and aged grows relative to **15–59** workers — the dividend fades as ageing advances.
43. Rural→urban migration is highly visible in metros, but **rural→rural** remains the largest internal stream in Census teaching.
44. Distribution factors include terrain, climate, soil, water, minerals, industry, transport and history — not density alone.
45. Census town ≠ always a municipality; statutory town and census town are separate urban doors into the urban count.

📋 Must-score facts — density, growth, Census tags

Terms

Term	Lock
Density	Population ÷ area
Physiological density	Population ÷ arable land
Agricultural density	Farmers ÷ arable land
Sex ratio	Females / 1000 males
Literacy	7+ who can read & write with understanding

India Census 2011 spine (teaching)

Item	Lock
Highest density state (सर्वाधिक घनत्व) (usual)	Bihar
Lowest density (large state)	Arunachal teaching
Highest literacy (usual)	Kerala
Phase model	High stationary → early expanding → late expanding → low stationary

Topic 12 — Human Geography (Settlements)

Consolidated — 45 Must-Score Facts

1. Human geography studies the **man–environment relationship** and the spatial patterns of human life, settlements and economy.
2. **Environmental determinism** (Ratzel / Semple) says nature controls culture. **Possibilism** (Vidal de la Blache) says humans choose among nature's options. **Neo-determinism** (Griffith Taylor) is stop-and-go determinism.
3. **Site** is the local ground of a settlement. **Situation** is its wider regional location relative to routes, resources and other places.
4. A **census town** needs population **≥5,000**, density **≥400/km²**, and **≥75%** of **male main workers** in non-agriculture. It need not be a municipality.
5. Rural settlements lean on primary activities and lower density. Urban places are statutory towns or census towns.
6. Village types are **clustered / nucleated**, **semi-clustered**, **hamleted** and **dispersed**. Clustered villages appear on plains, around Rajasthan water points, and for defence in Bundelkhand / Nagaland — do **not** call the Thar "dispersed by default."
7. Dispersed villages typify **Meghalaya, Uttarakhand, Himachal Pradesh, Kerala** and many North-East forest–hill tracts.
8. Hamlet local names include **panna, para, palli, nagla and dhani**. Semi-clustered villages often show a dominant-caste centre with lower strata on the flanks (Gujarat / Rajasthan).
9. Linear villages follow a road, river, canal or coast. Circular / radial sketches appear around tanks or forts in coaching diagrams.
10. Town functions are classed by the **dominant** job: administrative, industrial, transport, commercial, mining, garrison, educational, religious or tourist.
11. Class I towns have **≥1 lakh** people (468 towns in 2011 held about **60%** of urban population). Classes II–VI step down from 50–99 thousand to under 5 thousand.
12. In the standard Indian size ladder, metropolitan cities are **10 lakh–50 lakh** and mega cities are **above 50 lakh** — **six** in 2011. A UN megacity is **≥1 crore**.
13. 2011 million-plus UA rank starts **Mumbai > Delhi > Kolkata > Chennai > Bengaluru > Hyderabad**. There were **53** million-plus UAs. The **2001** order Mumbai > Kolkata > Delhi is a trap.
14. An urban agglomeration can be a town with outgrowths, two contiguous towns, or a city with adjoining towns and outgrowths. Outgrowths include railway colonies, campuses, ports and cantonments.
15. Settlement evolution pairs: ancient **Varanasi / Prayag / Madurai**; medieval **Delhi / Agra / Jaipur / Lucknow**; modern planned **Chandigarh** (Le Corbusier). Satellite towns include Ghaziabad.
16. The **Smart Cities Mission** launched on **25 June 2015** for **100** cities under **MoHUA**, implemented through an SPV. ABD means area-based development plus pan-city ICT.
17. Uttar Pradesh's **Central / Mission** Smart City list is **10**: Lucknow, Kanpur, Prayagraj, Agra, Varanasi, Aligarh, Bareilly, Jhansi, Moradabad and Saharanpur. **Ghaziabad is not** in that Central-10.
18. Uttar Pradesh's **State Smart Cities (2019)** are seven Nagar Nigams (including Ghaziabad, Meerut, Gorakhpur, Mathura, Ayodhya, Firozabad, Shahjahanpur). State list does not rewrite Mission-100 keys.
19. ISAC-2020 theme pairs: Culture **Indore**, Governance **Vadodara**, Social **Tirupati**, Urban environment **Bhopal**. Best State award went to **Uttar Pradesh**.
20. **HRIDAY** covers **12** heritage cities. In Uttar Pradesh the pair is **Varanasi and Mathura** — not Prayagraj or Ayodhya.
21. **SPMRM (Rurban)** was **launched on 21 February 2016** (Cabinet approval 2015). It is under **MoRD**, not MoHUA.
22. **Rurbanization** is linked to sociologist **G.S. Ghurye**. McLuhan's **Global Village** rests on **transport plus communication**.

23. Scheme chronology: **JNNURM 2005** → Urban Housing Policy **2007** → **AMRUT June 2015** → **Jal Jeevan Mission 2019**. **AMRUT 2.0** is **1 October 2021**. **SAGY** is **2014**.
24. Smart Cities = **MoHUA**; smart village / Rurban = **MoRD**. Keep the ministries separate.
25. **Bhopal** is the classic "not on a major river bank" city trap against Agra, Patna or Kolkata.
26. NCR slice of Uttar Pradesh includes Ghaziabad, Noida, Greater Noida and Meerut in the urban–industrial belt.
27. Economic activities: **primary** (farming, mining, fishing), **secondary** (manufacturing), **tertiary** (services), with **quaternary** for knowledge / R&D.
28. Approaches in human geography coaching sets include welfare, behavioural, radical and humanistic lenses on spatial problems.
29. **Primate city** means one city dominates the urban system far above the second city; **rank-size** expects a more regular step-down of city sizes.
30. Rural settlements lean on primary work and lower density; urban places are **statutory towns** or **census towns**.
31. Functional town tags stay classed by the **dominant** job — a transport town may also trade, but the leading function decides the label.
32. Planned modern city classic = **Chandigarh** (Le Corbusier). Do not park Lucknow or Jaipur as the Le Corbusier planned-city key.
33. Smart Cities Mission timeline extension note (works to **31 Mar 2025**) does not change the **100 cities / 25 June 2015** launch facts.
34. PMAY-U shares the **25 June 2015** launch day with Smart Cities — useful chronology glue with AMRUT (also June 2015).
35. Hamleted villages break the main settlement into secondary units; they remain a **rural** pattern, not a census-town category.
36. Dispersed settlement is hill/forest/high-rainfall logic (Meghalaya–UK–HP–Kerala), not the default Thar tag.
37. Clustered settlement on the Ganga plain is the high-density rural norm; defence clustering also appears in Bundelkhand / Nagaland notes.
38. Mega-city (NCERT >50 lakh) ≠ UN megacity (≥1 crore) ≠ metropolitan (10–50 lakh) — keep the three thresholds apart.
39. Smallest million-plus UA in the 2011 set is often taught as **Kota** — useful against inventing a Himalayan million city.
40. Outgrowths (railway colony, campus, cantonment, port colony) can sit inside an **urban agglomeration** without being separate Class-I towns.
41. HRIDAY is heritage revitalisation; Smart Cities is urban ICT/ABD modernisation — do not treat them as the same mission.
42. SAGY (2014) is village adoption / MP-linked rural development framing, distinct from SPMRM Rurban clusters.
43. Global Village is **transport + communication**, not a UN political category alone.
44. Site vs situation: a river-bank site can have a poor regional situation if routes bypass it — both must be read separately.
45. Uttar Pradesh Mission-10 Smart Cities never include Ghaziabad; Ghaziabad appears only on the **State 2019** list for that trap.

📋 Must-score facts — approaches, settlements, activities

Nature & scope

Item	Lock
Human geography	Study of human–environment relations / spatial organisation
Approaches	Welfare, behavioural, radical, humanistic (coaching set)

Item	Lock
Primary / secondary / tertiary	Extraction · manufacturing · services
Quaternary / quinary	Knowledge / high decision-making

Settlements

Item	Lock
Rural patterns	Compact, dispersed, linear, circular
Urban functions	Administrative, industrial, commercial, transport, garrison
Primate city / rank-size	Coaching definitions

Topic 13 — Disaster Geography

Consolidated — 45 Must-Score Facts

1. Coaching risk fact: **Risk ≈ Hazard × Vulnerability / Capacity**. A disaster happens when a hazard hits exposed people or assets and capacity is too weak.
2. The disaster-management cycle runs **mitigation → preparedness → response → recovery**. Mitigation is before impact; response is during and just after.
3. Earthquake **focus** is the point **inside** the Earth; **epicentre** is the point on the **surface** above it. **Richter** measures magnitude; **Mercalli** measures intensity / damage.
4. Seismic waves split into **body** (P, S) and **surface** (Love, Rayleigh). Body waves arrive **P then S**; surface waves come last and do most damage.
5. **P-wave** shadow is about **103–142°**; **S-wave** shadow lies beyond about **103°** because the outer core is liquid. **P** travels through solids, liquids and gases; **S** through **solids only**.
6. India uses seismic **Zones II–V** only (Zone I dropped in 2002). **Zone V is highest**. A draft Zone VI was **withdrawn in March 2026** — maps still follow the 2016 II–V zones.
7. Zone V covers the North-East, Himalayan pockets, **Kutch** and the **Andaman & Nicobar** belt. Much of Rajasthan and the Deccan sits in Zones II–III.
8. About **59%** of India's landmass is earthquake-prone in NDMA note. Famous shocks: Kangra 1905, Bihar–Nepal 1934, **Koyna 1967 (reservoir-induced)**, **Latur 1993 (peninsular / Killari)**, Bhuj 2001.
9. A **tsunami** (Japanese "harbour wave") is caused by **seafloor displacement**, not ordinary wind waves. India's warning hub is **INCOIS, Hyderabad**. Hall event: **26 Dec 2004**.
10. **Kilimanjaro** sits on the East African Rift and is **not** part of the Pacific Ring of Fire.
11. Landslide belts are the Himalaya, Western Ghats and North-East. Landmark pairs include Kedarnath **2013**, Joshimath subsidence **2023**, and Sikkim GLOF **2023**.
12. Flood types include riverine (Ganga–Brahmaputra), flash floods in hills, urban floods and coastal storm surge. Drought is meteorological, hydrological and agricultural. IMD drought criteria often use rainfall deficiency **>25%** of normal.
13. IMD **cloudburst** fact is rainfall **≥100 mm in one hour**, typically in Uttarakhand–Himachal–J&K–North-East hill belts around **1000–2500 m**.
14. IMD plains heat-wave gate: when **Tmax ≥40°C**, a departure of **4.5–6.4°C** is a heat wave and **>6.4°C** is severe; or actual **≥45°C / ≥47°C**. Cold-wave plains gate is **Tmin ≤10°C** with a sharp drop.
15. The **Bay of Bengal** produces **more** cyclones than the Arabian Sea. Peak seasons are roughly **May–June** and **October–December**.
16. Cyclone-name pairs: **Baguio = Philippines**; Hurricane = USA; Typhoon = China–Japan; **Willy-willy = Australia**; cyclone = North Indian Ocean.
17. **NDMA** is chaired by the **Prime Minister**. **SDMA** is chaired by the **Chief Minister**. **DDMA** is chaired by the District Magistrate / Collector.
18. Framework years: **DM Act 2005**, **NPDM 2009**, **NDRF 2006** (MHA), **NDMP 2016**, **Sendai Framework 2015–2030**, **CDRI 2019**. International Day for DRR is **13 October**.
19. NEC pair = Union Home Secretary. **NIDM** handles training from Delhi. Its ancestor **NCDM** was set up in **1995** under **IIPA**. India is **not** a disaster-free country.
20. India's multi-hazard note: ~**59%** quake-prone, ~**12%** flood-prone, ~**5,700 km** cyclone/tsunami coast, ~**68%** of cultivable area drought-vulnerable.
21. **Mangroves** cut cyclone and surge impact. **Bhopal 1984** is the classic man-made industrial / chemical disaster.
22. **DPAP** began in **1973–74**. The earlier Community Development Programme pair is **1952**.

23. Uttar Pradesh sits mainly in seismic **Zones III–IV** (not Zone V as a whole-State label). Eastern districts face river floods; Bundelkhand faces drought.
24. Plains heat waves (**Loo**) peak in **May–June**; cold-wave and fog risk in western UP and the Terai peaks in **December–January**.
25. In the 2018 UP paper, the **Gomati** carried the "biological disaster" pollution label among the given rivers.
26. Hazard is potential danger; disaster is when capacity fails. Do not treat the two words as identical.
27. Hazard classes include geophysical, hydrological, meteorological, climatological and technological / industrial.
28. Landslide triggers include steep slopes, heavy rain, earthquakes, deforestation and toe-cutting of slopes. Avalanche risk sits mainly in high Himalayan snow belts.
29. Early-warning chain: **IMD** (weather / cyclone / heat), **INCOIS** (tsunami / ocean), **CWC** (floods), GSI / NDMA guidance for landslides.
30. **Sendai Framework** has **seven** global targets and **four** priorities for action. Priority 4 includes **Build Back Better**. Adopted at the **Third** UN World Conference on DRR (Sendai, Japan).
31. Cyclone formation needs warm SST (about **27°C**), Coriolis force, and low vertical wind shear — Arabian Sea can still produce severe storms (e.g. Tauktae).
32. GLOF means glacial-lake outburst flood; Sikkim **2023** is the teaching CA pair, distinct from ordinary monsoon riverine floods.
33. Joshimath **2023** = subsidence; Kedarnath **2013** = rain–debris–flood — do not merge the mechanisms.
34. Pacific **Ring of Fire** holds most of the world's quakes and volcanoes; India's Himalayan quakes are mainly **Indian–Eurasian** convergence, not island-arc volcanoes.
35. Reservoir-induced seismicity classic = **Koyna 1967**; peninsular Stable Continental Region shock classic = **Latur / Killari 1993**.
36. Storm surge is a coastal cyclone hazard; flash flood is a hill / urban intense-rain hazard — do not label every flood as storm surge.
37. CDRI (**2019**) is India's coalition for disaster-resilient infrastructure — distinct from NDMA's statutory authority role.
38. Hyogo Framework (**2005–15**) preceded Sendai (**2015–30**); Yokohama is an older DRR conference tag, not the Sendai successor name.
39. NDRF was raised in **2006** under MHA; DM Act year **2005** is the statute, not the force-raising year.
40. Heat-wave plains need the **Tmax ≥40°C** gate before departure/actual cuts apply — a warm 36°C afternoon is not automatically a heat wave.
41. Willy-willy = Australia; Baguio = Philippines; Hurricane = Atlantic/USA naming — regional name swaps are frequent.
42. Tsunami warning for India is **INCOIS Hyderabad** (MoES), not NDMA HQ and not IMD Pune as the tsunami hub.
43. Mitigation reduces risk before impact; preparedness stages people and systems; response saves lives in the impact window; recovery rebuilds — including Build Back Better under Sendai Priority 4.
44. About **12%** flood-prone and **68%** drought-vulnerable cultivable area sit beside the **59%** quake-prone landmass note in multi-hazard stems.
45. Man-made / technological disasters include industrial chemical events (Bhopal) and can intersect natural triggers (Natech) — India is not "natural disasters only."

📋 Must-score facts — hazard types, DM cycle, India belts

Terms

Term	Lock
Hazard	Potential threat

Term	Lock
Vulnerability	Susceptibility
Risk	Hazard × vulnerability
DM cycle	Mitigation → Preparedness → Response → Recovery

India belts

Hazard	Lock
Earthquake	Himalayan / Indo-Gangetic / peninsular zones
Cyclone	Bay of Bengal > Arabian Sea frequency
Flood	Ganga–Brahmaputra plains; flash floods hills
Drought	Rain-shadow / arid west; monsoon failure
Landslide	Himalaya / WG steep slopes

Topic 14 — Earth & Universe

Consolidated — 50 Must-Score Facts

1. **Big Bang** is linked to **Lemaître** / **Gamow** (~**13.7–13.8 bn yr**) with **Hubble** redshift expansion. **Steady State** is **Hoyle–Bondi–Gold**. Solar-system **Nebular** theory is **Kant–Laplace**. **Otto Schmidt** = interstellar dust capture by the Sun.
2. The **Milky Way** is a **barred spiral**. **Andromeda (M31)** is the nearest major spiral and the farthest naked-eye object. **Chandrasekhar limit ≈ 1.44 M⊙** is the white-dwarf ceiling; above it the path runs to neutron star / black hole.
3. One **galactic year** for the Sun is ~**22.5–25 crore** years. IAU lists **88** constellations; **Saptarishi** points to the Pole Star. A **light-year** measures **distance**; star colour tracks **temperature** (blue hottest). **Brittle star** is a sea animal, not a star class.
4. Planet order from the Sun is **MVEMJSUN**. **Venus** is hottest (greenhouse / "veiled planet") and rotates **retrograde**. **Earth** has the highest mean density. **Jupiter** is largest. **Saturn** has the clearest rings. **Mercury** has the fastest orbit; Mercury and Venus have **no moon**.
5. **Copernicus** is the classic heliocentric name. **Kepler** gave planetary-motion laws; **Newton** explained orbits by universal gravitation. **Uranus** (Herschel, 1781) tilts ~**98°** ("planet on its side") and also rotates retrograde. **Neptune** is farthest and coldest among the eight and has the longest year (~**165** Earth years).
6. Largest moon is **Ganymede** (Jupiter — not Titan). **Titan** (Saturn) has a thick atmosphere. Mars moons are **Phobos** and **Deimos**. Asteroid belt sits between **Mars and Jupiter**; **Trojans** share Jupiter's orbit at L4/L5 (~60° ahead/behind).
7. A **meteor** burns in the **mesosphere**; an unburned fragment on ground is a **meteorite**. A **comet** tail always points **away from the Sun**; Halley's period is ~**76** years. **Pluto** became a **dwarf planet** (IAU, Prague, **24 Aug 2006**; MPC number **134340**).
8. A **solar eclipse** occurs at **New Moon**; a **lunar eclipse** at **Full Moon**. Sunlight takes ~**8 min 20 s** to reach Earth. Sun's energy is **nuclear fusion**. One **AU ≈ 149.6 million km**. Photosphere ≈ **6000°C**; corona is the outer solar atmosphere.
9. In the 2022 set, Saturn's atmosphere is linked with **methane**. Mars is the classic life-candidate planet (2018 subsurface-lake news). **Mangalyaan / MOM** = India's first Mars orbiter. **Aditya-L1** = solar; **Chandrayaan** = lunar — do not swap.
10. Earth **rotates** ~**24 h** (day–night, west→east) and **revolves** ~**365¼ days** (year / leap year). Shape is an **oblate spheroid**. **Perihelion ≈ 3–4 January**; **aphelion ≈ 4 July**.
11. Seasons come from **23½° axial tilt**, not perihelion distance. Equinoxes ≈ **21 March / 23 September**; solstices ≈ **21 June / 22 December**. On equinox, day and night are about equal everywhere; Arctic / Antarctic Circles mark 24-hour day or night at solstice.
12. One degree of latitude ≈ **111 km**. Tropics = **23½°**; polar circles = **66½°**. Heat zones: torrid between tropics, temperate to 66½°, frigid to the poles. **Great circles** = Equator + **all meridians**; a parallel such as **60°N is not** a great circle.
13. **0° lat and 0° long** meet in the **Atlantic** (Gulf of Guinea), not on land. **Not** on Equator: **Mexico**, **Equatorial Guinea**, **Rwanda**. **Not** on Cancer: **Iran**. **Not** on Capricorn: **Bolivia**.
14. India's area rank is **7th** (not 6th); land share ≈ **2.4%**. Tropic of Cancer runs through the **middle** of India — India is **not** wholly tropical. **Uttar Pradesh** does **not** lie on the Tropic of Cancer.
15. **IST** is based on **82°30′ E** near **Mirzapur (UP)** and is **GMT + 5 h 30 min** for the **whole** country. Time zones use ~**15° = 1 hour** (**1° = 4 min**). **IDL ≈ 180°** (zigzag); the **west** side is a day ahead.
16. Earth's spheres: **lithosphere** (crust + upper mantle), **atmosphere**, **hydrosphere**, **biosphere** (where the other three meet). Rocks cycle among igneous, sedimentary and metamorphic states.

17. Atmosphere upward order: **troposphere → stratosphere → mesosphere → thermosphere/ionosphere → exosphere**. Weather lives in the **troposphere**; **ozone UV shield** sits in the **stratosphere** (ideal for jet flight); mesosphere is coldest / meteors burn; ionosphere reflects radio waves and hosts auroras.
18. Dry-air abundance: **N_2 (~78%) → O_2 (~21%) → Ar (~0.93%) → CO_2 (~0.04%)**. Permanent gases = **N_2, O_2, Ar**; variable = vapour, CO_2, O_3, dust. **Argon is not** a greenhouse gas. **Ozone (O_3)** is variable.
19. Atmosphere is heated mainly by **long-wave terrestrial radiation**, not direct short-wave alone. Normal lapse ≈ **6.5°C/km**. **Albedo** = reflecting power (fresh snow highest). Max heating belt often ~**20°N**. Same-latitude **Agra vs Darjeeling** contrast is an **altitude** fact.
20. Heat budget (NCERT 100-unit picture): about **35**/100 insolation is reflected (planetary albedo) and about **51**/100 reaches the surface; surface returns **long-wave**. **Nuclear** and **geothermal** energy are **not** stored solar energy.
21. Greenhouse gases include water vapour, CO_2, CH_4, N_2O, O_3 and CFCs. **Montreal** = ozone / CFCs; **Kyoto** / **Paris** = climate / GHG path. Ozone-hole focus = **Antarctica** in spring.
22. **Weather** is short-term; **climate** is the long-term average. Light scattering / diffusion is classically linked to **dust**. Earth Day is **22 April**.
23. Cyclone regional names: **Hurricane**–USA, **Typhoon**–China Sea, **Willy-Willy**–Australia, **Baguio**–Philippines. Tornado scale = **Fujita**; hurricane = **Saffir–Simpson**. Cumulonimbus = thunder / intense rain.
24. **El Niño** = warm eastern Pacific / Peru phase with **less upwelling**, so **plankton and fish fall**; it can disturb the Indian monsoon. **La Niña** is the cold opposite phase.
25. **Coriolis** deflects to the **right in the NH** and **left in the SH**; it is **zero at the equator** (Ferrel's law).
26. Jet streams are mainly high-altitude **westerlies** (polar + subtropical). **Tropical Easterly Jet (TEJ)** is the summer easterly exception (~14°N). Speeds can reach **300–500 km/h**. "All jets are easterly" is false.
27. Pressure belts: **ITCZ** / doldrums = equatorial **low**; horse latitudes = subtropical **high** ~**30°**; subpolar low ~**60°**; polar high. Planetary winds = trades, westerlies, polar easterlies. Belts shift with the solstice seasons.
28. **Roaring Forties** = strong westerlies near **40°S** (Furious Fifties / Shrieking Sixties farther south). Mediterranean winter rain links to southward belt shift around NH winter solstice.
29. Interior shells: **Sial** (continental crust), **Sima** (oceanic crust), **Nife** (core). Discontinuities: crust–**Moho**–mantle–**Gutenberg** (~**2900 km**, S-wave stop)–**Lehmann**–inner core.
30. **P-wave** shadow ≈ **103–142°**; **S-wave** shadow beyond ~**103°** because the outer core is **liquid**. **P** waves are fastest (all states); **S** waves solids only; surface waves are slowest and most damaging.
31. **Wegener (1912)** proposed continental drift with **Pangaea** and **Panthalassa**. Plate margins: **divergent** (MOR / new crust), **convergent** (trenches / fold mountains), **transform** (side-slip). New ocean crust forms by **sea-floor spreading**. Himalaya = continent–continent collision.
32. **Fold mountains**: Himalaya, Alps, Andes, Rockies. **Fault / block mountains**: Vosges, Black Forest, Sierra Nevada, Rhine graben.
33. **Igneous** = cooled magma/lava (**no fossils**); **sedimentary** = layered + fossils; **metamorphic** = heat/pressure remake. Pairs: granite → **gneiss**; limestone → **marble**; sandstone → **quartzite**; shale → **slate**. Magma = underground; lava = surface melt.
34. Volcano types: shield (**Mauna Loa**), composite (**Fuji / Etna / Vesuvius**), cinder (**Paricutin**), fissure (**Deccan**), caldera (**Krakatoa**). Most abundant volcanic gas = **water vapour**. **Stromboli** = "lighthouse of the Mediterranean." **Barren Island** = India's only confirmed active volcano (Andaman).
35. Match pairs: **Rainier = USA**, **Etna = Italy**, **Paricutin = Mexico**, **Apo = Philippines**. **Kilimanjaro** = East African Rift (**dormant**), **not** Ring of Fire. **Aconcagua** is a peak, not an active-volcano match like Etna.
36. Earthquake–volcano belts: Pacific **Ring of Fire** (~**90%** of world quakes), Mid-Atlantic Ridge, Alpine–Himalayan belt, East African Rift. India's seismic map uses Zones **II–V**. NW India shakes mainly from **Indian–Eurasian** convergence.
37. Earthquake **focus / hypocentre** = rupture point inside Earth; **epicentre** = surface point above it. **Richter** = magnitude (~**10×** amplitude / ~**32×** energy per step); **Mercalli** = intensity. **Tsunami** = Japanese for harbour wave; classic cases = **2004** Indian Ocean and **2011 Fukushima**.

38. The Moon always shows roughly the same face because rotation ≈ revolution (~**27.3 days** sidereal). Weight on the Moon ≈ **1/6** of Earth weight. Apollo 11 (**1969**) landed in the **Sea of Tranquillity**. Moon has **no atmosphere**, no twilight, no sound.
39. Humidity terms: absolute vs relative humidity; dew point is the temperature where air becomes saturated. Fog is a cloud on the ground; frost forms when dew freezes.
40. Local mountain winds often asked: **chinook** (warm downslope, Rockies), **mistral** (cold, Rhône), **bora** (cold Adriatic), **foehn** (Alps warm downslope), **loo** (hot dry North Indian summer wind).
41. Homosphere (mixed major-gas ratios) roughly reaches ~**80–100 km**; above that the **heterosphere** sorts gases more by molecular weight.
42. Temperature **inversion** = temperature rises with height in a short layer; common on calm valley nights and can trap pollution near the ground.
43. Prime Meridian land set in coaching lists includes UK, France, Spain, Algeria, Mali, Burkina Faso, Togo, Ghana — **not Niger / Nigeria**. Cairo ≈ **30° E** ≈ **+2 h** on GMT.
44. China keeps **one** official time zone despite its width; India also keeps **one** (IST). Russia has the most time zones among large states in the usual teaching map.
45. If the Pole Star is on your **left**, you are walking **east**. If stars rise **perpendicular to the horizon**, the observer is on the **Equator**.
46. Inner planets are rocky terrestrials; outer set is gas giants (**Jupiter, Saturn**) plus ice giants (**Uranus, Neptune**). Sun holds ~**99.8–99.9%** of solar-system mass; H:He in the Sun is roughly **3:1**.
47. Kuiper Belt = icy leftovers beyond Neptune; **Oort Cloud** = far comet reservoir. A nebula is a star-forming cloud, **not** a solar-system "member" like planets/moons/asteroids/comets.
48. Earth's axis makes about **66½°** with the orbital plane (complement of 23½°). Sidereal day ≈ **23 h 56 min**; mean solar day = **24 h**. Century years are leap only if divisible by **400**.
49. India's Tropic of Cancer states matter for map stems; do not park UP on Cancer. Nuclear fission energy and Earth's internal heat (geothermal) are the classic "**not** from the Sun" pair in energy stems.
50. Plate-tectonic quick map: divergent = plates apart / MOR; convergent = trenches or collision mountains; transform = side-slip faults (e.g. San Andreas class). Himalaya ≠ fault-block Vosges type.

📋 Must-score facts — motions, structure, latitudes

Earth motions / shape

Item	Lock
Rotation	Day–night; west→east
Revolution	Seasons; elliptical orbit
Shape	Oblate spheroid
Axial tilt	~**23½°**

Structure / circles

Item	Lock
Crust / mantle / core	Silicates · denser rocks · iron–nickel
Tropic of Cancer	**23½°N** (cuts India)
Equator	0°
International Date Line	~180° (zigzag)
Atmosphere layers	Troposphere → Stratosphere → Mesosphere → Thermosphere

Topic 15 — Geomorphology & Landform Processes

Consolidated — 48 Must-Score Facts

1. **Endogenic** processes (folding, faulting, volcanism, uplift) build relief from inside the Earth. **Exogenic** processes wear and deposit at the surface. **Gradation** = degradation + aggradation.
2. Davis's cycle of erosion is framed as **structure–process–time** (youth–mature–old). Penck and King modify timing, but the landform vocabulary stays useful.
3. **Igneous** rocks are primary: **granite** is intrusive (acidic); **basalt** is extrusive (basic). Fossils belong to **sedimentary** rocks.
4. Igneous bodies: batholith, laccolith, lopolith, phacolith; a **sill** is parallel to beds; a **dyke** cuts across beds.
5. Metamorphic pairs: **limestone → marble**, **sandstone → quartzite**, **granite → gneiss**, **shale → slate / schist**. Marble and quartzite are non-foliated.
6. **Weathering** breaks rock **in situ**. **Erosion** picks up and carries material. **Denudation** is weathering plus erosion. Chemical routes include carbonation, oxidation, hydrolysis and hydration.
7. River load moves by traction, saltation, suspension and solution. Agents of erosion are river, wind, glacier, sea, groundwater (karst) and gravity (mass wasting).
8. Fluvial stages: youth = **V-valley / waterfall / pothole**; mature = **meander / floodplain / levee**; old = **ox-bow / delta / peneplain**. Rejuvenation adds knickpoints, terraces and incised meanders.
9. On a meander, the **concave** bank erodes and the **convex** bank builds a **point bar**. A **yazoo** stream is deferred behind a levee. Mountain-foot dumps are **alluvial fans**; coalescing fans form a **bajada**.
10. Stream genetic types include consequent, subsequent, obsequent and resequent. **Antecedent** streams cut rising land; **superimposed** streams inherit a course from a cover.
11. The **only** bird's-foot example is the **Mississippi**. **Arcuate** examples include the **Nile** and the **Ganga–Brahmaputra**. The usual cuspate example is the **Tiber**.
12. East-flowing Indian mouths often build deltas; **Narmada** and **Tapi** usually form estuaries into the Arabian Sea.
13. Drainage patterns: **folded** beds → **trellis**; homogeneous rock → **dendritic**; dome → **radial**; joints / faults → **rectangular**; inward drain → **centripetal**.
14. The classic fault-valley river tag is the **Damodar**, not the Chambal. **Narmada–Tapi** are rift / fault-line pairs as well.
15. The **Imphal** basin is **lacustrine** (lake-filled), not loess or glacial. Kashmir Vale also has lacustrine / structural pairs. **Loess** is wind-laid silt (China belt).
16. Structure landforms include cuesta, hogback, mesa and butte. A **mesa** is broader than a **butte**. Davis's humid old-age plain is a **peneplain** with leftover **monadnocks**; King's arid form is a **pediplain**.
17. Glacial erosional forms: cirque, arête, horn, **U-valley**, hanging valley, fjord, tarn. Depositional forms: moraine, drumlin, esker, kame, outwash, erratic.
18. A **fjord** is a drowned glacial trough. A **ria** is a drowned river valley. Do not treat them as the same estuary type.
19. Aeolian points: **barchan** horns point **downwind**; parabolic dunes often show horns upwind. Yardang, zeugen and mushroom rocks are wind-eroded forms.
20. Coastal deposition builds **spit, bar, tombolo and lagoon** with longshore drift. Coral reefs include fringing, barrier and atoll types. An **atoll** is horseshoe / ring around a lagoon.
21. Karst needs **limestone plus carbonation**. **Stalactite** hangs from the **ceiling**; **stalagmite** grows from the **floor**. Forms include doline, cave, uvala and polje.
22. River capture vocabulary includes pirate stream, beheaded stream, wind gap and elbow of capture.
23. Conglomerate has rounded pebbles; breccia has angular fragments. Organic sedimentary rock pair includes coal.
24. Intertrappean beds between Deccan lava flows hold **land and freshwater** fossils, not marine plant–animal assemblages.

25. Himadri is fossil-poor crystalline rock; Lesser Himalaya carries marine fossils; Shiwalik holds human remains — keep the three belts distinct.
26. A gorge is a steep mountain cut; a canyon is often the arid stair-step equivalent. U-valley = glacier; V-valley = youthful river.
27. The rock cycle links magma ⇄ igneous → sedimentary → metamorphic → melt, so the same material can reappear as a different rock type.
28. Mass wasting moves material downslope under gravity (creep, slide, flow, fall). It is an agent alongside river, wind, glacier, sea and groundwater.
29. If the stem calls a moribund delta a **subdivision**, the answer is the **Bengal Delta** (moribund, mature, active). If it asks which river's delta is the moribund **lobe**, the answer is the **Cauvery**.
30. World delta locations: **Irrawaddy–southern Myanmar**; **Nile–northern Egypt**; **Indus–Sindh (Pakistan)**; **Danube–Black Sea (Romania/Ukraine)**; **Rhine–Netherlands**; **Volga–Caspian**; **Mekong–Vietnam**.
31. The **Niger** has an inland delta in **Mali** and a sea delta in **Nigeria**. The **Amazon** meets the Atlantic in **Brazil** as a broad estuary. **Red River** delta = northern Vietnam.
32. Indian delta States: Godavari and Krishna → **Andhra Pradesh**; Mahanadi → **Odisha**; Cauvery → **Tamil Nadu**.
33. Dendritic rivers include Ganga and Indus plains, Godavari, Mahanadi, Krishna and Kaveri. From **Amarkantak**, Narmada west and Son toward Ganga give a **radial** spread (Mahanadi at Sihawa is outside that set).
34. Indian **trellis** ground is the old folded belt of **Singhbhum** on the Chotanagpur plateau. **Rectangular** streams follow joints on **Vindhyan** rocks.
35. **Centripetal** streams drain into **Loktak** and **Sambhar**. The **Sharavati** follows a **parallel** pattern down the Western Ghats.
36. Mekong source is Tibet; its delta is in **Vietnam**, not Cambodia.
37. Antecedent Himalayan streams keep courses while land rises; superimposed streams are let down from an older cover onto different structure below.
38. Point-bar deposition on the convex bank pairs with cut-bank erosion on the concave bank — the meander migrates laterally toward the cut bank.
39. Pediplain (King / arid) ≠ peneplain (Davis / humid old age). Monadnocks are resistant leftovers on a peneplain.
40. Tombolo links an island to the mainland by a depositional bar; a spit is a free depositional finger built by longshore drift.
41. Karst carbonation needs CO_2-charged water on limestone — sandstone country does not produce classic stalactite caves.
42. Bird's-foot delta grows finger distributaries into standing water (Mississippi); arcuate deltas have a bowed seaward front (Nile / Ganga–Brahmaputra).
43. Bajada is coalesced alluvial fans along a mountain front; a single fan is not yet a bajada.
44. Erratics are glacier-transported blocks left far from source; they are depositional evidence, not wind yardangs.
45. Hanging valleys mark tributary glaciers that did not cut as deep as the main U-trough — waterfalls often mark their mouths after ice retreat.
46. Rejuvenation signs — knickpoints, river terraces, incised meanders — mean renewed downcutting after uplift or base-level fall.
47. Loess is wind-laid silt; do not tag Imphal basin as loess when the teaching key is lacustrine.
48. West-flowing Narmada–Tapi estuary habit contrasts with east-coast delta habit — Arabian Sea macrotidal / structural factors sit behind the estuary tag.

📋 Must-score facts — agents, landforms, rocks

Endogenic / exogenic

Item	Lock
Endogenic	Diastrophism, volcanism, earthquakes
Exogenic	Weathering, erosion, deposition
Weathering types	Physical, chemical, biological

Agent ↔ landform

Agent	Landforms
River	V-valley, meander, oxbow, floodplain, delta
Glacier	U-valley, cirque, moraine, fjord
Wind	Yardang, dune, loess
Wave	Cliff, beach, spit, stack
Karst	Sinkhole, cave, stalactite

Rock cycle

Igneous → **Sedimentary** → **Metamorphic** (with melting / uplift loops)

Topic 16 — Oceans

Consolidated — 45 Must-Score Facts

1. By area the oceans rank **Pacific > Atlantic > Indian > Southern > Arctic**. The Indian Ocean was the Greek **Erythraean Sea** and straddles **both sides of the Equator**.
2. Ocean-floor order from the shore is **shelf → slope → rise (or trench) → abyssal plain**. A **guyot** is a flat-topped seamount.
3. The continental shelf is shallow (about **200 m**), rich in fish and oil, and covers roughly **7.5%** of the ocean floor. Abyssal plains are the most extensive floor province (~**76%**).
4. India's **west shelf is wider than the east**. The Gujarat–Mumbai belt is widest and holds **Bombay High**. The east shelf is narrower but carries large delta fans.
5. The **Telegraphic Plateau** is part of the **North Atlantic Ridge**, not a separate land plateau.
6. Trench–ocean pairs: **Mariana–Pacific**, **Puerto Rico–Atlantic**, **Sunda/Java–Indian**, and **Molloy–Arctic**.
7. **Agulhas** and **Brazil** are **warm** currents. **Humboldt (Peru)** and **California** are **cold**. Do not mark Agulhas or Brazil as cold.
8. The **Benguela** is an **Atlantic** cold current off south-west Africa. It is **not** a Pacific current.
9. Among classic options, the current linked to the **Indian Ocean** is the **Agulhas**. **Kuroshio** is warm (Japan); **Oyashio** is cold (Kurile).
10. **El Niño** sends warm water off Peru, weakens upwelling, and cuts plankton and fish catch. **La Niña** is the cold opposite phase.
11. Open-ocean **salinity** is about **35‰**. The maximum sits near the **Tropics of Cancer and Capricorn**, not on the Equator. The **Red Sea** is high; the **Baltic** is low.
12. Seawater density rises when water is **colder** and **saltier**. Cold salty water sinks and drives the **thermohaline** circulation. **NaCl** is about **77%** of dissolved salts.
13. Surface temperature peaks near the equator. The **thermocline** lies roughly **300–1000 m**. Deep water is cold in every ocean.
14. A **spring tide** forms at **syzygy** (new/full moon) and is **large**. A **neap tide** forms at **quadrature** and is **small**. Tidal period ≈ **12 h 25 min**.
15. A **tidal bore** is famous on the **Hooghly** and the **Amazon**. The **Bay of Fundy** has the world's greatest tidal range. India's tidal-energy coast fact is the **Gulf of Khambhat**.
16. **Upwelling** is strongest on **west coasts**: Peru, California, Canary, Benguela, and Somalia.
17. **Constructive** waves build beaches; **destructive** waves erode them. A **tsunami** is not a tide.
18. Ocean currents shape climate (mild or foggy coasts, west-coast deserts), concentrate fish where warm and cold meet or upwelling occurs, and affect navigation routes.
19. Deepest deposit is **red clay**. **Globigerina** and **pteropod** oozes are **calcareous**. **Diatom** and **radiolarian** oozes are **siliceous**. **Manganese nodules** sit on abyssal plains.
20. The **Suez Canal** (Egypt, **1869**) joins the **Mediterranean** and the **Red Sea**. It is a **sea-level** cut — ships do **not** climb stepped chambers. Ends: **Port Said** (N) and **Suez** (S).
21. Suez lakes north→south: **Manzala → Timsah → Great Bitter → Little Bitter**. India–Europe sea route is shorter by about **7000 km**. Suez does **not** itself touch the Atlantic or the open Indian Ocean.
22. The **Panama Canal** (**1914**) joins the **Atlantic/Caribbean** and the **Pacific**. Ships **climb stepped chambers** and use **Gatun Lake**. The **Kiel Canal** joins the **North Sea** and the **Baltic**.
23. UNCLOS belts: **territorial sea 12 nm**, **contiguous zone 24 nm**, **EEZ 200 nm**. The continental shelf starts at **200 nm** and may extend to **350 nm**.
24. Classic fishery centres: **Grand Banks** (Labrador meets Gulf Stream), **Dogger Bank**, Peru upwelling, and the **Kuroshio–Oyashio** mix.

25. India's **operational** Antarctic stations are **Maitri** and **Bharati**. **Dakshin Gangotri** is not the operational pair. **Himadri** is India's **Arctic** station.
26. The **Antarctic Treaty** dates to **1959/61**. India joined in **1983**. India is an **observer** in the Arctic Council (Arctic Policy **2022**).
27. Key straits: **Hormuz** (Persian Gulf ↔ Gulf of Oman; **Iran north / Oman south**; ~**25%** seaborne oil), **Malacca**, **Gibraltar**, **Bering**, **Bab-el-Mandeb**, **Palk**, and the **10° Channel**.
28. Persian Gulf coasts = Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE — **not Oman**. Oman sits on the Gulf of Oman / Musandam side of Hormuz.
29. The **Carlsberg Ridge** lies in the north-west Indian Ocean. The **Ninetyeast Ridge** is not the Mid-Atlantic Ridge.
30. The International Seabed Authority manages the seabed **Area** beyond national zones (HQ Jamaica).
31. **Drake Passage** is not the Strait of Magellan. Panama opened in **1914** with stepped chambers; Suez opened in **1869** as a sea-level cut.
32. Hydrosphere ≈ **71%** of Earth. Usable fresh ≈ **<1%** of all water. Of fresh water, ice ≫ groundwater ≫ rivers/lakes.
33. **Datum line** = mean-sea-level height/depth zero. **Halocline** = salinity gradient with depth.
34. **Sargasso** = North Atlantic, **no coast**. **Red Sea** = axial trough. **Diamantina** = Indian Ocean trench/fracture teaching tag.
35. Türkiye clock: **N = Black**, **S = Mediterranean**, **W = Aegean**, **NW = Marmara**. West→east ladder: **Mediterranean → Black → Caspian → Aral**.
36. **Gaza** faces the **Mediterranean**. Baltic coasts include Denmark, Germany, Poland, Baltic States, Russia, Finland, Sweden — **not Norway**. **Jordan** has no Mediterranean coast (only **Aqaba** / Red Sea).
37. Warm currents raise coastal temperature and humidity; cold currents cool and dry coasts and help west-coast deserts. Fog is common where warm and cold currents meet.
38. Coral needs warm clear shallow water; bleaching pairs with about **2°C** sea-surface warming. India's reef facts: Andaman & Nicobar, Lakshadweep, Gulf of Mannar, Gulf of Kachchh.
39. Hormuz oil/LNG chokepoint: roughly **20 mb/d** oil products (~**25%** seaborne oil) and ~**20%** world LNG in recent baseline years; most flows go to **Asia**.
40. Only **Saudi Arabia** and the **UAE** have meaningful pipeline routes that can skip Hormuz; Iraq, Kuwait, Qatar, Bahrain and Iran lean almost entirely on the strait.
41. **BBNJ 2023** is the high-seas biodiversity treaty path; UNCLOS **12 / 24 / 200 nm** belts remain the distance spine.
42. Canary and California are **cold** eastern-boundary currents; Gulf Stream and Kuroshio are **warm** western-boundary currents.
43. Somali upwelling is a seasonal Indian Ocean west-coast upwelling fact tied to monsoon winds.
44. North Atlantic Deep Water / thermohaline "conveyor" links dense sinking in high latitudes to global deep circulation.
45. India's inland waterway **NW-1** is the **Ganga** from Haldia to Prayagraj — a waterway fact often parked beside ocean chapters in mixed papers.

📋 Must-score facts — relief, currents, El Niño

Ocean relief

Feature	Lock
Continental shelf	Shallow margin
Slope / rise	Steep then gentle to floor
Abyssal plain	Deep flat floor
Trench	Deepest (e.g. Mariana)
Mid-ocean ridge	Divergent volcanism

Currents / climate

Item	Lock
Open-ocean salinity	~**35‰**; max near Tropics (not Equator)
Warm / cold currents	Affect coasts & fog / desert margins
Agulhas	Into Indian Ocean (S. Africa teaching)
El Niño	Warm water off Peru; weak upwelling; fish crash
Tides	Spring / neap; lunar–solar alignment

Topic 17 — World Rivers & Lakes

Consolidated — 45 Must-Score Facts

1. **New Orleans** stands on the **Mississippi**, not the Missouri. The Missouri joins the Mississippi at **St Louis**.
2. **Budapest** is on the **Danube**. **Cologne** is on the **Rhine**. India's **Hyderabad** is on the **Musi**, not the Godavari or Paleru.
3. The **Mekong** rises in Tibet, flows **south / south-east**, and builds its delta in **southern Vietnam** — not Cambodia and not south-west.
4. Direction pairs: **Amur** north-east; **Syr Darya** north-west into the Aral; **Angara** north out of Baikal; **Volga** from the Valdai Hills into the **Caspian**.
5. The main USA–Mexico border river is the **Rio Grande**. The **Colorado** is the usual trap option for that border stem.
6. **Lake Onega** is in **Russia**, not Canada. **Lake Michigan** lies wholly in the **USA**. **Maracaibo** is in Venezuela. **Baikal** is in Russia.
7. Superlatives: the **Nile** is longest for Prelims teaching; the **Amazon** has the largest discharge; the **Congo** is deepest and crosses the Equator **twice**; the **Yangtze** is Asia's longest; the **Volga** is Europe's longest.
8. Delta shapes: **Mississippi** = bird's-foot; **Nile**, **Hwang Ho**, and **Niger** = arcuate (bow).
9. The **Danube** crosses the most countries. The **Rhine** is Europe's busiest navigation artery.
10. **Khartoum** is where the White and Blue Nile meet. **Aswan** created **Lake Nasser**. **GERD** sits on the **Blue Nile** in Ethiopia.
11. Dam pairs: **Three Gorges–Yangtze**, **Aswan–Nile**, **Itaipu–Paraná**, **Hoover–Colorado**, **Kariba–Zambezi**.
12. The **Caspian** is the largest lake. **Superior** is the largest freshwater lake by area. **Baikal** is deepest, holds the most fresh volume, and is among the oldest.
13. **Tanganyika** is second deepest and the longest freshwater lake. **Titicaca** is the highest navigable lake. The **Dead Sea** is the lowest land surface and is hypersaline (~**34%** salt).
14. Great Lakes west to east: **Superior** → **Michigan** → **Huron** → **Erie** → **Ontario** (Super Man Helps Every One).
15. Falls: **Angel** (Venezuela) is highest; **Victoria** is on the Zambezi; **Niagara** is US–Canada; **Iguazu** is Brazil–Argentina.
16. The **St Lawrence** is the Great Lakes seaway. The **Rhine–Main–Danube** canal links the Black Sea system to the Rhine.
17. The **Aral Sea** shrinks mainly because the **Amu Darya** and **Syr Darya** were diverted. **Lake Chad** is also shrinking.
18. The **Niger** is the paradox river that first flows inland. The Congo is also known as the **Zaire**.
19. The **Darling Range** is a highland of south-west Australia. It is **not** the same as the **Murray–Darling** river system.
20. **Endorheic** lakes have no ocean outlet — Caspian, Aral, Dead Sea, Chad, and Eyre.
21. City–river pairs: **Baghdad–Tigris**, **Paris–Seine**, **London–Thames**, **Cairo–Nile**, Phnom Penh–Mekong, Hanoi–Red, Yangon–Irrawaddy, Bangkok–Chao Phraya, Basra–Shatt al-Arab.
22. Lake types include tectonic/rift, glacial, crater, lagoon, oxbow and artificial reservoir.
23. Continent spine: Nile (Africa), Amazon (South America), Yangtze (Asia), Mississippi–Missouri (North America), Volga (Europe), Murray–Darling (Australia).
24. **Limpopo** crosses the Tropic of Capricorn **twice**. **Congo/Zaire** crosses the Equator **twice**.
25. **Mahaweli** = longest Sri Lanka river; Sri Lanka drainage is **radial** from central highlands.
26. **Grand Canyon** = Colorado. **Chisapani Gorge** = Nepal (Karnali). Largest delta = **Ganga–Brahmaputra**. Inselberg ≠ glacier.
27. Delta mouths: Mekong → **South China Sea**; Nile → **Mediterranean**; Mississippi → **Gulf of Mexico**; Danube → **Black Sea**; Indus → **Arabian Sea**.
28. Border rivers beyond Rio Grande: **Amur** (Russia–China), **Orange** (South Africa–Namibia), **Zambezi** (Zambia–Zimbabwe at Victoria Falls).

29. More dam–river pairs: **Guri–Caroní** (Venezuela), **Tucuruí–Tocantins** (Brazil), **Grand Coulee–Columbia** (USA), **Tarbela–Indus** (Pakistan).
30. Yangtze chain: Tibetan Plateau → China → **Three Gorges** → Shanghai belt → **East China Sea**.
31. Nile chain: Victoria / Tana → Sudan–Egypt → **Aswan High Dam** → Nile Delta → **Mediterranean**.
32. Only a marked stretch of a river is the international border — do not treat the whole Mekong or Colorado course as one continuous border.
33. Navigation classics include the Rhine, Danube, Volga, St Lawrence, Yangtze and Mississippi.
34. Madrid–Manzanares is a European capital–river pair often used against inventing a Tagus-only Madrid key.
35. Victoria Falls is on the **Zambezi**; Niagara is US–Canada; do not park Angel Falls on the Zambezi.
36. Lake Michigan is wholly USA; Superior/Huron/Erie/Ontario are shared — Michigan's "wholly USA" tag is the trap.
37. Dead Sea hypersalinity (~34%) and lowest land surface tag differ from Baikal's fresh-volume / depth tag.
38. Murray–Darling is Australia's classic basin; Darling **Range** is a SW highland — name overlap is intentional trap fuel.
39. Itaipu is on the **Paraná**; Hoover on the **Colorado**; Kariba on the **Zambezi** — dam–river swaps are frequent.
40. Amazon discharge leadership does not make it the "longest river" key when the stem asks length (Nile teaching key).
41. White Nile / Blue Nile confluence at **Khartoum** precedes the Egyptian main Nile; GERD is Ethiopian Blue Nile.
42. Endorheic Aral shrinkage is diversion-driven; do not blame only climate without the Amu–Syr diversion story.
43. Rhine–Main–Danube canal creates a North Sea–Black Sea inland link via the Rhine and Danube systems.
44. Titicaca = highest navigable; Tanganyika = longest freshwater / second deepest — do not swap with Baikal depth leadership.
45. Inselberg is a residual / wind-related residual hill tag in teaching, not a glacial horn.

📋 Must-score facts — city–river, deltas, lakes

City ↔ river

City	River
New Orleans	**Mississippi** (≠ Missouri)
Budapest	**Danube**
Cologne	**Rhine**
Hyderabad (India)	**Musi** (≠ Godavari)
Khartoum	White + Blue Nile meet

Direction / border

River	Lock
Mekong	Tibet → SE; delta **S. Vietnam**
Amur	North-east
Syr Darya	North-west into Aral
Rio Grande	Main USA–Mexico border (≠ Colorado trap)

Superlatives / deltas

Item	Lock
Longest (Prelims)	**Nile**
Largest discharge	**Amazon**

Item	Lock
Bird’s-foot delta	**Mississippi**
Arcuate	Nile, Hwang Ho, Niger
Onega	**Russia** (≠ Canada)
Michigan	Wholly **USA**

Topic 18 — World Landforms

Consolidated — 45 Must-Score Facts

1. Area order (largest → smallest): **Asia** → **Africa** → **North America** → **South America** → **Antarctica** → **Europe** → **Australia**. Australia is the **smallest** continent; Africa has the **most countries (~54)**.
2. Mean elevation leader is **Antarctica** (~2300 m). Europe has the **highest share of plains** in its area. **Guyana** is in **South America**, not Africa.
3. The **Andes** are the world's **longest** fold chain. The **Himalaya** are the **highest**. **Aconcagua** stands in the Andes of Argentina.
4. Length ladder pair: Andes > Rockies > Great Dividing Range > Himalaya.
5. **Pyrenees** = Spain–France; **Alps** = Switzerland / central Europe (**not England**); **Apennine** = Italy; **Balkan** = Bulgaria; **Urals** mark Europe–Asia.
6. **Toubkal** (Atlas) is in **Morocco**. **Hoggar** is in **Algeria**. **Stanley / Rwenzori** is in Uganda. **Kilimanjaro** is in **Tanzania**. **Darling Range** = SW Australia.
7. **Chimborazo** is in **Ecuador**. The Atlas system is North African. **Sierra Nevada** = block mountain (not young fold like Rockies/Alps/Himalaya).
8. **Kilimanjaro** sits on the **East African Rift** and is **not** in the Pacific Ring of Fire. **Etna** is in Italy / the Mediterranean.
9. Tertiary young folds: Alps, Andes, Himalaya, Rockies, Atlas; Appalachians = **Caledonian / old** residual fold belt.
10. **Tibet** is the highest large plateau (~4500 m). The **Pamir** is the "**Roof of the world**". The **Altiplano** is Bolivia–Peru. The **Meseta** / Madrid is Spain.
11. The **Colorado Plateau** holds the Grand Canyon. The **Columbia Plateau** is a lava plateau. Do not swap the two. **Telegraphic Plateau** sits on the **North Atlantic Ridge**.
12. Grassland names: **Pampas** = Argentina; **Campos** = Brazil; **Llanos** = Venezuela–Colombia; **Puszta** = Hungary; **Prairie** = North America; **Steppe** = Eurasia; **Veld** = South Africa; **Downs** = Australia.
13. **Borneo** is shared by three countries and is **not** a volcanic dump. **Java** and **Sumatra** sit on a volcanic arc.
14. Island rank by size: **Greenland > New Guinea > Borneo > Madagascar > Baffin > Sumatra > Honshu > Victoria > Great Britain > Ellesmere**. Australia is a continent, not ranked as an island here.
15. **Honshu** = Japan's largest island; **Faroe** = Sheep Islands (Denmark).
16. The **Gobi** lies in **Mongolia and China** only. It is a cold desert, not Russia or Kazakhstan.
17. Hot-desert size order: **Great Sandy < Gobi < Arabian < Sahara**. The **Sahara** is the largest hot desert. **Thar** = densest populated desert; **Atacama** = driest.
18. **Gibson** = Australia (not Brazil). **Sonoran** = USA. **Taklamakan** = China. **Karakum** = Turkmenistan.
19. The **Atacama** (N Chile / S Peru) is the driest. The **Namib** is a fog coast. **Patagonia** is Argentina's rain-shadow / temperate desert.
20. Hot deserts favour **west coasts** because of Horse Latitudes plus cold ocean currents (~15–30°).
21. "Land of Big Games" points to the tropical **savanna**, not the Sahara.
22. Sclerophyll scrub: **Maquis** = Mediterranean; **Fynbos** = South Africa; **Chaparral** = California; **Matorral** = Chile.
23. **Epiphytes** mark equatorial forest; **baobab** marks savanna; **cedars** mark the Mediterranean; **acacia** marks the Sahara fringe.
24. In Brazil, **Selva** is the rainforest and **Terra Roxa** is the famous coffee soil.
25. The **Mediterranean** climate has **winter rain** on five **west coasts** near **30–45°**. Summer is dry under the subtropical high.
26. **Taiga** is boreal conifer forest. **Tundra** is treeless. Climate letters: Savanna **Aw**, Steppe **BS**, Tundra **ET**.
27. Volcano pairs: **Rainier** USA; **Etna** Italy; **Paricutin** Mexico; **Apo** Philippines; also Fuji (Japan), Pinatubo (Philippines), St Helens (USA).

28. Peak continents: **Elbrus** = Europe (Caucasus / Russia); **Mont Blanc** = Alps; **Denali / McKinley** = North America; **Kosciuszko** = Australia mainland; **Cook / Aoraki** = New Zealand.
29. Temperate grasslands (Prairie, Steppe, Pampas, Veld, Downs) sit on **chernozem**-type wheat soils.
30. **Death Valley** (California) = rift valley, extreme heat. **Silicon Valley** = California chip belt. **Great Artesian Basin** = Australia. Blind valley / sinkhole = **karst**.
31. **Arakan Yoma** = Myanmar. **Golan Heights** = Middle East (SW Syria / occupied note). **Black Forest** = Germany (east of Rhine; Vosges west).
32. Plateau types: intermontane (**Tibet**), piedmont, and volcanic / lava (**Columbia**, Deccan).
33. Valley tags: rift valleys (East Africa, Rhine), glacial U-valleys, and structural vales.
34. Residual and dome mountains are worn or laccolith leftovers, not young fold belts like the Himalaya.
35. British Columbia = "Sea of Mountains." **Patagonia Plateau** = mineral storehouse pair in coaching notes.
36. Europe has the least desertification problem among continents in the usual teaching contrast.
37. Young fold vs block: Rockies/Alps/Himalaya/Andes = fold; Sierra Nevada / Vosges–Black Forest class = block / rift flanks.
38. Equatorial rainforest ≠ Mediterranean sclerophyll ≠ taiga conifers — vegetation belts follow climate, not continent labels alone.
39. Savanna Aw climate supports large herbivore / "big game" parks; hot desert does not get that tag.
40. Island size list excludes Australia as an island; Greenland remains largest island.
41. Atacama dryness pairs with cold current + subtropical high on a west coast — same logic as Namib fog-coast desert.
42. Pamir phrase "Roof of the world" ≠ Tibet "highest large plateau" — keep the two plateau tags distinct.
43. Appalachians are old/Caledonian; do not park them with Tertiary Alps–Himalaya–Andes–Rockies young folds.
44. Pampas–Argentina vs Campos–Brazil vs Llanos–Venezuela/Colombia is the South American grassland triangle.
45. Maquis–Mediterranean vs Chaparral–California vs Fynbos–South Africa vs Matorral–Chile is the sclerophyll west-coast set.

📋 Must-score facts — continents, mountains, grasslands

Continent area (large → small)

Asia → Africa → N. America → S. America → Antarctica → Europe → Australia

Mountain / plateau tags

Item	Lock
Longest fold chain	**Andes**
Highest chain	**Himalaya**
Pyrenees	Spain–France
Alps	Central Europe (≠ **England**)
Kilimanjaro	East African Rift (≠ **Ring of Fire**)
Roof of the world	**Pamir**
Highest large plateau	**Tibet**
Colorado vs Columbia Plateau	Grand Canyon vs lava plateau

Grasslands

Name	Region
Pampas	Argentina

Name	Region
Campos	Brazil
Llanos	Venezuela–Colombia
Puszta	Hungary
Prairie / Steppe / Veld	N. America / Eurasia / S. Africa

Topic 19 — World Regional Geography

Consolidated — 45 Must-Score Facts

1. Continent area order: **Asia > Africa > North America > South America > Antarctica > Europe > Australia/Oceania**.
2. The **Nobi** and **Kanto** plains are in **Japan**, not Korea.
3. Iraq's **Sunni Triangle** is **Baghdad, Tikrit, and Ramadi**. **Basra** (Shia south) is the trap.
4. West Asia mountains west to east: **Pontic → Zagros → Hindu Kush → Karakoram**.
5. Central Asia capitals: Uzbekistan **Tashkent**, Tajikistan **Dushanbe**, Kyrgyzstan **Bishkek**, Turkmenistan **Ashgabat**, Kazakhstan **Astana**.
6. The **Kara Kum** desert is in **Turkmenistan**. **Kyzylkum** is the neighbouring "red sand" pair often contrasted with Kara Kum.
7. **Borneo** is shared by Indonesia, Malaysia, and Brunei and is **not** a volcanic island dump.
8. Philippines cane and coconut history fact: **Spanish and Americans** (not British/Dutch as the first key).
9. Korea: **Seoul** south, **Pyongyang** north, roughly the **38th parallel**. Nobi/Kanto are not Korean plains.
10. Palestine map: **Gaza** on the Egypt side, **West Bank** on the Jordan side, with the **Jordan River / Dead Sea** belt.
11. Western Europe has **westerlies** and rain in **all months** (not Mediterranean winter-only).
12. The **Suez Canal** shortened the India–Europe sea route by about **7000 km**.
13. **Cape Verde**'s capital is **Praia**. **Bamako** is Mali's capital — do not swap them.
14. Maghreb capitals: Morocco **Rabat**, Algeria **Algiers**, Tunisia **Tunis**. Maghreb ≠ Sahel.
15. Brazil points: **Selva** rainforest and **Terra Roxa** coffee soil are both true.
16. Australia's **interior is desert**. The **north is tropical**, not temperate. The **Darling Range** lies in **south-west** Australia.
17. India's operational Antarctic stations are **Maitri** and **Bharati**. **Dakshin Gangotri** is not the operational answer.
18. **Madeira** is Atlantic Portugal and is **not** Caribbean.
19. Indonesia west to east: **Sumatra → Java → Bali → Lombok**.
20. The **Mekong** delta is in **southern Vietnam**, not Cambodia.
21. South America's landlocked pair is **Bolivia** and **Paraguay**. Uruguay, Peru, and Suriname have coasts.
22. **Igarka** is in **Russia**, not China.
23. New Zealand's capital is **Wellington**. NZ has North and South Islands with **Cook Strait** and is **not** an Australian state.
24. **Ethiopia** is landlocked after Eritrea's secession; **Eritrea** holds the Red Sea coast. The Horn is Ethiopia–Somalia–Eritrea–Djibouti.
25. Only **double-landlocked** states in the usual set are **Uzbekistan** and **Liechtenstein**.
26. Capitals that are not the country's most-famous tourist city: Turkey **Ankara**, Australia **Canberra**, Brazil **Brasília**, UAE **Abu Dhabi**, NZ **Wellington**.
27. Boundary parallels: **38th** ≈ Koreas; **49th** ≈ Canada–USA (western land border).
28. South Asia map: Nepal and Bhutan are landlocked; Maldives and Sri Lanka are islands; **Kabul** is Afghanistan's capital.
29. The Caucasus trio is Georgia–Armenia–Azerbaijan; Istanbul is not Turkey's capital (**Ankara** is).
30. **Esperanto** is an artificial world auxiliary language. **Tamil** is an official language of **Singapore**. **Bahasa** = Indonesia (not Thailand).
31. Spanish is official in Chile/Colombia/Cuba — **not** Congo. Mandarin leads classic L-1 speaker counts; English often leads total L-1+L-2.
32. Great Lakes west→east: **Superior → Michigan → Huron → Erie → Ontario**. The St Lawrence is the seaway outlet.
33. Sahel is the semi-arid belt south of the Sahara — not the same as Maghreb (Morocco–Algeria–Tunisia).

34. Alaska is USA; Greenland is Denmark politically / North America geographically.
35. East Asia flash cards: Japan's **Honshu** is the main island; Korea split near the **38th parallel**; China's capital is **Beijing** (not Shanghai).
36. Pontic = northern Turkey; Zagros = western Iran — keep the west→east mountain ladder ordered.
37. ASEAN map diet often tests mainland SE Asia vs island SE Asia; Mekong countries vs maritime chokepoints (Malacca).
38. Caribbean vs Atlantic island trap: Madeira/Cape Verde are Atlantic African/Portuguese tags, not West Indies.
39. Landlocked Africa after Eritrea: Ethiopia; South America: Bolivia and Paraguay — do not add Uruguay.
40. Astana (Kazakhstan) and Ashgabat (Turkmenistan) capital swaps are common Central Asia traps with Tashkent/Dushanbe/Bishkek.
41. Palestine neighbours include Israel, Jordan, Egypt, Lebanon and Syria — map units remain West Bank and Gaza.
42. Northern Australia tropical climate ≠ temperate — a frequent Australia climate stem.
43. Cook Strait separates NZ North and South Islands; it is not an Australian state boundary.
44. Double-landlocked means every neighbour is landlocked — Uzbekistan (Central Asia) and Liechtenstein (Europe).
45. Igarka's Russia tag pairs with other "not China" Siberian place traps in world regional stems.

📋 Must-score facts — capitals, deserts, regional traps

Central Asia capitals

Country	Capital
Uzbekistan	**Tashkent**
Tajikistan	**Dushanbe**
Kyrgyzstan	**Bishkek**
Turkmenistan	**Ashgabat**

Regional locks

Item	Lock
Nobi / Kanto plains	**Japan** (≠ Korea)
Sunni Triangle	Baghdad–Tikrit–**Ramadi** (≠ Basra)
W Asia mountains W→E	Pontic → Zagros → Hindu Kush → Karakoram
Kara Kum	**Turkmenistan**
Borneo	Indonesia + Malaysia + Brunei
Korea	Seoul south / Pyongyang north (~38th parallel)
Gaza / West Bank	Egypt side / Jordan side

Topic 20 — World Agriculture

Consolidated — 45 Must-Score Facts

1. In plantation MCQs the classic plantation crop is **tea**, not wheat, rice, or maize. Plantation means estate, capital, monoculture, hired labour, and export. **Fazenda** = Brazil plantation; it is **not** jhum.
2. Coffee producer order often taught for **2016** is **Brazil > Vietnam > Colombia > Indonesia**. Brazil is mainly **arabica** on Terra Roxa; Vietnam is bulk **robusta**. **Mocha** = Yemen; **Kona** = Hawaii. Ethiopia is the arabica **origin** story, not that tonnage king.
3. Major cocoa producers are **Côte d'Ivoire, Ghana, and Cameroon**. **Latvia** is not a cocoa producer. Cocoa originated in the Amazon; chocolate manufacturing centres (Switzerland, Belgium) are not the same as growers.
4. India's rubber fact is **Kerala**. Rubber (*Hevea*) originated in the Amazon; **Henry Wickham** moved seed stock to Ceylon / Malaya. Malaysia's **Kinta Valley** is **tin**, not rubber. Tree tapping starts after about **6–7 years**. Recent world rubber volume often tags **Thailand**.
5. Shifting-cultivation names: **Jhum** (NE India), **Podu** (Andhra/Odisha), **Ladang** (Malaysia), **Milpa** (Mexico), **Roca** (Brazil), **Chena** (Sri Lanka), **Caingin** (Philippines), **Taungya** (Myanmar). It needs low density, short crop years and long fallow — not estate tea.
6. The world **citrus** / vine / olive belt is **Mediterranean** on five **west coasts** near **30–45°** (Med basin, California, C Chile, Cape, SW Australia), not equatorial. **Olive** majors = Spain, Italy, Greece.
7. Philippines cane and coconut history fact: **Spanish and Americans**. Dutch = Indonesia estates; British = Malaya rubber / Ceylon tea. Wrong Indian crop–state dumps include Gujarat–tea, UP–jute and Assam–wheat. Right pairs: **Kerala–rubber**; **Assam–tea** volume.
8. **Oil palm** majors are **Indonesia and Malaysia**. World **sugarcane** leader is **Brazil**; **sugar beet** belongs to temperate **Europe** (France / Russia / Ukraine belt).
9. For rice and wheat, **China and India** lead in **volume**, but classic **exporters** are often other countries (Thailand / Vietnam / India / Pakistan / USA for rice; Russia / USA / Canada / France / Australia for wheat). **Golden rice** carries **Vitamin A**.
10. Rice types: **Indica** (long-grain South / SE Asia), **Japonica** (stickier East Asia), **Javanica** (Indonesia). **IRRI** sits at **Los Baños, Philippines**; classic HYV rice = **IR-8**. Rice climate ≈ **20–27°C** and **100–200 cm** rain or irrigation.
11. Wheat needs cool growth and bright ripening (~**50–75 cm** rain). **Winter wheat** = autumn sow in mild-winter belts; **spring wheat** = spring sow in harsh-winter Prairie / Siberia; **durum** = pasta wheat of the Mediterranean.
12. **Intensive subsistence** = monsoon wet rice. **Extensive commercial grain** = Prairie / Pampas / Downs / steppe wheat. **Mixed farming** = crops plus livestock (West Europe / US Midwest). Ranching belts = Pampas, Prairie, Veld, Downs.
13. **Black tea** is fermented; **green tea** is not; **oolong** is semi-fermented. The pluck fact is **two leaves and a bud**. Tea likes **20–30°C**, **150–300 cm** rain, slope and acid soil. China leads tea volume; Assam leads Indian volume; Kenya is a classic African export name.
14. **Von Thünen** = market rings around a city (distance rent: dairy/veg → forestry → intensive crops → grain → ranching). **Whittlesey** = about **thirteen** world agricultural region types. Do not merge the two models.
15. Cotton: long staple = **Egypt / Sudan** (and Sea Island); medium = USA Upland; short staple = India–Pakistan–China belt. Maize volume order often **USA > China > Brazil**; **US Corn Belt** is the classic tag.
16. Banana: India produces heavily; **Ecuador** is a classic shipper. Silk and wool volume often centre on **China**; Australia is famous for **Merino** wool. **Shahtoosh** = Chiru wool. Pig numbers often tag **China**.
17. Soy export triangle = **USA–Brazil–Argentina**. India leads world **milk volume**; New Zealand and the Netherlands dominate the dairy-**export** story. World Food Day = **16 October**.
18. **IRRI** (Philippines) drove HYV rice; **CIMMYT (Mexico)** and **Norman Borlaug** drove HYV wheat — Nobel for **Peace** (**1970**), not "agriculture." **FAO** HQ = **Rome**. Green Revolution package = HYV + water + fertiliser + pesticide.

19. Term cards: **apiculture** = bees; **viticulture** = grapes; **sericulture** = silk (**China** leads volume); **floriculture** hub = Netherlands; **olericulture** = vegetables; **pisciculture** = fish rearing; **horticulture** = fruit / veg / flowers.
20. India often leads milk + banana + castor; China leads tea / tobacco / silk volume stories; Brazil leads cane + coffee #1; Thailand is a rubber story. **Golden Crescent** = Afghanistan–Iran–Pakistan opium.
21. Jute ("golden fibre") world is almost entirely **India and Bangladesh** on Ganga–Brahmaputra alluvium — do not treat UP as a jute state.
22. Plantation vs shifting: plantation = capital estate monoculture for export; shifting = humid-tropics slash-and-burn subsistence. They are not the same system.
23. Subsistence farming feeds the family with little surplus; **commercial** farming grows for market / export. **Sedentary** stays on fixed fields; nomadic herding and shifting plots **move**.
24. Mediterranean agriculture grows citrus, vine, olive, winter wheat and vegetables under **winter rain** and dry bright summers — not under equatorial cloud.
25. Dairy farming classics = NW Europe, Great Lakes, New Zealand, Denmark and the Netherlands. Truck / market gardening sits on **urban fringes** for perishable vegetables, fruit and flowers.
26. Commercial wheat belts: **Prairies** (Canada–USA), **Pampas** (Argentina), **Downs** (Australia), Ukrainian–Russian **steppe**, and N European plain (France export story).
27. Millets / coarse grains belong to **drought** belts of Africa and India — low rain, not 200 cm tea slopes.
28. Coffee climate wants **15–28°C**, **150–250 cm** rain, shade and no frost. Coffee rust (*Hemileia*) is a famous plantation disease linked to Ceylon's historic tea shift.
29. Cocoa wants equatorial heat, shade and high rain; pods grow on the trunk (**cauliflory**). Growers ≠ chocolate manufacturers.
30. Oilseeds map: groundnut tropics; rapeseed / canola cool temperate (Canada–EU–China–India); sunflower temperate (Russia–Ukraine–Argentina). Oil palm ≠ olive; oil palm ≠ rubber.
31. India coffee belt = Karnataka (Kodagu / Chikmagalur / Hassan) ahead of Kerala and Tamil Nadu — not Assam as coffee king.
32. Nomadic herding = Sahara–Arabia–Central Asia–tundra subsistence with camel, sheep, yak or reindeer. Livestock **ranching** on Pampas / Downs / Veld is commercial — do not merge the two.
33. Spice Islands / Maluku story = Indonesia cloves and nutmeg. Vietnam / India / Indonesia share the pepper trade map; cardamom classics = Guatemala and India's Kerala–Karnataka hills.
34. Collective / cooperative teaching examples include former USSR kolkhoz, Israel kibbutz and Denmark dairy co-ops — secondary to the main Whittlesey systems.
35. Producer ≠ exporter remains the chapter rule for rice, wheat, tea, banana and milk: China / India volume leaders often do **not** own the classic export nickname.
36. Rubber climate wants **25–35°C**, evenly high rain (**>200 cm**), humidity and no frost. Natural rubber ≠ synthetic neoprene chemistry fact.
37. Tea processing ladder: black = fully fermented; green = not fermented; oolong = semi-fermented; CTC = crush–tear–curl bulk; orthodox = whole-leaf quality.
38. Whittlesey's rice-dominant intensive subsistence belts = East / Southeast / South Asia deltas and coasts; intensive without paddy = North China, interior India wheat–millet and Nile-type belts.
39. Commercial plantation belts in the Whittlesey map include Caribbean estates, NE India–Sri Lanka tea, Malaysia–Indonesia rubber/palm, Brazil coffee and West Africa cocoa.
40. India cattle inventory is often #1, but beef-export leaders are usually Brazil / Australia / USA — do not equate herd size with beef-ship leadership.
41. Groundnut and tobacco volume stories often put **China** high; castor oilseed leadership often tags **India** in coaching frames.
42. Philippines coconut + cane credit is Spanish–American; do not dump Dutch (Indonesia) or British (Malaya) onto that pair.

43. Von Thünen's outer ring is livestock ranching; forestry sits near the city because wood is bulky — climate-region names belong to Whittlesey, not Thünen rings.
44. Hybrid-rice volume association often tags **China**; India's harvested rice **area** is often the largest even when China leads rice **output**.
45. World Food Day is **16 October**; FAO HQ is **Rome**; Borlaug's Nobel is **Peace (1970)** — three separate identity facts that papers still mix.

📋 Must-score facts — plantation, shifting, leaders

Crop / system tags

Item	Lock
Classic plantation crop (MCQ)	**Tea** (estate/capital/labour)
Coffee order (2016 teaching)	Brazil > Vietnam > Colombia > Indonesia
Cocoa majors	Côte d'Ivoire, Ghana, Cameroon
India rubber	**Kerala** (*Hevea* Amazon origin)
Oil palm majors	Indonesia, Malaysia
Olive majors	Spain, Italy, Greece
Sugarcane leader	**Brazil**; sugar beet = temperate Europe
Citrus belt	**Mediterranean**

Shifting cultivation names

Name	Region
Jhum	NE India
Podu	Andhra / Odisha
Ladang	Malaysia
Milpa	Mexico

Topic 21 — World Minerals & Energy

Consolidated — 46 Must-Score Facts

1. Coalfield–country pairs: **Appalachian–USA**, **Lancashire–England**, **Ruhr–Germany**, **Kuzbass–Russia**. **Donetsk / Donbas** = Ukraine's coal centre. **Karaganda** = Kazakhstan coal city.
2. **Mount Newman / Pilbara / Hamersley** = **iron** in Western Australia. **Krivoy Rog** = Ukraine iron. **Lorraine** and **Normandy** = France (Germany–Normandy iron is wrong). **Kiruna** = Sweden magnetite. **Mesabi** = USA Lake Superior.
3. Iron exporters = Australia (**Pilbara**) and Brazil (**Carajás** / Minas). China mines iron and still imports heavily. **Hematite** = red bulk ore; **magnetite** = black highest grade.
4. **Chile** leads copper with northern Andes **porphyry** deposits (**Chuquicamata**, **El Teniente**, Escondida). Also fact Katanga (DRC), Copperbelt (Zambia), and Bingham Canyon (USA).
5. Malaysia's **Kinta Valley** = **tin** (cassiterite). Also tin centres: Bangka–Belitung (Indonesia) and the Andes of Bolivia. Myanmar's **Pegu Yoma** = **mineral oil**, not tin (tin sits in Tenasserim).
6. **Kashagan** oil is in **Kazakhstan**, not Kuwait. **Burgan** = Kuwait. Also fact Ghawar / Dhahran (Saudi), Kirkuk / Zubair (Iraq), Haft Kel (Iran), Baku (Azerbaijan). **Brent** = North Sea light crude.
7. Natural gas's main constituent is **methane**. **LPG** (propane–butane) is not the same as **CNG**. Qatar's **North Field** and Iran's **South Pars** are one continuous Gulf gas giant.
8. **Postmasburg** (South Africa) = **manganese** (not uranium / mica / bauxite). Also Mn: Moanda (Gabon), Groote Eylandt (Australia), Nikopol (Ukraine).
9. Bauxite → aluminium: **Weipa / Gove / Darling Range** (Australia); **Guinea** for reserves renown; Les Baux (France) is the name origin. Mount Newman is **iron**, not bauxite.
10. Gold: **Witwatersrand** (South Africa historic); Kalgoorlie (Australia); modern volume often **China**. Diamond: **Kimberley** / kimberlite (South Africa); Golconda is historical **India**.
11. Nickel: **Sudbury** (Canada) and **Norilsk** (Russia). Chromite: **Bushveld** (South Africa) and Great Dyke (Zimbabwe). Lead–zinc: **Broken Hill** / Mount Isa (Australia).
12. Phosphate rock king = **Morocco**. Potash names include Canada (Saskatchewan) and the Dead Sea belt. **Iodine** (2018 spelling "Iodine") fact = **Chile** Atacama caliche.
13. Uranium volume often facts on **Kazakhstan**; Canada **Athabasca**; Australia **Olympic Dam** (U + Cu + Au). **Uranium City** = **Canada**. Thorium / monazite sands = India (Kerala–TN), Brazil, Australia.
14. **Lithium Triangle** = Chile–Argentina–Bolivia (Brazil is **not** in it). Cobalt volume = **DRC**. REE processing fact = **China**. Tungsten volume often **China**.
15. Nuclear and geothermal energy are **not** "stored solar". Wind, biomass and hydro are solar-linked renewables. Coal, oil and gas are conventional non-renewables. Tidal energy is linked to the Moon's gravity in usual teaching.
16. **OPEC** HQ = **Vienna**. Original OPEC five (**1960**): Iran, Iraq, Kuwait, Saudi Arabia, Venezuela. North Sea oil/gas = UK–Norway; West Siberia = Russia's giant hydrocarbon province. IAEA HQ is also Vienna — different job.
17. **Itaipu** = Brazil–Paraguay hydel on the Paraná. **Three Gorges** = China on the Yangtze. Steel tonnage top = **China**.
18. Ancient **shields** host many metals. Sedimentary **basins** host coal, oil and gas. Andean porphyry belts host copper, with lithium brines nearby in the same broad story.
19. **German silver** contains **no silver** (Cu–Ni–Zn). **Peace Pipeline** = Iran–Pakistan. Sheet **mica** classic = **India**. Platinum = Bushveld / Norilsk.
20. Coal rank rises in carbon from peat → lignite → bituminous → anthracite. Coking coal is for steel; lignite is thermal-only. Pennsylvania anthracite and South Wales are high-rank names.
21. Coal producer ≠ exporter: China is the huge volume miner–consumer; classic shippers include **Indonesia, Australia, Russia, USA and South Africa**.
22. Brass = copper + zinc; bronze = copper + tin. Chalcopyrite is the chief copper ore. Do not swap brass / bronze alloys.

23. Oil reserves names often put Venezuela / Saudi Arabia high; production names often put USA / Saudi / Russia high — freeze a year only when the paper quotes it.
24. Gas reserves names often tag Russia / Iran / Qatar; LNG export classics = Qatar / Australia / USA. Groningen = historic Netherlands giant; Urengoy = West Siberia volume gas.
25. Geothermal classics: Iceland (power + heating), New Zealand (Rotorua / Taupo), Philippines (Ring of Fire), USA The Geysers, Italy Larderello (first plant).
26. Tidal classics: **La Rance** (France) and Sihwa (Korea). India's potential teaching often ranks Gulf of Khambhat ahead of Kutch.
27. Nuclear electricity **share** classic = **France**; large **capacity** classic = USA. Main nuclear fuel = uranium.
28. Solar and wind capacity leadership often tags **China**; high wind-share teaching names include Denmark and the UK.
29. Saar = Germany–France border coal story; Silesia Upper = Poland coal–steel; Lusatia / Rhineland lignite = Germany — do not dump Saar as an iron-field only.
30. Australia Hunter Valley and Bowen Basin are export coal names; South Africa Witbank / Highveld are coal tags; India's Damodar (Jharia coking / Raniganj) is the India-map contrast.
31. Carajás = Brazil iron (Amazon-side giant); Itabira / Minas Gerais = Brazil iron quadrangle; Cerro Bolívar = Venezuela iron — keep them off the Pilbara list.
32. Sudbury yields nickel **with** copper; Grasberg (Indonesia) is copper **with** gold — polymetallic traps sit next to pure Cu/Ni dumps.
33. Morocco phosphate ≠ Moroccan oil; Broken Hill Pb–Zn ≠ Broken Hill iron; Mesabi = iron, not copper.
34. Silver volume often tags **Mexico** in recent frames; primary aluminium volume often tags **China**; diamond volume often tags **Russia**.
35. Non-ferrous teaching set = aluminium, copper, zinc, nickel, tin — pig iron and carbon steel are ferrous products.
36. Ras Tanura = Saudi export / refining terminal; Masjid-i-Suleiman = early Iran commercial oil story; Yenangyaung–Chauk sit with Myanmar's Pegu Yoma oil belt.
37. IEA HQ = Paris; IRENA HQ = Abu Dhabi; OPEC and IAEA share Vienna as a city but not as a job description.
38. Gasohol teaching ≈ 90% gasoline + 10% ethanol — USA classic producer/consumer frame.
39. First commercial oil teaching often parks Romania (1857 register) beside Drake 1859 Titusville (USA) — two classic "first well" stories, not one exclusive key.
40. Camphor tree native belt in coaching lists = China / Japan (*Cinnamomum camphora*) — not a mineral, but a GC extras neighbour on the same sheet.
41. Producer vs exporter for iron and coal: China can lead volume and still import; Australia / Brazil / Indonesia often own the shipping nickname.
42. Anthracite = highest-rank hard coal; lignite = brown soft thermal coal — swapping them flips coking / thermal logic.
43. Katanga (DRC) and Zambia Copperbelt are the African copper–cobalt pair opposite Chile's Andes porphyry story.
44. Olympic Dam (Australia) is the multi-metal U + Cu + Au deposit — do not park it as a pure gold or pure iron dump.
45. Hydro needs fall + discharge: Three Gorges / Itaipu / Grand Coulee / Aswan High / Guri are the usual world match bank.
46. Energy classification check: coal / oil / gas = conventional non-renewable; wind / biomass / hydro = renewable solar-linked; nuclear and geothermal stand apart from "stored solar" wording.

📋 Must-score facts — coalfields, ores, energy types

Coalfield ↔ country

Field	Country
Appalachian	USA
Lancashire	England
Ruhr	Germany
Kuzbass	Russia

Ore / oil tags

Item	Lock
Pilbara / Newman / Hamersley	Australia **iron**
Krivoy Rog	Ukraine iron
Chile copper	Andes porphyry (Chuquicamata)
Kinta Valley	Malaysia **tin**
Pegu Yoma	Myanmar **oil** (≠ tin)
Kashagan	**Kazakhstan** oil (≠ Kuwait)
Burgan	Kuwait
Postmasburg	South Africa **manganese**
Hematite / magnetite	Red bulk / black high-grade

Energy

Item	Lock
Natural gas main	**Methane**
LPG ≠ CNG	Different fuel forms
Nuclear / geothermal	**Not** “stored solar”

Topic 22 — World Industries

Consolidated — 45 Must-Score Facts

1. **Weight-losing** industries (iron–steel, sugar, timber) sit near bulky raw materials. **Weight-gaining** industries (bottling, brewing) sit near the **market**.
2. **Aluminium** and electro-chemicals seek cheap **hydel**. **Oil refining** often sits at a port or pipeline end. **Cement** follows bulky limestone.
3. **Footloose** industries (electronics, software, diamond cutting) are light and high-value — not tied to a coalfield. **Silicon Valley** (California) = electronics / IT, not Detroit.
4. Classic location factors are raw material, power, labour, market, transport, agglomeration and government/SEZ — not "any city works."
5. City–industry pairs: **Osaka–cotton**, **Detroit–auto**, **Cuba–cigar**, **St Petersburg–shipbuilding**.
6. **Akron** = tyres / rubber (USA). **Toulouse** = Airbus. **Seattle** = Boeing. **Hollywood / Los Angeles** = films. Swiss Jura = watches.
7. Japan nicknames: **Osaka** = Manchester of Japan (cotton); **Nagoya** = Detroit of Japan (autos); **Kawasaki** = Pittsburgh of Japan (steel). **Ivanovo** = Russian Manchester.
8. Japan belts: **Keihin** = Tokyo–Yokohama; **Hanshin** = Osaka–Kobe; **Chukyo** = Nagoya autos.
9. **Ruhr** (Germany) = coal, steel, heavy engineering (Essen–Dortmund–Duisburg). **Lancashire** = cotton; **Yorkshire** = wool.
10. **Pittsburgh–Great Lakes** = steel (Mesabi ore + Appalachian coal story). **Detroit** = automobiles and parts agglomeration.
11. Italy's industrial triangle is the **Po Basin / Milan–Turin–Genoa** belt. **Wolfsburg** and **Turin** are classic auto centres.
12. China's classic coastal manufacturing story includes the **Pearl River Delta**; **Shanghai** leads common container-port MCQs. Shipbuilding volume leaders = **China–South Korea–Japan**.
13. Europe coal–steel belt tags: **Saar**, **Lorraine** (France iron/steel), **Sambre–Meuse** (Belgium), **Upper Silesia** (Poland), **Randstad** (Netherlands port + manufacturing).
14. **Aberdeen** = oil capital of Europe. Japan's steel is largely **port / market-based** because ore and coal are imported.
15. **Igarka** is in **Russia** (Yenisei timber), not China. **Rotterdam** is in the **Netherlands**. **Montevideo** = Uruguay. **Jakarta** = Indonesia.
16. **Duisburg** is an inland **Rhine** port in **Germany**, not a Dutch sea mouth. Break-of-bulk points concentrate at ports and lake/rail junctions.
17. **Entrepôt** classics = **Singapore, Rotterdam, Hong Kong** — import, store/sort, re-export.
18. The **Suez Canal** joins the **Mediterranean** and **Red Sea**, cuts India–Europe distance by about **7000 km**, and is a **sea-level** cut (no stepped chambers).
19. Suez lakes north to south: **Manzala** → **Timsah** → **Great Bitter** → **Little Bitter**. **Port Said** = north end; **Suez** town = south end.
20. **Panama** joins Atlantic / Caribbean and Pacific **with stepped chambers** and **Gatun Lake**. **Kiel** joins the **North Sea** and **Baltic**.
21. Shipbuilding likes deep estuaries: Glasgow, St Petersburg, Yokohama, Busan, Belfast. Textiles seek labour + market clusters.
22. **Chinook** = warm dry Rockies wind; **Foehn** = Alps equivalent. Both can be true together in a multi-statement stem.
23. **Mistral** = southern France / Rhône (**not** Australia). **Shamal** = Arabia / Persian Gulf (**not** Austria). **Brickfielder** = Australia.

24. A **willy-willy** is an Australian **cyclone** name, not a local wind like Brickfielder.
25. More local-wind pairs: **Santa Ana** = California; **Haboob** = Sudan; **Yamo** = Japan; **Leveche** = Spain; **Bora** = Adriatic cold; **Harmattan** = West Africa; **Sirocco** = Sahara→Med; **Khamsin** = Egypt.
26. Cotton textiles favour humid labour districts (Lancashire, Osaka, Mumbai–Ahmedabad). Wool follows sheep hinterland + mill town (Yorkshire).
27. Iron–steel classic locations: coalfield, ore field, or lake/port mixing both. Sugar and sawmills pull to bulky perishable cane/timber.
28. Aircraft needs tech + large airfield (**Toulouse** Airbus; **Seattle** Boeing). Pharma / electronics pull capital and R&D.
29. **Least-cost** idea: plant sits where assembly + processing + distribution cost is lowest. Ubiquitous inputs (air, water) do not pull location; localised minerals do.
30. Agglomeration keeps auto parts around Detroit / Nagoya / Turin. Government / SEZ can create planned estates.
31. Do not swap Suez and Panama: Suez = Med–Red, sea-level cut; Panama = Atlantic–Pacific, with stepped chambers / Gatun.
32. Do not dump Duisburg into the Netherlands or call Igarka Chinese. Do not park Silicon Valley on Detroit.
33. Port–country traps: Rotterdam = Netherlands; Montevideo = Uruguay (not Argentina); Jakarta = Indonesia.
34. **Manchester of the East** tag for Osaka means cotton, not autos. Nagoya holds the auto nickname.
35. Container / entrepôt story: Shanghai for volume MCQs; Singapore–Rotterdam–Hong Kong for re-export hubs; Rhine mouth = Rotterdam.
36. Lorraine = France iron–steel; do not park Normandy iron under Germany (that belongs with France in mineral chapters).
37. Film industry classic = Hollywood / Los Angeles. Watch industry classic = Switzerland (Alps–Jura skilled labour).
38. Soft teaching still separates **primary** (mines/fields) from **secondary** (this chapter's factories) — world industries = secondary activities plus Lucent port/canal/wind lists.
39. **Break-of-bulk** industries (oil refining, flour, lake-steel) concentrate where cargo changes mode — ports, lakes, railheads.
40. Cold vs warm wind traps: Bora / Mistral are cold or cold-dry Mediterranean stories; Chinook / Foehn / Santa Ana are warm-dry downslope / desert-margin winds.
41. Harmattan = dusty West African winter wind; Sirocco / Khamsin / Leveche are hot dusty Sahara-outflow families toward the Med / Spain / Egypt.
42. Industrial region map spine: Ruhr, UK Lancashire/Yorkshire, US Great Lakes–Detroit, Po Basin, Japan Pacific belts, Pearl River Delta, Chotanagpur (India mineral heartland contrast).
43. When a stem says "Manchester of …", map cotton. When it says "Detroit of …", map autos. When it says "Pittsburgh of …", map steel.
44. Canal distance / geometry traps: Suez ~7000 km India–Europe saving; Panama uses stepped chambers; Kiel is North Sea–Baltic only.
45. Local-wind vs cyclone trap: Brickfielder = local wind (Australia); willy-willy = cyclone name (Australia) — do not merge them.

📋 Must-score facts — city–industry, canals, winds

City ↔ industry

City	Industry
Osaka	Cotton
Detroit	Auto
Cuba	Cigar
St Petersburg	Shipbuilding

City	Industry
Ruhr	Heavy industry (Germany)
Lancashire / Yorkshire	Cotton / wool
Pittsburgh–Great Lakes	Steel

Japan belts

Belt	Lock
Keihin	Tokyo–Yokohama
Hanshin	Osaka–Kobe
Chukyo	Nagoya autos

Canals / winds / industry type

Item	Lock
Suez	Med ↔ Red Sea; ~7000 km shorter India–Europe
Panama	Atlantic/Caribbean ↔ Pacific
Kiel	North Sea ↔ Baltic
Chinook / Foehn	Rockies / Alps warm dry winds
Mistral	S. France (≠ **Australia**)
Footloose	Electronics; aluminium seeks hydel; steel seeks ore+coal
Entrepôt	Singapore, Rotterdam, Hong Kong

Topic 23 — Political & Map-Based Geography

Consolidated — 48 Must-Score Facts

1. UNCLOS: **territorial sea 12 nm**, **contiguous zone 24 nm**, **EEZ 200 nm**. The continental shelf may extend to **350 nm**, but that does **not** push EEZ water beyond 200.
2. **Innocent passage** applies in the territorial sea. **Transit passage** applies in international straits. One nautical mile ≈ **1.852 km**.
3. **McMahon Line** = India–China (1914, Simla / Henry McMahon). **Durand Line** = Pakistan–Afghanistan (1893). **Radcliffe Line** = 1947 India–Pakistan/Bangladesh.
4. India has **seven** land neighbours (Pakistan, Afghanistan via Wakhan/PoK, China, Nepal, Bhutan, Myanmar, Bangladesh). Maritime neighbours are **Sri Lanka** and the **Maldives**.
5. Longest Indian **state** coastline = **Gujarat**. Longest land border = **Bangladesh**. Shortest land border = **Afghanistan**. Mainland plus islands coastline ≈ **7516 km**. Nine coastal states; **Telangana** is not coastal; shortest coastal state often **Goa**.
6. The **Suez Canal** joins Med and Red Sea, shortens India–Europe by about **7000 km**, and lakes run Manzala → Timsah → Great Bitter → Little Bitter.
7. **Panama** = Atlantic–Pacific canal (with stepped chambers / Gatun Lake). **Kiel** = North Sea–Baltic.
8. Straits: **Hormuz** = Persian Gulf oil chokepoint; **Malacca** = Indian Ocean–South China Sea; **Gibraltar** = Med–Atlantic; **Bosporus** = Black Sea–Marmara; **Bering** = Russia–USA.
9. Area ladder: Russia > Canada > USA > China > Brazil > Australia > **India (7th)**. Longest world coastline = **Canada** (then Indonesia, Russia, Philippines, Japan, Australia in the usual set).
10. Central Asia capital set: Uzbekistan **Tashkent**, Tajikistan **Dushanbe**, Kyrgyzstan **Bishkek**, Turkmenistan **Ashgabat**; Kazakhstan capital = **Astana** (Nur-Sultan phase), not Almaty.
11. **Bolivia** is landlocked among common South America traps. **Laos** = only SE Asia landlocked. **Nobi/Kanto** = Japan. **Igarka** = Russia.
12. **Thornthwaite** is the true **vegetation** climate index. **Köppen** gives letter-code climate classes. Mediterranean = **winter rain** (**Cs**). Western Europe = rain **all months** plus westerlies (**Cfb**).
13. India is the **seventh**-largest country, about **2.4%** of world land, with the Tropic of Cancer through the middle — so India is **not** wholly tropical. The Tropic does **not** cross **Uttar Pradesh**.
14. **Cape Verde** capital = **Praia**. **Bamako** = Mali. Only common double-landlocked states: **Uzbekistan** and **Liechtenstein**.
15. Uttar Pradesh's only foreign neighbour is **Nepal**. Mainland India latitudinal extent ≈ **8°4′N to 37°6′N**; **6°4′N** is Indira Point (islands).
16. Largest landlocked country by area = **Kazakhstan**. Most populous landlocked = **Ethiopia**. **Lesotho** is an enclave inside South Africa.
17. **49th Parallel** ≈ USA–Canada. **38th Parallel** ≈ Koreas. Maginot ≈ France–Germany; Rio Grande ≈ USA–Mexico; Oder–Neisse ≈ Germany–Poland.
18. Equator traps: Egypt and Mexico are **not** equatorial countries in the usual MCQ sense. North America and Oceania have no classic landlocked sovereign states.
19. Capitals ≠ famous cities: Australia **Canberra**; Brazil **Brasília**; Nigeria **Abuja**; Myanmar **Naypyidaw**; Tanzania **Dodoma**; Türkiye **Ankara**; Slovenia **Ljubljana** (Bratislava = Slovakia).
20. Old names: Siam→Thailand; Formosa→Taiwan; Gold Coast→Ghana; Dutch Guiana→**Suriname**; Southern Rhodesia→Zimbabwe; Ceylon→Sri Lanka; Burma→Myanmar; Abyssinia→Ethiopia.
21. Horn of Africa = Djibouti, Eritrea, Ethiopia, Somalia (**not Sudan**). Balkans include Slovenia/Greece/etc. — **Austria not** Balkan. Oceania excludes **Indonesia**. Caspian five exclude Armenia/Iraq.
22. Greenland = Denmark politically / N America geographically. Gaza borders **Egypt + Israel**. Afghanistan does **not** border Russia. Dead Sea shores = Israel / West Bank / Jordan — not Lebanon.

23. **Norway** = Land of the Midnight Sun (Arctic Circle). **Japan** = Land of the Rising Sun. **Finland** = Thousand Lakes. **(South) Korea** = Morning Calm. **Thailand** = White Elephants.
24. **South America** = Bird Continent. **Sri Lanka** = Mistress of the Eastern Sea / Pearl of the Indian Ocean. **Singapore** = Gateway to Asia. **Istanbul** = Gateway to the West.
25. City tags: **Venice** = canals; **Osaka** = Manchester of the East; **San Francisco** = Golden Gate; **Chicago** = City of Smoke; **Buenos Aires** = Paris of South America; **St. Petersburg** = Venice of the North.
26. **Pamir** = Roof of the World. **Baikal** = Pearl of Siberia. **Bahrain** = Island of Pearls. **Aberdeen** = Oil Capital of Europe. **Ninety East Ridge** = Indian Ocean.
27. LoC = India–Pakistan (J&K); LAC = India–China actual control; Sir Creek = India–Pakistan creek / maritime Kutch–Sindh story — do not dump them onto McMahon / Durand.
28. India's extremes teaching: south **Indira Point** (islands) / mainland Kanyakumari; west **Ghuar Mota** (Gujarat); east **Kibithu** (Arunachal); IST meridian **82°30′ E**.
29. Sri Lanka sits across **Palk Strait / Gulf of Mannar**; Maldives across the **8° Channel**. Tajikistan is **not** an India land neighbour.
30. UNCLOS timeline: adopted **1982**, in force **1994**, India party from **1995**. High seas beyond national zones keep freedom of navigation; the Area seabed is common heritage (ISA HQ Kingston).
31. Continent peaks match bank: Asia **Everest**; Africa **Kilimanjaro**; South America **Aconcagua**; North America **Denali**; Europe **Elbrus** (usual coaching); Antarctica **Vinson**; Australia **Kosciuszko** (continent) / Carstensz if Oceania framing.
32. Range–region pairs often asked: Andes–South America; Rockies–North America; Alps–Europe; Atlas–NW Africa; Great Dividing Range–Australia; Urals–Europe/Asia divide teaching.
33. River mouths / seas: Nile→Mediterranean; Amazon→Atlantic; Congo→Atlantic; Volga→Caspian; Danube→Black Sea; Rhine→North Sea; Indus→Arabian Sea; Ganga–Brahmaputra→Bay of Bengal.
34. Lakes: **Baikal** = deepest / largest freshwater by volume (Siberia); **Caspian** = largest lake / inland sea; **Superior** = largest freshwater by area among Great Lakes set; Dead Sea = hypersaline rift low.
35. Port–country map: Rotterdam–Netherlands; Shanghai–China; Singapore–Singapore; Hamburg–Germany; Santos–Brazil; Mombasa–Kenya; Karachi–Pakistan; Chittagong–Bangladesh.
36. Dependent territories: Greenland–Denmark; Falklands–UK; Puerto Rico–USA; Canaries–Spain; Azores–Portugal; Christmas Island–Australia.
37. Great Britain = England + Wales + Scotland (**not** Northern Ireland). UK adds Northern Ireland. Chechnya = republic of **Russia**.
38. Arabian Peninsula core = Saudi Arabia, Yemen, Oman, UAE, Bahrain, Qatar, Kuwait — **Syria not**. UAE has **seven** emirates.
39. Afghanistan neighbours = Iran, Pakistan, China, Tajikistan, Uzbekistan, Turkmenistan (+ India SE teaching note) — not Russia / Azerbaijan / Kyrgyzstan as the usual wrong trio.
40. Myanmar neighbours = India, Bangladesh, China, Laos, Thailand — not Vietnam / Cambodia / Malaysia as that trio.
41. Israel land borders = Lebanon, Syria, Jordan, Egypt. Chile's extreme N–S length is a common South America trap option.
42. Indonesia land borders = Malaysia, Papua New Guinea, Timor-Leste — **not Brunei**. Japan is the largest among common "island state without land border" options vs NZ / Philippines / Cuba.
43. South Asia area: India largest; **Maldives** smallest. South Asia set ≈ Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, Sri Lanka, Afghanistan.
44. Köppen quick tags: **Af** equatorial rainforest; **Am** monsoon; **Aw** tropical savanna; **BWh** hot desert; **Cs** Mediterranean winter rain; **Cfb** marine west-coast all-year rain; **ET/EF** tundra / ice.
45. Mountain pass / strait neighbours often revised with canals: Hormuz for Gulf oil; Malacca for East Asia trade; Gibraltar for Med access; Bosporus for Black Sea access.

46. Most megacities are coastal as **ocean gateways**. Alaska = USA; Malta = Mediterranean island state; Baikonur = **Kazakhstan**.
47. Mindanao = Land of Promise (Philippines teaching). Bird Continent = South America — not Australia. Morning Calm = Korea — not Japan.
48. Shelf rights beyond 200 nm (up to 350) are seabed rights — they do not create a wider EEZ water column. Contiguous zone is for customs / fiscal / immigration / sanitary control, not full sovereignty like the territorial sea.

📋 Must-score facts — UNCLOS, borders, straits

UNCLOS distances

Zone	Distance
Territorial sea	**12 nm**
Contiguous zone	**24 nm**
EEZ	**200 nm**
Continental shelf max	Up to **350 nm** (conditions)
Innocent passage	Territorial sea
Transit passage	International straits

Borders / lines

Line	Lock
McMahon	India–China (1914)
Durand	Pakistan–Afghanistan (1893)
Radcliffe	1947 India–Pakistan
Longest Indian state coastline	**Gujarat**
Longest land border	**Bangladesh**
Shortest land border	**Afghanistan** (teaching)
Area ladder	Russia > Canada > USA > China > Brazil > Australia > **India (7th)**

Straits

Strait	Lock
Hormuz	Gulf oil
Malacca	Indian Ocean–South China Sea
Gibraltar	Med–Atlantic
Bosporus	Black Sea–Marmara

Topic 24 — Geography of Uttar Pradesh

Consolidated — 28 Must-Score Facts

1. UP is India's most populous state and has the **largest rural** population. Area rank is **4th** (~**2.4 lakh km²**). There are **75** districts and **18** divisions. Capital is **Lucknow**; High Court is at **Prayagraj**. The State name date is **24 January 1950**; **Uttarakhand** was carved on **9 November 2000**.
2. Latitudes run roughly **23°52′–31°28′ N**; longitudes **77°–84° E**. The **Tropic of Cancer does not** enter UP. Present UP has **no Himalayan high peaks** after 2000 — **Nanda Devi** is in **Uttarakhand**.
3. Neighbours are Uttarakhand, Haryana, Delhi, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, Bihar and **Nepal**. The only foreign land neighbour is **Nepal**. Seven Nepal-border districts are Pilibhit, Lakhimpur Kheri, Bahraich, Shravasti, Balrampur, Siddharthnagar and Maharajganj.
4. District extremes: north **Saharanpur**, south **Sonbhadra** (सोनभद्र), east **Ballia**, west **Shamli**. **Lalitpur** has only **one** UP-district neighbour (**Jhansi**). **Devipatan** division HQ is **Gonda** (Gonda–Bahraich–Shravasti–Balrampur) — **not Basti**.
5. Physiographic map: **Bhabar** = pebble soak where streams sink; **Terai** = marshy re-emergence belt (Dudhwa / Katarniaghat / Suhelwa). **Doab** = Ganga–Yamuna Green Revolution heart; **Awadh** = Gomti–Ghaghara; **Rohilkhand** = Ramganga / mentha; **Purvanchal** = eastern rice plain; **Bundelkhand** = drought plateau; **Vindhyan** = Sonbhadra–Mirzapur energy fringe.
6. Köppen tag for the Gangetic plain is **Cwg** (monsoon with dry winter). Rain generally rises **west to east** (west doab ~**60–80 cm**; east / Terai ~**100–140 cm**). Classic wettest belt tag = **Gorakhpur**; driest = **Mathura**.
7. May–June **Loo** hits the western plain. Hottest summer belt tags include **Jhansi, Banda and Agra**. **Western Disturbances** give winter rain that supports **rabi wheat**. After Uttarakhand's creation, UP has **no alpine / snow climate**.
8. About **90%** of UP is **alluvial**. **Khadar** = new fertile floodplain; **Bangar** = older alluvium with **kankar**; **Bhur** = sandy western-doab ridges. **Usar / reh / kallar** on west–central canal tracts needs **gypsum plus drainage**, not lime alone.
9. Bundelkhand soils: **Mar / Kabar** = **black**; **Parua / Rakar (pathari)** = **red**. **Bhonta / Monta** = Vindhyan red–yellow (Mirzapur–Sonbhadra). Ravines mark the **Agra–Etawah–Jalaun** Chambal–Yamuna belt.
10. **Ramganga** meets the Ganga at **Kannauj**. **Betwa** meets the Yamuna at **Hamirpur** (not Prayagraj). **Ken** joins near **Banda**. Yamuna right-bank order west to east: **Chambal–Sind–Betwa–Ken**.
11. **Gomti** rises at **Madho Tala / Fulhar, Pilibhit** — Lucknow stands on it (2018 pollution / "biological disaster" tag). **Ghaghara / Saryu** passes Ayodhya; **Rapti** passes Gorakhpur. **Hindon** is a Yamuna **left**-bank NCR river.
12. National **Chambal** Sanctuary spans **UP–MP–Rajasthan** (Etawah–Agra) for gharial — **not** Haryana. **Badua** dam is **not** in UP. **Matatila** is on the **Betwa** in **Lalitpur**. **Ken–Betwa** is the live Bundelkhand interlinking project. Eastern UP hydel is blocked because the Himalaya to the north lies in **Nepal**.
13. Irrigation is dominated by **tubewells** (~**75%**; canals ~**15%**). Western UP faces groundwater **dark-zone** pressure. UP leads absolute tubewell area and groundwater resources for irrigation.
14. Canal classics: Upper Ganga offtake **Bhimgoda / Haridwar** (**1854**, P.T. Cautley); Lower Ganga **Narora** (**1878**); **Sharda** at **Banbasa** (longest UP system length); East Yamuna at Tajewala / Hathinikund; Agra Canal at Okhla; **Saryu Canal** National Project for eastern UP; plus Sarda Sahayak.
15. Crop map: **wheat** = west–centre rabi GR belt; **sugarcane** = west–centre (UP often **#1 quantity**, Maharashtra often higher **productivity**); **rice** = east and Terai; **potato** = UP leads quantity; **mentha** = Rohilkhand; oilseeds / pulses = Bundelkhand. UP–jute / UP–tea / UP–rubber are **wrong** pairs.
16. CIP potato centre fact = **Agra (Singna)**, not Aligarh. **IRRI-SARC** rice centre = **Varanasi**. Malihabad pairs with **Dasheri** mango.
17. **Sonbhadra** is the energy / mineral capital (non-coking Singrauli-side coal, limestone, thermal). **HINDALCO Renukoot** = aluminium. **Shankargarh** silica feeds **Firozabad** glass. **Chunar–Churk–Dalla** = cement limestone.

Sonrai (Lalitpur) = copper occurrence. **Mathura** has the oil refinery. Obra, Anpara and Rihand are thermal hubs.

18. City–industry pairs: **Kanpur** textile / leather; **Moradabad** brass; **Aligarh** padlocks / hardware; **Firozabad** glass; **Meerut** sports goods; **Bhadohi** carpets; **Khurja** pottery; **Saharanpur** woodcraft; **Varanasi** silk; Noida–Ghaziabad = NCR engineering. ODOP Moonj pair = **Amethi + Sultanpur**.
19. Forest cover is about **6%** of TGA (not 12.8%). ISFR >20% districts: **Chandauli** and **Shravasti**. Jaunpur–Ghazipur–Ballia have near-nil forest land. Terai–bhabar vegetation is moist deciduous (**sal**); Vindhyan south is dry deciduous.
20. UP's **only National Park** is **Dudhwa** (Lakhimpur Kheri). Tiger Reserves are Dudhwa (NP + Kishanpur + **Katarniaghat**), **Pilibhit**, **Amangarh (Bijnor)** and **Ranipur (Chitrakoot)** — do not invent extra NPs. **Katarniaghat** WLS = Bahraich, **1975**. **Chandraprabha** WLS = **Chandauli**.
21. Other sanctuary tags: **Hastinapur** (Meerut–Bijnor–Amroha; swamp deer); **Kaimoor** (Sonbhadra–Chandauli); **Suhelwa** (Balrampur–Shravasti–Gonda); **Sohagibarwa** (Maharajganj); **Mahavir Swami** (Lalitpur).
22. UP has **13** Ramsar sites; India's **100th** was **Surha Tal / JP Narayan Bird Sanctuary, Ballia**. Sites **not** in UP: Rudrasagar (Tripura), Sultanpur (Haryana), Surinsar–Mansar (J&K). **Sarsai Nawar** and **Samaspur are** in UP.
23. State animal = **swamp deer**; state bird = **Sarus crane**; state flower = **Butea / Dhak / Palash** (*Butea monosperma*).
24. Census 2011 freeze: UP ~**16.5%** of India; density **829**; sex ratio **912**; literacy ~**67.7%**; urban ~**22.3%**. Lowest female literacy among common facts = **Shrawasti**. Highest literacy belt = **Gautam Buddha Nagar**. Smallest-area trap among listed four = **Hapur**. Largest-area standard = **Lakhimpur Kheri**.
25. Density extremes: highest **Ghaziabad**; lowest belt **Lalitpur / Sonbhadra**. UA 2011 order **Kanpur > Lucknow > Ghaziabad > Agra**. Kanpur crossed a million in **1971**; Lucknow in **1981**.
26. Central / Mission Smart Cities = **10** (Lucknow, Kanpur, Prayagraj, Agra, Varanasi, Aligarh, Bareilly, Jhansi, Moradabad, Saharanpur). **Ghaziabad is not** in that Central-10. **HRIDAY** in UP = **Varanasi and Mathura** only.
27. NER HQ = **Gorakhpur**; NCR HQ = **Prayagraj**. No major seaport; waterway fact = **NW-1** on the Ganga. **Kushinagar** international airport opened **24 June 2020**. Jewar = Noida International. Amrit Bharat UP stations fact = **149**.
28. Expressways: **Yamuna** (Noida–Agra) was the first; also Agra–Lucknow, **Purvanchal** (Lucknow–Ballia via **Mau**), Bundelkhand, Ganga Expressway and Gorakhpur Link. Built Purvanchal does **not** run through Basti.

📋 Must-score facts — parks, rivers, soils, Census

Parks / wetlands

Item	Lock
Only National Park	**Dudhwa** (Lakhimpur Kheri)
Chandraprabha	Sanctuary — **Chandauli**
Forest cover	~**6%** (not 12.8%)
Ramsar count (teaching)	**13** in UP
Not in UP	Rudrasagar, Sultanpur, Surinsar–Mansar

Rivers / soils / agri

Item	Lock
Ramganga–Ganga meet	**Kannauj**
Betwa–Yamuna meet	**Hamirpur** (≠ Prayagraj)
Bundelkhand soils	Mar/Kabar = **black**; Parua/Rakar = **red**
Alluvial share	~**90%** of UP

Item	Lock
Matatila	Betwa — **Lalitpur**
Potato	UP leads quantity; CIP **Agra (Singna)**
Mentha	Rohilkhand / Tarai belts

Admin / Census traps

Item	Lock
Devipatan HQ	**Gonda** (≠ Basti)
Lowest female literacy (common)	**Shrawasti**
NER HQ	**Gorakhpur**
Smart Cities	**10**; **Ghaziabad** out of central list teaching

Uttarakhand — Topic 1 — Location, Relief & Structure

Consolidated — 22 Must-Score Facts

1. Uttarakhand occupies the **central Himalayan** sector of India, roughly **28°43′–31°27′ N** and **77°34′–81°02′ E**.
2. **Himachal Pradesh** borders the west; **Nepal** the east; the **Tibetan** frontier the north; **Uttar Pradesh** plains the south.
3. Historic–administrative divisions: **Garhwal** is west; **Kumaon** is east.
4. Garhwal districts include Dehradun, Tehri, Uttarkashi, Chamoli, Rudraprayag, and Pauri with related hill tehsils.
5. Kumaon districts include Nainital, Almora, Bageshwar, Champawat, and Pithoragarh with related hill tehsils.
6. **Haridwar** and **Udham Singh Nagar** sit on the southern Tarai fringe of this dual map.
7. South→north relief stack: **Tarai** → **Bhabar** → **Shiwalik** → **Dun** → **Lesser Himalaya** → **Great Himalaya**.
8. Compact four-name lists often read **Tarai** → **Shiwalik** → **Lesser Himalaya** → **Great Himalaya**; placing Tarai last is the usual inversion trap.
9. National plains teaching reads the same foothill belts southward as **Shiwalik** → **Bhabar** → **Terai** — do not reverse either list without checking where the stem starts.
10. **Bhabar** is porous gravel–boulder at the Himalayan foot where streams often **sink** underground.
11. **Tarai** is the marshy belt south of Bhabar where groundwater **reappears** and soils support dense cultivation and forest.
12. **Shiwaliks** are young, soft foothills of unconsolidated sediments and are landslide-prone.
13. **Duns** are longitudinal flat-floored valleys in the Shiwalik belt; **Dehradun** is the classic dun.
14. **Main Boundary Thrust (MBT)** separates the Shiwaliks from the Lesser Himalaya.
15. **Main Central Thrust (MCT)** separates the Lesser Himalaya from the Higher / central crystalline belt.
16. **South of the MBT** = Shiwalik foothills and duns; **between MBT and MCT** = **Lesser Himalaya**.
17. **North of the MCT** = Higher / central crystalline **Great Himalaya** (snow line, major peaks, glacier heads).
18. Tarai and Bhabar sit **south of the Shiwalik**, not between the thrusts.
19. Farther north, **Trans-Himalaya / Tethys Himalaya** is a separate structural family beyond the Great Himalayan crest.
20. **Nanda Devi** is the highest major UK peak at about **7816–7817 m** in **Chamoli**; **Kamet** is next among major peaks at about **7756 m**.
21. Named Garhwal crest peaks also include **Trisul**, **Dunagiri**, and **Chaukhamba**.
22. Great Himalayan glaciers (Gangotri, Pindari, Milam and related ice fields) feed Bhagirathi–Alaknanda headstreams and shape pilgrimage–tourism geography.

Uttarakhand — Topic 2 — Climate & Drainage

Consolidated — 28 Must-Score Facts

1. **Kali (Sharda)** is the longest river in Uttarakhand at **252 km**; source Lipulekh / Kalapani; it forms the international boundary with Nepal.
2. **Bhagirathi (205 km)** and **Alaknanda (195 km)** unite at **Devprayag** to form the **Ganga**.
3. West→east river alignment: **Yamuna → Bhilangana → Alaknanda → Gori Ganga**.
4. Bhagirathi rises at **Gaumukh** from the **Gangotri Glacier** (Uttarkashi); Alaknanda rises from **Satopanth** and **Bhagirath Kharak** (Chamoli).
5. **Yamuna** rises from the Yamunotri Glacier (Bandarpunch, Uttarkashi); **Tons** carries about **twice the water volume** of the Yamuna.
6. **Western Ramganga** rises in the Dudhatoli range and traverses Corbett National Park into UP; **Kosi (Kaushiki)** drains Almora–Ramnagar and joins Ramganga in UP.
7. **Saryu** rises at Sarmool (Bageshwar) and meets Kali at **Pancheshwar**; **Mandakini** rises from Chaurabari Glacier and meets Alaknanda at Rudraprayag.
8. **Bhilangana** rises from Khatling Glacier (Tehri) and is submerged in the Tehri Dam reservoir.
9. Panch Prayag north→south: **Vishnuprayag → Nandaprayag → Karnaprayag → Rudraprayag → Devprayag**.
10. Vishnuprayag = Alaknanda + Western Dhauliganga; Nandaprayag = Alaknanda + Nandakini; Karnaprayag = Alaknanda + Pindar.
11. Rudraprayag = Alaknanda + Mandakini; Devprayag = Alaknanda + Bhagirathi → Ganga.
12. Longest river = **Kali (252 km)**; river with greatest water volume among common traps = **Alaknanda**.
13. **Gangotri Glacier** (Uttarkashi) is the largest in Uttarakhand (~**30 km** long, 2 km wide); snout **Gaumukh** (~3,900 m).
14. **Milam Glacier** (Pithoragarh) is the largest in Kumaon (~**16 km**); source of **Gori Ganga**.
15. **Pindari Glacier** (Bageshwar) feeds the **Pindar**; discovered / crossed by **G. W. Traill** via Traill's Pass (1830).
16. **Khatling Glacier** (Tehri) feeds Bhilangana; **Chaurabari Glacier** (Rudraprayag) feeds Mandakini and Chaurabari Tal (Gandhi Sarovar) — linked to the June 2013 Kedarnath disaster.
17. **Roopkund** (Chamoli, ~5,029 m) is the "Mystery / Skeleton Lake" with 9th-century skeletal remains on the Nanda Devi Raj Jat route.
18. **Naukuchiatal** (Nainital) is the deepest lake in Kumaon (~40 m, nine corners); **Bhimtal** is the largest lake in Kumaon.
19. **Nainital Lake** is pear-shaped, discovered by **P. Barron** in 1841; **Dodital** (Uttarkashi) is the six-cornered Ganesha lake feeding Asiganga.
20. **Hemkund (Lokpal)** (Chamoli) is sacred among seven snow peaks; **Deoria Tal** (Rudraprayag) reflects the Chaukhamba peaks.
21. Panch Kedar body parts: Kedarnath = hump; Madhyamaheshwar = navel; **Tungnath = arms** (highest Shiva temple, **3,680 m**); Rudranath = face; Kalpeshwar = hair (open all year).
22. All **Panch Badri** seats are in **Chamoli**: Vishal Badri (Badrinath), Yogdhyan Badri, Bhavishya Badri, Vridha Badri, Adi Badri.
23. Kali major junctions include Gori Ganga at **Jauljibi** and Saryu at **Pancheshwar**; Gori Ganga is the eastern Pithoragarh trunk in west→east lists.
24. Glacier-fed Himalayan rivers and steep gradients shape both drainage and hydropower potential of the state.
25. High-altitude lakes and glacial tarns (Sahasratal on the Khatling ridge; Roopkund; Hemkund) sit on pilgrimage and adventure circuits.
26. Do not swap Gangotri (largest in UK) with Milam (largest in Kumaon).
27. Do not swap Vishnuprayag (Alaknanda + Western Dhauliganga) with Devprayag (Alaknanda + Bhagirathi).

28. Do not swap Tungnath (highest Shiva temple / arms) with Kedarnath (supreme Jyotirlinga / hump).

Uttarakhand — Topic 3 — Vegetation & Wildlife

Consolidated — 24 Must-Score Facts

1. Forest richness of Uttarakhand is explained primarily by **topography and climate** packing many belts into a short horizontal distance.
2. Altitude stack: foothill **sal** → **chir** → oak–rhododendron → **deodar / fir** → alpine **birch–juniper** → **bugyal**.
3. Foothill **sal** and mixed deciduous forests occupy Tarai–Bhabar and the lower Shiwalik face.
4. Sub-tropical **chir** pine commonly sits about **900–2000 m** on outer slopes.
5. **Deodar** is abundant in the western Himalayan temperate belt at about **1500–2500 m**.
6. Among deodar / pine / sal / teak options, **deodar** grows at the **highest** elevation from sea level.
7. **Birch** belongs to Himalayan alpine / moist alpine scrub and has paper-like bark once used for writing.
8. Above the tree-line, alpine meadows called **bugyals** support seasonal pastures and high-altitude herbs.
9. Southern Himalayan slopes get more rain and sunlight than the rain-shadow north face, so vegetation is denser on the south.
10. Tree-line is generally **lower** in the Western Himalaya than in the Eastern Himalaya.
11. **Lisa** is pine resin; distillation yields turpentine and residual resin used in paint, varnish, and related industries.
12. **Khair** is a **sub-tropical** forest tree of Shiwalik / western Himalaya scrub and also yields **katha**.
13. **Burans** is the rhododendron flowering hill shrub — do not confuse it with Lisa or deodar.
14. **Nanda Devi Biosphere Reserve** is a UNESCO World Heritage property with Valley of Flowers; fauna includes **snow leopard, musk deer, and blue sheep (bharal)**.
15. **Askot** sanctuary = **Pithoragarh**; **Binsar** sanctuary = **Almora** — classic district swap trap.
16. **Govind Pashu Vihar / NP** = **Uttarkashi**; **Nandhaur** sanctuary = **Nainital**.
17. National Park year spine: **Corbett 1936**, **Rajaji 1983**, **Gangotri 1989**, **Govind 1990** (Govind is not 1989).
18. Corbett carries India's first national-park tradition; keep year pairs stable across sanctuary–NP match stems.
19. Forests remain the major natural resource of the western Himalayan region and support rural livelihoods through resin, timber, and NTFP.
20. Oak–rhododendron and mixed broad-leaved forests mark the temperate mid-hill transition below the deodar–fir belt.
21. Higher belts carry **fir and spruce** before alpine birch–juniper scrub toward the tree-line.
22. Keekar and acacia appear with khair in sub-tropical scrub above about **1000 m**.
23. Swapping Askot and Binsar districts loses sanctuary–district match codes.
24. Protected-area size crowns and Ramsar detail live mainly in the Environment UK parks chapter; this card keeps vegetation belts, products, and district–sanctuary pairs.

Uttarakhand — Topic 4 — Minerals, Power & Industry

Consolidated — 18 Must-Score Facts

1. Uttarakhand has scattered Himalayan–foothill minerals; it does **not** have one giant coal–iron belt.
2. **Dolomite** pairs with **Dehradun** and **Pithoragarh**.
3. **Soapstone** pairs with **Chamoli** and **Bageshwar**.
4. **Graphite** pairs with **Almora** and **Nainital**.
5. **Silica sand** pairs with **Dehradun, Nainital, and Tehri Garhwal** — **not Pauri Garhwal**.
6. Pauri Garhwal is the planted wrong match for silica sand in the standard pairing.
7. Limestone and magnesite appear as minor coaching occurrences; the four named pairs above carry the heaviest recall weight.
8. Hydropower dominates the power story because of steep gradients and glacier-fed rivers.
9. Major project belts follow the **Bhagirathi**, **Alaknanda**, and **Yamuna** systems.
10. The **Tehri** hydropower complex on the Bhagirathi near the Bhilangana confluence is the flagship multipurpose project (power, flood moderation, storage).
11. Named coaching heads also include run-of-river and storage schemes such as **Vishnuprayag** and **Srinagar** on the Alaknanda belt, plus Yamuna-side projects in western Garhwal.
12. High alpine sites face glacial and seismic constraints; **thermal coal is not** the core identity of the UK power map.
13. Industry concentrates in **Tarai / plains** districts and the **Dehradun** dun more than in high alpine tehsils.
14. **SIIDCUL** anchors planned industrial estates in the plains–dun belt.
15. Coaching industrial nodes include **Haridwar**, **Pantnagar / Rudrapur** (Udham Singh Nagar), and **Sitarganj**-side Tarai estates.
16. **Udham Singh Nagar** hosts agro-based and light manufacturing linked to the Tarai farm belt.
17. **Dehradun** carries administration, education, pharma, IT, and service-industry pockets; forest-based and tourism services matter in mid-hill towns.
18. Heavy smokestack industry is limited by terrain, ecology, and transport cost — light manufacturing and services dominate instead.

Uttarakhand — Topic 5 — Agri, AH & Irrigation

Consolidated — 18 Must-Score Facts

1. Cropped area concentrates in the southern plains and Tarai where continuous fields are possible on gentler slopes.
2. **Haridwar** and **Udham Singh Nagar** form the granary / high cultivated-area belt.
3. Series B for 2025 marks **Haridwar** as the district with the **highest agricultural land**; many notes also name **Udham Singh Nagar** as the Tarai rival — keep both names.
4. Hill agriculture is fragmented and **terrace-based** on steep slopes.
5. Mid-hill farms grow cereals, millets, pulses, and oilseeds on small holdings.
6. Horticulture matters in suitable altitudes — apples, stone fruits, citrus, and off-season vegetables.
7. Plains and Tarai emphasise **rice**, **wheat**, **sugarcane**, and oilseeds under denser irrigation.
8. Hills emphasise **mandua (finger millet)** and **jhangora**, with barley and other hardy cereals on terraces.
9. Pulses and potatoes appear widely as catch or kitchen crops in mid-hills; orchard and vegetable belts add cash income where roads allow.
10. Crop choice follows altitude, slope, and water availability — the state does not run on a single statewide monoculture.
11. Animal husbandry is a **supplementary** occupation to agriculture through milk, wool, manure, and household nutrition — not the sole livelihood for most households.
12. Cattle and buffalo support milk and draught in plains and mid-hills; sheep and goats matter in higher pastures.
13. In the 2025 multi-statement AH stem, Series B accepts **Only statement 1** (supplementary occupation); the sheep-rearing environmental-degradation claim is not accepted in that key.
14. **Tarai / plains** irrigation relies on **canals and tubewells** (Haridwar and Udham Singh Nagar type districts).
15. Mid-hill systems rely more on **springs** and small gravity channels called **guls**; tanks appear in some mid-hill systems.
16. Steep terrain limits large canal networks in the higher Himalaya, so irrigation intensity falls from Tarai granary districts toward high alpine valleys.
17. Plains lead cultivated-area stories; hills lead terrace / horticulture stories — do not invert the contrast.
18. Fresh land-use releases can shuffle absolute Haridwar vs Udham Singh Nagar ranks; prefer the Series B letter when that paper is cited.

Uttarakhand — Topic 6 — Population, SC/ST & Settlements

Consolidated — 26 Must-Score Facts

1. Census 2011 total population of Uttarakhand is **1,00,86,292** (~1.01 crore).
2. Highest population district = **Haridwar** (~18.90 lakh); lowest = **Rudraprayag** (~2.42 lakh).
3. State decadal growth (2001–11) = **18.81%** (national 17.70%); highest growth = **Udham Singh Nagar (33.45%)**.
4. Two districts registered **negative** decadal growth: **Pauri Garhwal (−1.41%)** and **Almora (−1.28%)** — the ghost-village / *palayan* story.
5. State density = **189** persons/sq km; highest = **Haridwar (801)**; lowest = **Uttarkashi (41)**.
6. State sex ratio = **963** (national 943); highest = **Almora (1142)**; lowest = **Haridwar (880)**.
7. Sex-ratio descending trap set: **Almora (1142) → Rudraprayag (1114) → Tehri Garhwal (1077) → Haridwar (880)**.
8. Seven hill districts have sex ratio above **1000** due to heavy male out-migration; plains districts Haridwar (880) and Dehradun (889) are among the lowest.
9. Child sex ratio (0–6) = **890** (national 919); Almora highest (922); Pithoragarh lowest (816).
10. Overall literacy = **78.82%**; highest = **Dehradun (84.25%)**; lowest = **Udham Singh Nagar (73.10%)**.
11. Highest male literacy = **Rudraprayag (93.90%)**; lowest female literacy extreme = **Uttarkashi (59.74%)** (only district under 60% female literacy).
12. SC share = **18.76%** of total population; highest SC % = **Bageshwar (27.73%)**; lowest SC % = **Dehradun (14.5%)**.
13. ST share = **2.89%** of total population; highest ST % = **Udham Singh Nagar (7.46%)**; lowest ST % = **Rudraprayag (0.15%)**.
14. **Udham Singh Nagar** holds about **42.1%** of the state's tribal population (mainly **Tharu** and **Buksa**).
15. **Dehradun** holds about **38.2%** of the tribal population (mainly **Jaunsari** in Jaunsar-Bawar).
16. **Pithoragarh** holds **Bhotia** (Johar/Darma) and the primitive **Raji (Vanrawat)**; **Chamoli** holds Marchha and Tolchha Bhotias; **Uttarkashi** holds Jad Bhotias.
17. **Vanrawats / Raji** are forest-dwelling groups concentrated in **Pithoragarh** and historically practised **shifting cultivation** until restricted by forest laws.
18. Urban population = **30.23%** (rural 69.77%); most urbanized = **Dehradun (55.52%)**; least urbanized = **Bageshwar (3.49%)**.
19. Mid-Himalayan settlements are often nucleated on southern spurs for sunlight with terraced *khet* cultivation.
20. Doon and Tarai valleys show dispersed and linear settlement ribbons along highways and canal banks.
21. Alpine zones retain transhumant seasonal dwellings (*Mait* in summer pastures; *Gunda/Munsa* in winter valleys).
22. Highest SC population count is in **Haridwar**; highest ST population count is in **Udham Singh Nagar**.
23. Do not swap highest overall literacy (**Dehradun**) with highest male literacy (**Rudraprayag**).
24. Do not swap highest SC % (**Bageshwar**) with highest ST % (**Udham Singh Nagar**).
25. Do not swap highest density (**Haridwar**) with lowest density (**Uttarkashi**).
26. Negative growth belongs to **Pauri** and **Almora** — not to Chamoli/Rudraprayag alone.

Uttarakhand — Topic 7 — Transport, Tourism & Hazards

Consolidated — 28 Must-Score Facts

1. **Silkyara–Barkot** bi-directional tunnel is **4.531 km** long on NH-134 under the Char Dham project and shortens the Gangotri–Yamunotri highway distance by more than **20 km**.
2. Airports / airstrips: **Jolly Grant** (Dehradun), **Pantnagar** (Udham Singh Nagar), **Naini Saini** (Pithoragarh), **Chinyalisaur** (Uttarkashi), **Gauchar** (**Chamoli**).
3. **Gauchar** is in **Chamoli** — not Rudraprayag (classic airport mismatch).
4. **Lipulekh Pass** (~5,334 m) links Pithoragarh with Tibet on the Kailash–Mansarovar route (India–Nepal–China tri-junction).
5. **Mana (Dungri-La)** (~5,608 m) and **Niti** (~5,068 m) link Chamoli with Tibet; Niti closed after the 1962 war.
6. **Sinla Pass** connects **Darma** and **Byans** valleys inside Pithoragarh (internal inter-valley pass).
7. **Traill's Pass** (~5,312 m) links Bageshwar (Pindari) with Pithoragarh (Milam); crossed by G. W. Traill in **1830**.
8. **Kalindi Pass** (~5,950 m) is a severe glaciated mountaineering pass linking **Gangotri (Uttarkashi)** with **Badrinath (Chamoli)**.
9. **Shringkanth Pass** connects Uttarkashi with Kinnaur / Himachal Pradesh.
10. Other Tibet-facing Garhwal–Kumaon passes include Thang-La, Unta Dhura, Chorhoti, and related high *la / dhura* corridors.
11. **Bugyals** are alpine meadows above the tree-line (about **3,000–4,500 m**); called Bugyal in Garhwal and Payar / Bugyal in Kumaon.
12. **Bedni Bugyal** (Chamoli) is the **largest** bugyal in the state and lies on the Nanda Devi Raj Jat route; Brahma Kamal flourishes here.
13. **Auli Bugyal** (Chamoli) is the premier winter skiing destination with a ropeway from Joshimath.
14. **Dayara Bugyal** (Uttarkashi) hosts the **Anduri Utsav / Butter Festival** on Bhado Sankranti.
15. **Chopta** (Rudraprayag) is called the "Mini Switzerland of Uttarakhand" and is the base for Tungnath–Chandrashila treks.
16. State Council for Climate Change (**SCCC**) was established in **2011**; UAPCC drafting belongs to the **Forest & Environment** side — not the Revenue Department.
17. Uttarakhand lies in **Seismic Zones IV and V** with high vulnerability along the Central Seismic Gap and MBT–MCT belt.
18. **Uttarkashi earthquake** of **20 October 1991** (~6.6 M) killed over **768** people along the MCT belt.
19. **Malpa landslide** of **17–18 August 1998** (Pithoragarh, Kali valley) killed about **221**, including Kailash–Mansarovar pilgrims and dancer **Protima Bedi**.
20. **Chamoli earthquake** of **29 March 1999** (~6.8 M) caused over **100** fatalities around Chamoli / Gopeshwar / Rudraprayag.
21. **Kedarnath** disaster of **16–17 June 2013** combined cloudburst with **Chaurabari Lake (Gandhi Sarovar)** breach and Mandakini deluge; over **5,700** fatalities and later SDRF strengthening.
22. **Tapovan / Rishiganga** disaster of **7 February 2021** (Chamoli) was a rock-ice avalanche that destroyed Rishi Ganga hydel and damaged Tapovan-Vishnugad; over **204** dead / missing.
23. **Joshimath land subsidence** (January 2023) affected wards built on ancient landslide moraine debris and forced large evacuations.
24. Do not swap **Sinla** (internal Darma–Byans) with **Lipulekh** (international to Tibet).
25. Do not swap **Dayara** (Butter Festival, Uttarkashi) with **Bedni** (largest bugyal, Chamoli).
26. Do not swap **Malpa 1998** (landslide / rockfall on Kali) with **Kedarnath 2013** (GLOF / cloudburst on Mandakini).
27. Silkyara exact length trap distractors are **3.531 km** and **6.531 km** — correct is **4.531 km**.
28. Char Dham corridor, border roads, and alpine tourism all sit on the same fragile slope–seismic hazard map.